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DEVICES AND METHODS FOR EXCLUDING THE LEFT ATRIAL APPENDAGE

Abstract

Devices and methods are described for occluding the left atrial appendage (LAA). The device excludes the LAA from blood flow to prevent blood from clotting within the LAA and subsequently embolizing, particularly in patients with atrial fibrillation. The implantable device is delivered via transcatheter delivery into the LAA and secured within the LAA. The implant comprises an expandable and compliant frame and an expandable and conformable tubular foam body carried by the frame. The device may have a thromboresistant cover at a proximal end and a thromboresistant coating on the foam body. The frame may have recapture struts inclining radially outwardly in the distal direction from a central hub. The frame may have axially extending side wall struts, with adjacent pairs of side wall struts joined at one or more apexes. Anchors extend from the frame to engage tissue. The anchors can also be reversible to allow retraction of the anchors and repositioning or retrieval of the device.

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Background/Summary

INCORPORATION BY REFERENCE TO ANY RELATED APPLICATIONS [0001] Any and all applications for which a foreign or domestic priority claim is identified in the Application Data Sheet as filed with the present application are hereby incorporated by reference under 37 CFR 1.57. [0002] This application is a continuation of U.S. application Ser. No. 18/486,974, entitled DEVICES AND METHODS FOR EXCLUDING THE LEFT ATRIAL APPENDAGE and filed Oct. 13, 2023, which is a divisional of U.S. application Ser. No. 17/338,520, entitled DEVICES AND METHODS FOR EXCLUDING THE LEFT ATRIAL APPENDAGE and filed Jun. 3, 2021, which is a continuation of U.S. application Ser. No. 15/795,083, entitled DEVICES AND METHODS FOR EXCLUDING THE LEFT ATRIAL APPENDAGE and filed Oct. 26, 2017, which claims the benefit of priority to U.S. provisional patent application No. 62/459,503, entitled DEVICES AND METHODS FOR EXCLUDING THE LEFT ATRIAL APPENDAGE and filed Feb. 15, 2017, and to U.S. provisional patent application No. 62/413,632, entitled MULTIFUNCTIONAL LEFT ATRIAL APPENDAGE (LAA) OCCLUDER and filed Oct. 27, 2016, the disclosure of each of which is hereby incorporated by reference herein in its entirety for all purposes and forms a part of this specification.

BACKGROUND

Field

[0003] This development relates generally to systems, devices and methods for excluding the left atrial appendage (LAA). In particular, systems, devices and methods for excluding the LAA using an expandable foam implant with a deployable and compliant frame are described herein.

Description of the Related Art

[0004] Atrial fibrillation (Afib) is a condition in which the normal beating of the left atrium (LA) is chaotic and ineffective. The left atrial appendage (LAA) is a blind pouch off the LA. In patients with Afib blood stagnates in the LAA facilitating clot formation. These clots (or clot fragments) have a tendency to embolize or leave the LAA and enter the systemic circulation. A stroke occurs when a clot/clot fragment embolizes and occludes one of the arteries perfusing the brain.

Anticoagulants, e.g. Coumadin, have been shown to significantly reduce the stroke risk in Afib patients. These drugs reduce clot formation but also increase bleeding complications including hemorrhagic strokes, subdural hematoma, and bleeding in the gastrointestinal tract.

[0005] There are about eight million people in the US and EU with Afib. About 4.6 million of these patients are at a high risk for stroke and would benefit from anticoagulation. A large portion of these patients cannot take anticoagulants due to an increased bleeding risk, leaving their stroke risk unaddressed. The prevalence of Afib increases with age.

[0006] Existing devices for occluding the LAA have drawbacks. Existing devices are offered in many sizes and must be closely matched to the highly variable LAA anatomy. This is difficult to do using fluoroscopy and often requires adjunctive imaging in the form of transesophageal echocardiography (TEE), cardiac CT and MRI, all with three dimensional reconstructions. If the device is significantly oversized, the LAA ostium may become overstretched leading to tearing, resulting in bleeding into the pericardial space. If the device is too small, it will not adequately seal the ostium and may be prone to embolization. Even if sized correctly, the device forces the oval LAA ostium to take the round shape of the device, often resulting in residual leakage at the edges due to poor sealing.

[0007] Existing devices require sufficient spring force or stiffness to seal and anchor to surrounding tissue. If too stiff, these devices may lead to leaking of blood through the tissue into the pericardial space which may lead to cardiac tamponade. Furthermore, the geometry of these devices limits repositioning once the implant is fully expanded. Existing devices also complicate delivery by requiring positioning in the LAA coaxial to the axis of the LAA.

[0008] There is therefore a need for an improved LAA occlusion device.

SUMMARY

[0009] The embodiments disclosed herein each have several aspects no single one of which is solely responsible for the disclosure's desirable attributes. Without limiting the scope of this disclosure, its more prominent features will now be briefly discussed. After considering this discussion, and particularly after reading the section entitled "Detailed Description," one will understand how the features of the embodiments described herein provide advantages over existing systems, devices and methods for left atrial appendage (LAA) occlusion.

[0010] The following disclosure describes non-limiting examples of some embodiments. For instance, other embodiments of the disclosed systems and methods may or may not include the features described herein. Moreover, disclosed advantages and benefits can apply only to certain embodiments and should not be used to limit the disclosure.

[0011] Devices and methods are described for occluding the LAA (LAA) to exclude the LAA from blood flow to prevent blood from clotting within the LAA and subsequently embolizing, particularly in patients with atrial fibrillation. An LAA occlusion device is delivered via transcatheter delivery into the LAA and anchored using a compliant frame and foam body. The device conforms to the oval shape of the LAA with superior sealing effect, does not require an excessive number of sizes and thus negates the need for extensive pre-procedure imaging, and can be delivered off-axis thereby allowing for simpler delivery procedure, among other advantages.

[0012] A foam body, which can be tubular in shape, and a compliant frame inside or within the foam body, are described that are collapsed for delivery and then expand in place within the LAA. The foam body may have a coating at least partially on the outer surface(s) of the foam body. The coating may be a layer of Polytetrafluoroethylene (PTFE). The device is anchored by structural

anchors of the frame and/or by tissue ingrowth from the left atrium (LA) and LAA into the foam. Some embodiments are additionally or alternatively anchored by independent or integrated repositionable anchors, by barbs, and/or by distal anchoring elements. For example, anchors extending from a compliant frame are described which deploy through the compressible foam plug. In some embodiments, repositionable atraumatic anchor system embodiments are also disclosed which can be independent structures or integral to the foam plug and/or skin.

[0013] The foam body may be at least partially covered by a proximal end cover. The cover may be an expanded Polytetrafluoroethylene (ePTFE) cover. The cover provides several advantages, such as the following: sufficiently strong to enable handling of the plugs without tearing; allow for repositioning and retrieval of the plugs; provides a thromboresistant surface within the LA which will encourage formation of a neointima; assist in the creation of occlusion zones designed to encourage thromboresistance and endothelialization from the blood and adjacent tissue and anchoring zones designed to promote fast and tenacious tissue ingrowth into the compressible implant from the adjacent non-blood tissue; and can assist in closure at the ostium. The cover, e.g. a layer, jacket or skin, etc., can be independent or can be attached to the foam body, for example with sutures, adhesives, etc. In some embodiments, retrieval finials can be attached at one or more points to aid in retrieval of an embolized device and to increase radiopacity.

[0014] Some embodiments are tracked over a guidewire and have a guidewire lumen within the foam that is expandable, to allow for placement of the guidewire, and then is self-closing upon removal of the guidewire. Some embodiments do not require a guidewire lumen. Further, some embodiments may be multi-functional and include features for ablation, pressure-sensing, drug-elution, pacing, electrical isolation, etc.

[0015] In one aspect, a left atrial appendage occlusion device is described. The device comprises a conformable, tubular foam body, a compressible side wall and an expandable support. The conformable, tubular foam body has a closed proximal end and a distal end. The compressible side wall extends between the proximal end and the distal end, and defines a central cavity. The expandable support is within the body and configured to compress the side wall against a wall of a left atrial appendage.

[0016] In some embodiments, the side wall may have an uncompressed thickness of at least about 0.5 mm. The compressible side wall may have an uncompressed thickness of at least about 1.5 mm. The compressible side wall may have an uncompressed thickness of about 2.5 mm. The compressible side wall may extend in a distal direction beyond a distal end of the support by at least about 2 mm in an unconstrained, expanded state. The compressible side wall may extend in a distal direction beyond a distal end of the support by about 5 mm in an unconstrained, expanded state. The compressible side wall may comprise a foam having a plurality of interconnected reticulations and voids, and further comprising a PTFE coating on at least some of the interconnected reticulations. The closed proximal end may comprise a foam end wall. The foam end wall may further comprise a cover. The cover may comprise ePTFE. The expandable support may be self-expandable. The expandable support may be in the central cavity. The tubular foam body may be substantially cylindrical in an unconstrained, expanded state.

[0017] In another aspect, a self-expandable, atraumatic occlusion device is described. The device is configured to conform to the side wall of a left atrial appendage. The device comprises a compressible open cell foam body, a self-expandable support and a proximal end wall. The compressible open cell foam body has a tubular foam side wall and a central cavity. The expandable support is within the cavity. The proximal end wall is on the foam body. The proximal end wall is positioned proximally of the proximal end of the support, and the foam side wall extends distally beyond the distal end of the support to form a distal, atraumatic bumper for preventing contact between the support and a wall of the left atrial appendage in an implantation in which a central longitudinal axis of the occlusion device is non-parallel to a primary longitudinal axis of the left atrial appendage.

[0018] In another aspect, a left atrial appendage occlusion device is described. The device comprises an expandable tubular foam cup and an expandable frame. The expandable tubular foam cup has a proximal end, a distal end, a tubular side wall and a proximal end wall. The side wall has a thickness of at least about 1.0 mm and a porosity of at least about 85% open void content. The expandable frame is configured to press the side wall into conforming contact with a wall of the left atrial appendage.

[0019] In some embodiments, the tubular side wall may have a thickness of at least about 2 mm. The tubular side wall may have a void content of at least about 90%. The tubular side wall may have an average pore size of at least about 100 microns. The tubular side wall may have an average pore size of at least about 200 microns. The tubular side wall may be provided with a thromboresistant coating. The thromboresistant coating may comprise PTFE. The proximal end wall may be provided with a thromboresistant cover. The frame may further comprise at least three recapture struts inclining radially inwardly in the proximal direction to a hub. The frame may comprise a plurality of axially extending side wall struts, with adjacent pairs of side wall struts joined at an apex. The frame may comprise at least six proximally facing apexes and at least six distally facing apexes. Each recapture strut may be joined to a unique proximally facing apex on the frame. The recapture struts may be integrally formed with the frame. The device may further comprise a lumen through the hub. The device may further comprise anchors to secure the device to tissue. The anchors may be flexible anchors configured to extend through the foam side wall at an inclined angle.

[0020] In another aspect, a conformable LAA occlusion device is described. The device comprises a compressible tubular foam wall. The wall comprises a reticulated, cross linked matrix having at least about 90% void content, an average cell size within the range of from about 250-500 microns, a wall thickness of at least about 2 mm and a compressive strength of at least about 1 psi. In some embodiments, the compressive strength is within a range of from about 1 psi to about 2 psi. In some embodiments, the device may have an expandable support configured to compress the side wall against a wall of a left atrial appendage.

[0021] In another aspect, a LAA occlusion device is described. The device includes an open cell foam body and an internal locking system. The body has a proximal end, a distal end and an outer skin. The proximal end is configured to face a left atrium and the distal end is configured to face the LAA following implantation in the LAA. The body can be compressed for delivery within a delivery catheter and can self-expand when removed from the delivery catheter. The internal locking system is coupled with the body and comprises at least one deployable tissue anchor. The deployable anchor is configured to deploy from a constrained configuration within the body to a deployed configuration where a tissue engaging segment of the anchor extends outside the body to secure the body within the LAA. The deployable anchor is configured to deploy to the deployed configuration after the body expands within the LAA. The deployable anchor may be retractable from the deployed configuration to a retracted configuration within the body.

[0022] In some embodiments, the internal locking system further comprises a plurality of the deployable anchors rotatably coupled with the body, wherein the plurality of anchors are configured to rotate to the deployed and retracted configurations. The internal locking system may comprise four of the deployable anchors. In some embodiments, the body further comprises a plurality of axially extending slots corresponding to the plurality of anchors, wherein each of the plurality of anchors is configured to deploy and retract through the corresponding axial slot.

[0023] In some embodiments, the internal locking system further comprises a restraint that restrains the anchor in the constrained configuration, and the anchor is deployed from the constrained configuration to the deployed configuration by removing the restraint from the anchor. The restraint may be a sheath that restrains the anchor in the constrained configuration by covering the anchor, wherein the anchor is deployed from the constrained configuration to the deployed configuration by removing the sheath from covering the anchor. The restraint may be a lasso that restrains the

anchor in the constrained configuration by surrounding the anchor, and the anchor is deployed from the constrained configuration to the deployed configuration by removing the lasso from surrounding the anchor.

[0024] In some embodiments, the internal locking system further comprises a moveable mount coupled with an end of the anchor, and the anchor is deployed from the constrained configuration to the deployed configuration by axially moving the mount.

[0025] In some embodiments, the internal locking system further comprises a constraint configured to move over the anchor to cause the anchor to retract. The constraint may be a ring configured to slide over the anchor to cause the anchor to retract.

[0026] In some embodiments, the skin comprises ePTFE.

[0027] In some embodiments, the device further comprises at least one tissue ingrowth surface on a sidewall of the body.

[0028] In some embodiments, the device further comprises a plurality of openings in the skin to permit tissue ingrowth into the open cell foam body. The plurality of openings of the skin may be located in an anchoring region of the device located at least between the proximal and distal ends of the device, and the device may further comprise an occlusion region located at the proximal end of the device and configured to encourage thromboresistance and endothelialization from the blood and adjacent tissue.

[0029] In another aspect, a LAA closure system is described. The system comprises a delivery catheter and a LAA occlusion device. The delivery catheter comprises an elongate flexible tubular body, having a proximal end and a distal end and at least one lumen extending therethrough. The LAA occlusion device is configured to be compressed within the delivery catheter and to self-expand upon deployment from the delivery catheter. The device comprises a self-expandable open cell foam body coupled with an internal locking system. The internal locking system comprises a deployable anchor configured to deploy from a constrained configuration to a deployed configuration after the body expands within the LAA and is configured to retract from the deployed configuration to a retracted position within the body.

[0030] In some embodiments, the system further comprises an axially movable deployment control extending through a lumen of the body, for deploying the deployable anchor. The system may further comprise an axially movable deployment control extending through a lumen of the body, for deploying the foam body from the distal end of the closure system. The internal locking system may further comprise a restraint that restrains the anchor in the constrained configuration, and the anchor is actively deployed from the constrained configuration to the deployed configuration by removing the restraint from the anchor using an axially movable deployment control extending through a lumen of the body. The internal locking system may further comprise a moveable mount coupled with an end of the anchor, and the anchor is actively deployed from the constrained configuration to the deployed configuration by axially moving the mount using an axially movable deployment control extending through a lumen of the body.

[0031] In another aspect, a method of excluding a LAA is described. The method comprises advancing a guidewire into the LAA, advancing a distal end of a delivery catheter over the guidewire and into the LAA, and deploying a LAA occlusion device from the distal end of the delivery catheter. The device comprises an expandable foam body coupled with an internal locking system having a deployable anchor, and the body expands within the LAA upon deploying from the distal end of the delivery catheter. The method further comprises actively deploying the deployable anchor after the body expands within the LAA. The deployable anchor is configured to retract from the deployed configuration to a retracted position within the body. In some embodiments, the method further comprises retracting the deployable anchor from the deployed configuration to the retracted position.

[0032] In another aspect, a LAA occlusion device is described. The device comprises an expandable foam body and an internal locking system. The body can be compressed for delivery

within a delivery catheter and can self-expand when removed from the delivery catheter. The internal locking system is coupled with the body and comprises a deployable anchor configured to deploy from a constrained configuration within the body to a deployed configuration where the anchor extends outside the body to secure the body within the LAA. The body is configured to expand upon removal from the delivery catheter, and the deployable anchor is configured to deploy to the deployed configuration after the body expands.

[0033] In another aspect, a LAA occlusion device is described. The device comprises an expandable foam body and an internal locking system. The body can be compressed for delivery within a delivery catheter and can self-expand when removed from the delivery catheter. The internal locking system is coupled with the body and comprises a deployable anchor configured to deploy from a constrained configuration within the body to a deployed configuration where the anchor extends outside the body to secure the body within the LAA. The deployable anchor is configured to retract from the deployed configuration to a retracted configuration within the body such that the body can be repositioned within the LAA.

[0034] In another aspect, a LAA occlusion device is described. The device comprises an expandable tubular frame, an expandable tubular foam layer and a tissue scaffold. The expandable tubular frame has a proximal end, a distal end and a central lumen. The expandable tubular foam layer is carried by the frame and has a thickness of at least about 0.5 mm. The emboli retention layer is carried by the frame and encloses the lumen at the proximal end.

[0035] In some embodiments, the foam layer may have a thickness of at least about 1 mm. The foam layer may have a thickness of at least about 2.5 mm. The foam layer may have a void content of at least about 80%. The foam layer may have a void content of at least about 90%. The foam layer may have an average pore size of at least about 100 microns. The foam layer may have an average pore size of at least about 200 microns. The foam layer may extend across the proximal end of the frame to form the tissue scaffold. The tissue scaffold may be provided with a thromboresistant coating. The tissue scaffold may be provided with a thromboresistant layer. The thromboresistant coating or layer may comprise PTFE. The thromboresistant coating or layer may comprise ePTFE. The frame may further comprise at least three recapture struts inclining radially inwardly in the proximal direction to a hub.

[0036] In some embodiments, the foam layer extends across the proximal end of the frame to form the tissue scaffold, and the frame may comprise a plurality of axially extending side wall struts, with adjacent pairs of side wall struts joined at an apex. The device may comprise at least six proximally facing apexes and at least six distally facing apexes. The device may comprise at least three recapture struts joined at a proximal hub, where each recapture strut has a distal end joined to the frame. Each recapture strut may be joined to a unique proximally facing apex on the frame. The recapture struts may be integrally formed with the frame. The device may further comprise a lumen through the hub.

[0037] In some embodiments, the device may comprise anchors to secure the device to tissue. The anchors may be static anchors that are configured to deploy upon deployment of the device from a delivery catheter. The anchors may be constrained anchors that are configured to be controllably released into a deployed configuration after expansion of the foam. The anchors may be dynamic anchors that are configured to be deployed from a contracted configuration to a deployed configuration and are further configured to be retracted from the deployed configuration back to a retracted configuration. The anchors may be further configured to be retracted from the deployed configuration back to the contracted configuration.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0038] The foregoing and other features of the present disclosure will become more fully apparent from the following description and appended claims, taken in conjunction with the accompanying drawings. Understanding that these drawings depict only several embodiments in accordance with the disclosure and are not to be considered limiting of its scope, the disclosure will be described with additional specificity and detail through use of the accompanying drawings. In the following detailed description, reference is made to the accompanying drawings, which form a part hereof. In the drawings, similar symbols typically identify similar components, unless context dictates otherwise. The illustrative embodiments described in the detailed description, drawings, and claims are not meant to be limiting. Other embodiments may be utilized, and other changes may be made, without departing from the spirit or scope of the subject matter presented here. It will be readily understood that the aspects of the present disclosure, as generally described herein, and illustrated in the drawing, can be arranged, substituted, combined, and designed in a wide variety of different configurations, all of which are explicitly contemplated and make part of this disclosure.

[0039] FIG. **1** shows the anatomy of the left atrium (LA) and left atrial appendage (LAA).

[0040] FIG. **2** shows an LAA with an embodiment of an LAA occlusion device implanted in the LAA and that uses adhesive.

[0041] FIG. **3** shows an x-ray image of an embodiment of an LAA occlusion device.

[0042] FIG. **4** shows an LAA with an embodiment of an LAA occlusion device and distal anchor implanted in the LA.

[0043] FIG. **5** shows an embodiment of a screw anchor that may be used with the various LAA occlusion devices described herein.

[0044] FIG. **6** shows a longitudinal cross section of an embodiment of an LAA occlusion device.

[0045] FIGS. **7-15** are sequential schematic cross section views of an embodiment of an LAA and delivery system showing delivery and anchoring techniques that may be used with the various LAA occlusion devices described herein, including but not limited to the devices of FIGS. **85A-90D**.

[0046] FIG. **16** is a side cross section view of an embodiment of an LAA occlusion device having a foam body, frame and proximal cover.

[0047] FIG. **17** is a side cross section view of an embodiment of an LAA occlusion device having metal coils and foam.

[0048] FIG. **18** is a side view of an embodiment of an LAA occlusion device having a single metal coil.

[0049] FIG. **19** is a side view of an embodiment of an LAA occlusion device having a dilating distal tip.

[0050] FIG. **20** is a side cross section view of an embodiment of an LAA occlusion device having proximal and distal caps.

[0051] FIG. **21** is a schematic of an embodiment of an implant delivery system that may be used with the various LAA occlusion devices described herein, including but not limited to the devices of FIGS. **85A-90D**.

[0052] FIG. **22** is a schematic of an embodiment of a delivery of an expanding foam system that may be used with the various LAA occlusion devices described herein, including but not limited to the devices of FIGS. **85A-90D**.

[0053] FIG. **23** is a side view of a plug with barbs that may be used with the various LAA occlusion devices described herein, including but not limited to the devices of FIGS. **85A-90D**.

[0054] FIG. **24** shows an embodiment of an LAA occlusion device having a retrieval suture attachment that may be used with the various LAA occlusion devices described herein, including but not limited to the devices of FIGS. **85A-90D**.

[0055] FIGS. **25A-26** show embodiments of distal anchoring systems that may be used with the various LAA occlusion devices described herein, including but not limited to the devices of FIGS.

85A-90D.

[0056] FIGS. **27A-27G** are various views of an embodiment of an LAA occlusion device with an internal locking system that may be used with the various LAA occlusion devices described herein, including but not limited to the devices of FIGS. **85A-90D**.

[0057] FIGS. **28A-28D** are various views of an embodiment of an internal locking system that may be used the device of FIGS. **27A-27G**.

[0058] FIGS. **29A-29B** are sequential side views of an unlocking mechanism that may be used with the device of FIGS. **27A-27G**.

[0059] FIG. **30** is a side view of an embodiment of an LAA occlusion device having flexible anchors that may be used with the various LAA occlusion devices described herein, including but not limited to the devices of FIGS. **85A-90D**.

[0060] FIGS. **31-32** are side views of an embodiment of an LAA occlusion device having flexible anchors and stiffening tubular members configuration that may be used with the various LAA occlusion devices described herein, including but not limited to the devices of FIGS. **85A-90D**.

[0061] FIG. **33** is a side view of an embodiment of an LAA occlusion device having discrete attachments of an outer skin to an internal foam that may be used with the various LAA occlusion devices described herein, including but not limited to the devices of FIGS. **85A-90D**.

[0062] FIG. **34** is a side view of the device of FIG. **34** including an outer rim.

[0063] FIG. **35** is a side view of an embodiment an LAA occlusion device having anchors with deployed V-tips that may be used with the various LAA occlusion devices described herein, including but not limited to the devices of FIGS. **85A-90D**.

[0064] FIG. **36** is a side view of an embodiment an LAA occlusion device having deployed anchors with V-tips that may be used with the various LAA occlusion devices described herein, including but not limited to the devices of FIGS. **85A-90D**.

[0065] FIGS. **37A-37C** are side views of various embodiments of anchors that may be used with the various LAA occlusion devices described herein, including but not limited to the devices of FIGS. **85A-90D**.

[0066] FIG. **38** is a side view of an embodiment of an LAA occlusion device implanted inside an LAA.

[0067] FIGS. **39A-39B** are perspective views of an embodiment of a deployable anchor that may be used with the various LAA occlusion devices described herein, including but not limited to the devices of FIGS. **85A-90D**.

[0068] FIGS. **40A-40B** are perspective views of an embodiment of a deployable anchor that may be used with the various LAA occlusion devices described herein, including but not limited to the devices of FIGS. **85A-90D**.

[0069] FIGS. **41A-41B** are perspective views of an embodiment of a deployable anchor that may be used with the various LAA occlusion devices described herein, including but not limited to the devices of FIGS. **85A-90D**.

[0070] FIGS. **42A-42D** are various views of embodiments external deployable anchors that may be used with the various LAA occlusion devices described herein, including but not limited to the devices of FIGS. **85A-90D**.

[0071] FIGS. **43A-43C** are sequential side views of an embodiment of a deployment constraint that may be used with the various LAA occlusion devices described herein, including but not limited to the devices of FIGS. **85A-90D**.

[0072] FIGS. **44A-44C** are side views of an embodiment of an adjustable two stage anchor system that may be used with the various LAA occlusion devices described herein, including but not limited to the devices of FIGS. **85A-90D**.

[0073] FIG. **45A** is a cross-sectional view of an embodiment of an LAA occlusion device shown in an expanded configuration and having a foam cup body, proximal cover, and a deployable frame that includes a hub, recapture struts and a tubular body.

[0074] FIGS. **45B** and **45C** are, respectively, a distal end view and a proximal perspective view of the device of FIG. **45A**.

[0075] FIG. **46** is a distal end view of the LAA occlusion device of FIG. **45A**.

[0076] FIGS. **47A-47B** are perspective and side views respectively of the LAA occlusion device of FIG. **45A** having a single piece internal frame.

[0077] FIG. **48** is a perspective view of the device of FIG. **45A**.

[0078] FIGS. **49-50** are side views of the device of FIG. **45A** as attached to a delivery catheter shown, respectively, in embodiments of an expanded configuration and a partially collapsed configuration.

[0079] FIGS. **51-55** are schematics showing various embodiments of static barbs that may be used with the various occlusion devices described herein, such as the devices of FIG. **45A** or **85A**.

[0080] FIGS. **56-58** are schematics showing various embodiments of constrained barbs that may be used with the various occlusion devices described herein, such as the devices of FIG. **45A** or **85A**.

[0081] FIGS. **59-65** are schematics showing various embodiments of dynamic barbs that may be used with the various occlusion devices described herein, such as the devices of FIG. **45A** or **85A**.

[0082] FIGS. **66A-66C** are side views of the implant of FIG. **45A** having a proximal cover.

[0083] FIG. **67** shows side and end views of an embodiment of an implant having grappling hook anchors.

[0084] FIGS. **68A-68B** are side and end views respectively of an embodiment of an implant having a thicker distal bumper.

[0085] FIG. **69** is a side view of an embodiment of an implant having a constrained anchor deployed in a secondary step.

[0086] FIGS. **70-72** depict embodiments of an implant having distal anchors and proximal speed bumps.

[0087] FIGS. **73-75** depict embodiments of an implant having distal loops.

[0088] FIGS. **76-77** depict embodiments of an implant having perfusion elements.

[0089] FIG. **78** is a side view of an embodiment of an LAA occlusion device having ablative features that may be incorporated with the various LAA occlusion devices described herein.

[0090] FIG. **79** is a side view of an embodiment of an LAA occlusion device having pressure-sensing features that may be incorporated with the various LAA occlusion devices described herein.

[0091] FIG. **80** is a side view of an embodiment of an LAA occlusion device having drug elution features that may be incorporated with the various LAA occlusion devices described herein.

[0092] FIG. **81** is a side view of an embodiment of an LAA occlusion device having pacing/defibrillatory features that may be incorporated with the various LAA occlusion devices described herein.

[0093] FIGS. **82-84** depict various systems and methods for electrically isolating the LAA that may be used with the various LAA occlusion devices described herein.

[0094] FIGS. **85A-85C** are proximal, distal and side views, respectively, of an embodiment of an LAA occlusion device having a compressible foam body, an expandable frame, and a proximal cover.

[0095] FIGS. **86A-86B** are side and cross-section views, respectively, of the compressible foam body of FIGS. **85A-85C**.

[0096] FIG. **86C** is a cross-section view of the foam body of FIGS. **85A-85C** with the expandable frame.

[0097] FIGS. **87A-87C** are top perspective, side, and cross-section views of another embodiment of an LAA occlusion device.

[0098] FIG. **88** is a top view of an embodiment of a proximal cover shown in a flat configuration that may be used with the various LAA occlusion devices described herein.

[0099] FIGS. **89A** and **89B** are side perspective and proximal perspective views, respectively, of

the frame of FIGS. **85B** and **86C** shown in a deployed configuration.

[0100] FIGS. **90A-90C** are sequential proximal perspective views of an embodiment of a frame showing assembly of a cap and pin with the frame that may be used with the various LAA occlusion devices described herein.

[0101] FIG. **90D** is a distal perspective view of the cap of FIGS. **90A-90C**.

[0102] FIG. **91** is a side view of an embodiment of a loading system for loading the device of FIGS. **85A-85C** into a delivery catheter.

[0103] FIG. **92** is a side view of a schematic of a transcatheter delivery system for delivering the device of FIGS. **85A-85C** via an artery or vein.

[0104] FIGS. **93A** and **93B** are proximal and distal perspective views respectively of a tether release system that may be used with the device of FIGS. **85A-85C**.

[0105] While the above-identified drawings set forth presently disclosed embodiments, other embodiments are also contemplated, as noted in the discussion. This disclosure presents illustrative embodiments by way of representation and not limitation. Numerous other modifications and embodiments can be devised by those skilled in the art which fall within the scope and spirit of the principles of the presently disclosed embodiments.

DETAILED DESCRIPTION

[0106] The following detailed description is directed to certain specific embodiments of the development. In this description, reference is made to the drawings wherein like parts or steps may be designated with like numerals throughout for clarity. Reference in this specification to “one embodiment,” “an embodiment,” or “in some embodiments” means that a particular feature, structure, or characteristic described in connection with the embodiment is included in at least one embodiment of the invention. The appearances of the phrases “one embodiment,” “an embodiment,” or “in some embodiments” in various places in the specification are not necessarily all referring to the same embodiment, nor are separate or alternative embodiments necessarily mutually exclusive of other embodiments. Moreover, various features are described which may be exhibited by some embodiments and not by others. Similarly, various requirements are described which may be requirements for some embodiments but may not be requirements for other embodiments. Reference will now be made in detail to embodiments of the invention, examples of which are illustrated in the accompanying drawings. Wherever possible, the same reference numbers will be used throughout the drawings to refer to the same or like parts.

[0107] The devices and related methods are described herein in connection with use in occluding, i.e. excluding, a LAA (LAA). The various figures show various embodiments of LAA occlusion devices, systems and methods for delivery of the LAA occlusion devices, and/or methods of using the device to occlude a LAA. The various systems, devices and methods described herein may include the same or similar features and/or functionalities as other LAA occlusion systems, devices and methods as described, for example, in U.S. application Ser. No. 14/203,187 entitled “DEVICES AND METHODS FOR EXCLUDING THE LAA” and filed on Mar. 10, 2014, and/or as described in U.S. Provisional Application No. 62/240,124 entitled “DEVICES AND METHODS FOR EXCLUDING THE LAA” and filed on Oct. 12, 2015, the entire disclosure of each of which is incorporated herein by reference for all purposes and forms a part of this specification.

[0108] Some embodiments of an LAA occlusion device **3000** include a foam body **3002**, a deployable and compliant frame **3040**, and a proximal cover **3100**, as primarily shown and described for example with respect to FIGS. **85A-90D**. Other features and functionalities that the device **3000** may include and employ are shown and described with respect to FIGS. **1-84** and **91-93B**.

[0109] The heart **100** is shown in FIG. **1** with the left atrial appendage (LAA) **102**, which is a cavity emanating from the left atrium (LA) **104**. The LAA **102** is quite variable in shape in all dimensions. If the heart is not beating normally, a condition called atrial fibrillation, blood within the LAA becomes stagnant which promotes clot formation. If blood clots within the LAA, the clots

may pass from the LAA **102** to the LA **104**, to the left ventricle **106** and out of the heart **100** into the aorta. Vessels that bring blood to the brain branch off the aorta. If the clot passes to the brain via these vessels, it may get stuck and occlude a small vessel in the brain which then causes an ischemic stroke. Strokes have severe morbidities associated with them. The opening of the LAA **102** to the LA **104** is called an ostium **110**. The ostium **110** is oval, highly variable and dependent on loading conditions, i.e., left atrial pressure. An object of the LAA occlusion devices described herein is to occlude the ostium **110** thereby sealing off the LA **104** from the LAA **102**.

[0110] One embodiment of an LAA occlusion device is shown in FIG. **2**. The occlusion device or plug **204** is placed within the LAA **200** at its opening to the LA **202**. It is understood that the “plugs” described herein, such as the plug **204**, may have the same or similar features as other implantable “devices” or “implants” described herein, such as the device **10**, device **1020**, device **3000**, foam body **3002**, etc., and vice versa. The plug **204** comprises an expandable media such as an open cell foam which enables collapse and expansion of the plug **204** and also to enhance ingrowth of tissue into the foam. The foam plug **204** is at least partially encapsulated within a thin strong layer **206** such as ePTFE (expanded polytetrafluoroethylene), polyolefin or polyester. The layer **206** may be referred to herein as a “skin” or “cover” and the like. Alternatively, bioabsorbable materials could be utilized such as PLA, PGA, PCL, PHA, or collagen. This thin encapsulating layer **206** can be oriented or otherwise modified to be elastomeric in at least one direction, such as radially. The layer **206** may have the same or similar features and/or functionalities as the cover **3100**, and vice versa.

[0111] The plug **204** may be made of polyurethane, polyolefin, PVA, collagen foams or blends thereof. One suitable material is a polycarbonate-polyurethane urea foam with a pore size of 100 μm -250 μm or 250 μm -500 μm and 90-95% void content. The foam could be non-degradable or use a degradable material such as PLA, PGA, PCL, PHA, and/or collagen. If degradable, the tissue from the LAA will grow into the foam plug and replace the foam over time. The plug **204** may be cylindrical in shape in an unconstrained expansion, but it may also be conical for example with its distal end smaller than the proximal end or reversed. It could also be oval in cross section to better match the opening of the LAA.

[0112] The foam plug **204** is oversized radially in an unconstrained expansion to fit snugly into the LAA and may be 5-50 mm in diameter depending on the diameter of the target LAA. In a free, unconstrained state, the axial length “L” of the plug is less than its outer diameter “D” such that the L/D ratio is less than 1.0. In some embodiments, this ratio may be greater than 1.0. The compliance of the foam material is designed such that it pushes on the walls of the LAA with sufficient force to maintain the plug **204** in place but without overly stretching the LAA wall. The foam and/or skin also conforms to the irregular surfaces of the LAA as it expands, to provide a complementary surface structure to the native LAA wall to further enhance anchoring and promote sealing. Thus, the expandable foam implant described herein conforms to the native configuration of the LAA. In one embodiment, the structure of the foam may be fabricated such that squeezing axially on the opposing ends of the foam causes the foam to increase in diameter.

[0113] The ePTFE or foam material may be provided with one or two or more radiopaque markers such as radiopaque threads **210** or be filled with or impregnated with a radiopaque filler such as barium sulfate, bismuth subcarbonate, or tungsten which permit the operator to see under x-ray the plug for proper positioning in the anatomy. An x-ray image is shown in FIG. **3** where one cannot see a foam plug **300** but can clearly see threads **302** and a crimp **304** (discussed below). This thread **302** or ribbon may be made from a radiopaque metallic wire or tube such as platinum, platinum-iridium or tungsten or a polymer with a radiopaque filler such as barium, bismuth, tantalum, tungsten, titanium or platinum.

[0114] An outer ePTFE layer may be formed from a tube with a diameter about the same diameter of the foam plug and a wall thickness between about 0.0001” and about 0.001” thick and serves to allow one to collapse and pull on the plug **204** without tearing the foam material. The ePTFE

material also serves as the blood contacting surface facing the LA **206** and has pores or nodes such that blood components coagulate on the surface and an intimal or neointimal covering of tissue grows across it and anchors tightly to the material. Pore sizes within the range of from about 4 μ to about 110 μ , ideally 5-35 μ are useful for formation and adherence of a neointima.

[0115] The outer covering **206** may be constructed of materials other than ePTFE such as woven fabrics, meshes or perforated films made of FEP, polypropylene, polyethylene, polyester or nylon. The covering **206** should have a low compliance (non-elastic), at least longitudinally, be sufficiently strong as to permit removal of the plug, a low coefficient of friction, and be thromboresistant. The outer covering **206** serves as a matrix to permit plug removal as most foams are not sufficiently strong to resist tearing when pulled. The plug **204** can also be coated with or contain materials, such as PTFE. Such materials may enhance the plug's **204** ultrasonic echogenic profile, thromboresistance, and/or lubricity. The plug **204** can also be coated with or contain materials to facilitate echocardiographic visualization, promote cellular ingrowth and coverage.

[0116] The outer covering **206** has holes in it to permit contact of the LAA tissue with the foam plug **204** to encourage ingrowth of tissue into the foam plug pores and/or allow blood flow therethrough. These holes may be 1 to 5 mm in diameter or may also be oval with their long axis aligned with the axis of the foam plug, the length of which may be 80% of the length of the foam plug and the width may be 1-5 mm. The holes may be as large as possible such that the outer covering maintains sufficient strength to transmit the tensile forces required for removal. The holes may be preferentially placed along the device. In one embodiment, holes are placed distally to enhance tissue ingrowth from the LAA wall.

[0117] In one implementation, the implant is provided with proximal and/or distal end caps of ePTFE, joined together by two or three or four or more axially extending strips of ePTFE. The axially extending strips are spaced apart from each other circumferentially, to provide at least two or three or four or more laterally facing windows through which the open cell foam body will be in direct contact with the tissue wall of the LAA. This outer covering could be a mesh or netting as well. As shown in FIG. **20**, the covering **2004** is only on the proximal and distal faces of the plug **2000**. They may be glued to the foam plug and then crimped to the center tube **2002**.

[0118] The implantable plug **204** or devices **10**, **1020**, **3000** (as described below) may be anchored and secured in place in the LAA by tissue ingrowth and/or with additional anchoring features. In some embodiments, the plug **204** or devices **10**, **1020**, **3000** may be anchored by tissue ingrowth alone.

[0119] In some embodiments, other anchoring means may be implemented. One means of adhering the foam plug in place within the LAA is to use an adhesive, such as a low viscosity cyanoacrylate (1-200 cps). The adhesive is injected into place along the sidewall near the distal end of the foam plug **208**. Holes in the ePTFE covering permit the adhesive to interact between the foam plug **204** and the LAA wall **200**. Injection of the adhesive may be accomplished with several means, one of which is to inject through the catheter into the center lumen **212**. Passages **214** serve to guide the adhesive to the correct location. The distal end of the foam plug may be restricted at that time to prevent the adhesive from exiting the distal crimp **216**. Alternatively, FIG. **21** shows tubes **2104** that are pre-placed through the guide catheter **2102**, through the center lumen of the plug **2106** and bend backwards in the LAA to the distal end of the plug **2100**. These tubes **2104** pass all the way to the proximal end of the guide catheter **2102** where a fitting is attached to permit injection of the adhesive which then exits the small tubes **2104** at the desired location of the plug. These tubes are made of polyethylene, polypropylene or FEP so that the adhesive will not adhere to the tubes. The tubes **2104** are withdrawn after injection through the guide catheter out of the patient.

[0120] Other one part adhesives including aqueous cross linking adhesives, polyurethane, PEG, PGA, PLA, polycaprolactone or a lycine-derived urethane may be used. In addition, these adhesives may be made in two components such that one component is adherent to the foam and the second injected in vivo. Also, these two component adhesives may be injected simultaneously

to mix in vivo to prevent fouling of injection tubes.

[0121] An alternative anchoring means for plug **400** or device **3000**, etc. is one or two or more distal anchors as shown in FIG. **4**. A wire **404** is passed through a center lumen **410** into the LAA and attached to the distal wall of the LAA. In this case, a screw wire **408** is threaded into the wall of the LAA **406**. A closer detail of this is seen in FIG. **5** as screw **502** is shown embedded into the LAA wall **504** but not all the way through the epicardial surface **506**.

[0122] Additional means of anchoring include the use of a plurality of hooks or barbs or graspers to grab the distal wall and baskets, malecots, distal foam plugs and Nitinol wire birds nests that open within the LAA and push outward on the wall or engage the protrusions of the LAA. It may be desirable to place the plug then engage the anchor as a secondary step. One such embodiment could include a multitude of nitinol wires with a ball or catch welded proximal to the anchor tip. These could be gathered with the delivery catheter then released when the ideal plug position has been confirmed.

[0123] A cross section of one embodiment is shown in FIG. **6** with foam plug **600** and the LA face **602** and the LAA face **610**. The ePTFE material **604** encapsulates the foam plug **600** and its open ends are connected with an attachment structure such as a wire, suture or tubular crimp **606** over an inner tube **608**. The inner tube **608** may be made of an implant grade stainless steel such as 304 or 316 grades or a cobalt-chromium alloy such as MP35n and the crimp **606** may be made of annealed 304 or 316 stainless steel or a cobalt-chromium alloy such as MP35n. This crimp also serves as an element which can be snared should the device need to be removed.

[0124] Referring to FIG. **6**, the tubular ePTFE layer **604** extends along an inner layer **612** which lines the guidewire lumen, and everts out around the left atrial face **602** to form outer layer **614**. In some embodiments, the layer **604** may cover the entire proximal face and/or part of the sidewall, such as the cover **3100** or the cover over proximal face **1064'**, as further described herein. As further shown in FIG. **6**, a first end **616** of inner layer **612** is disposed concentrically within a second end **618** of outer layer **614**. The first end **616** and second end **618** are clamped between inner tube **608** and outer crimp **606**. In this manner, the implant can be encapsulated in a manner that presents a seamless left atrial face **602**, and preserves the integrity of the guidewire lumen with inner tube **608**.

[0125] An embodiment of a technique for placement of an LAA occlusion device is shown in FIGS. **7** through **15**. To close the LAA, the LA is first accessed from the venous system. One approach is to use a Brockenbrough-style needle to puncture the atrial septum to access the LA from the right atrium (RA). The basic needle-puncture technique is performed obtaining venous access typically via the right femoral vein. A Mullins sheath and dilator are then tracked over a 0.025" or 0.032" guidewire previously placed in the superior vena cava (SVC). Fluoroscopic and echocardiographic imaging, such as transesophageal echo (TEE) or intracardiac echo (ICE), are typically utilized. If echo is not utilized, it is common to also place a pigtail catheter in the aortic root to define the location of the aortic valve, a step not necessary when using echo.

[0126] Once the Mullins sheath and dilator are in the SVC, the guidewire is removed and a trans-septal needle is placed through the dilator. The needle contains a stylette to prevent skiving off of polymeric material from the dilator lumen as it traverses to the tip. Once the needle is near the dilator tip, the stylette is removed and the needle is connected to a manifold and flushed. The Mullins sheath/dilator set and the needle (positioned within the dilator tip) are retracted into the SVC toward the RA as a unit. As the system is withdrawn down the wall of the SVC into the RA and positioned in the *Fossa ovale*, the preferred puncture location.

[0127] Once proper position in the *Fossa ovale* is observed, the needle is advanced across the *Fossa ovale* into the LA. Successful trans-septal puncture can be confirmed by echo, pressure measurement, O.sub.2 saturation and contrast injection. Once the needle position is confirmed to be positioned in the LA, the sheath and dilator can be advanced over it into the LA. In some cases, the user will first pass a guidewire through the needle into the LA and into an upper pulmonary vein

(typically the left) prior to crossing. Alternative options include the use of radiofrequency trans-septal needles, which are useful for crossing very thick or hypertrophic septa, or the use of a safety wire placed through the needle and utilized for the initial puncture.

[0128] Referring to FIGS. **8** through **15**, a guide catheter **802** is placed through the femoral vein into the right atrium of the heart and across the intra-atrial septum into the LA as described above and positioned near the LAA ostium **804**. A guidewire **902** usually of 0.035" diameter is placed through guide catheter **900** and into the LAA **904**. This guidewire **1002** may have attached to its distal end a balloon **1006** which is inflated in the LAA and serves as a bumper to prevent guide catheter **1100** from perforating the wall of the LAA. The guide catheter **1100** is then advanced over the guidewire **1108** into the LAA **1104**. A radiopaque marker **1102** is used to guide catheter placement under fluoroscopy.

[0129] The foam plug **1204** is then pushed through the guide catheter **1200** with pusher **1202** and is shown exiting the guide catheter **1300** slowly in FIG. **13** until it is fully deployed as shown in FIG. **14**. The foam plug **1404** position may then be adjusted in place using the distal balloon **1408** and the guide catheter **1400**, sliding the foam plug proximally by pulling on the balloon **1408** through shaft **1412** or sliding it distally by pushing guide catheter **1400** distally. The guidewire may also contain a pressure sensor within it such that sealing of the LAA is monitored and confirmation of a sufficient seal is made. Once the placement is satisfactory, the adhesive **1514** may be injected and/or mechanical anchors be deployed anchoring the plug **1404** to the wall. The guidewire balloon **1508** is deflated, after which the guidewire is removed. In an alternative embodiment, a binary adhesive system can be used where one component of the binary system is bonded to the outer surface of the skin covering the foam plug. The second component can be injected at the interface between foam plug and the wall of the LAA such that bonding happens only at the interface minimizing the risk of adhesive embolization. In some embodiments, adhesives and balloons may or may not be used, for example with the device **3000** further described herein.

[0130] An alternative to pushing the plug through the entire length of the guide catheter is that the plug **1204** may be initially located at the distal end of the guide catheter **1200** as shown in FIG. **12**. The guidewire **1210** passes through the center of the plug **1204** and in this mode, the pusher **1202** only needs to push the plug a short ways to deploy it into the LAA.

[0131] For alternative anchors, they may be deployed, the shafts disconnected and removed. Disconnection mechanisms may be any of several types, such as threaded, electrolytic detachment, or others known in the art. In some embodiments, a suture attachment may be implemented, for example as described with respect to FIG. **24**.

[0132] As shown in FIG. **16**, in some embodiments, a foam body **1600** and metal frame such as stent **1602** may be included. The foam body **1600** and stent **1602** may have the same or similar features and/or functionalities as respectively the foam body **3002** and tubular body **3080** (see FIGS. **85A-90D**), and vice versa. The foam **1600** is designed to provide ingrowth of tissue and also to provide a cushion of the metal stent **1602** onto the tissue of the LAA. The proximal face **1604'** of the plug is covered in ePTFE, polyester or another thromboresistant tissue scaffold material to facilitate sealing with the desired pore size to encourage overgrowth.

[0133] The stent **1602** may be made of Nitinol to enable it to pack into a 10, 12, 14, 16, 18 or 20 F delivery catheter and expand to its desired diameter. The stent **1602** may be braided, laser cut or wire formed. Any of a variety of stent wall patterns may be utilized, depending upon the desired performance. The stent **1602** may be a balloon expandable stent, or self-expandable stent. In the illustrated embodiment, the self-expandable stent **1602** comprises a plurality of proximal apices **1608** and distal apices **1610** connected by a plurality of zig zag struts **1612**. A hole **1606** allows passage of the guidewire for delivery. This design may be advantageous in that the expansion force exerted by the plug on the LAA can be controlled separately from the foam characteristics. Also, it may be easier to pack this concept into a smaller geometry. For example, the plug can be packed into a smaller geometry by reducing the amount of foam that must be compressed into the delivery

catheter while maintaining sufficient dilation force.

[0134] Alternatively, the foam plug may be constructed of two foams. One denser core to provide force, for example radial force, and an outer softer foam to engage the tissue irregularities. The softer foam could also be located on the proximal and/or distal ends to facilitate retrieval.

[0135] Another means of adding stiffness to the foam plug is shown in FIG. 17 where a cavity **1704** in the foam plug **1700** is made and a coil of wire **1702** may be advanced from the guide catheter at the proximal end **1706** into the cavity **1704**. As the wire enters the cavity, it expands to its predetermined size and exerts force on the foam radially outwards. The type and amount of wire may be determined in vivo using x-ray guidance to examine the radial expansion of the foam into the LAA.

[0136] Instead of wires as shown in FIG. 17, a balloon may be passed into the foam and inflated to provide radial force while the outer foam serves to engage the tissue irregularities and tissue ingrowth. Following inflation, the balloon may be detached from a deployment catheter and the deployment catheter withdrawn. The balloon is preferably provided with a valve, to prevent the escape of inflation media. Inflation media may be any of a variety of media which is convertible between a first, flowable state and a second, hardened state such as by cross linking or polymerization in situ.

[0137] Another LAA plug is shown in FIG. 18 as a spring like implant wire **1800** that is covered with foam **1802** to encourage ingrowth. The proximal face of the implant is covered with a sheet of ePTFE or other tissue scaffolding material. This implant may be stretched out for delivery and released in place.

[0138] Rather than using a foam, a low porosity outer bag without perforations could be placed in the LAA and then filled with a substance to provide the radial expansion. This substance may be a hydrogel, cellulose or polyvinylacetate.

[0139] Rather than requiring the use of a separate dilation device to cross the septum, the distal crimp element **1902** may be formed in a tapered manner such that it extends from the distal end of the catheter **1200** and serves as a dilating tip to dilate the opening in the septum as the catheter is advanced. See FIG. 19.

[0140] An alternative plug design uses a foam such as cellulose sponge material that is compacted and dehydrated such that it can be packed into the guide catheter. This foam material **2202** may be packed into the guide catheter as shown in FIG. 22. The foam plug **2202** is then advanced from the distal end of the guide catheter **2204** with a plunger **2206** into the LAA. The plug exits the guide catheter and opens to a disc shape **2210**. As the foam absorbs fluid in the blood, its length expands to form a cylinder **2220** filling the LAA. Expansion ratios for compressed cellulose materials may be as high as 17:1, expanded to compressed length.

[0141] It may be advantageous to use small barbs **2302** in FIG. 23 to further engage the plug **2204** into the LAA. Barbs may be unidirectional or bidirectional to resist movement in either the proximal or distal direction. These barbs are embedded into the foam plug and may be 0.1 to 1 mm in height. It may be desirable to place the plug then engage the barbs as a secondary step. One such embodiment could include a multitude of nitinol barb wires with a ball or catch welded proximal to the barb tip. These could be gathered with the delivery catheter within a sleeve or suture then released when the ideal plug position has been confirmed.

[0142] One means of removing a device that is not functioning properly is to releasably attach a retrieval suture **2400** to the implant, such as to the proximal cap **2402** which also passes proximally throughout the entire length of the guide catheter **2404** in FIG. 24. If the device is to be removed, pulling on both ends of the suture **2400** will pull the outer covering into the guide catheter **2404** which can then be removed from the patient. If the device is properly placed, the suture **2400** may be cut and removed leaving the plug in place.

[0143] Deployment of the occlusion device has been discussed primarily in the context of a transvascular access. However, the implants may alternatively be deployed via direct surgical

access, or various minimally invasive access pathways (e.g. jugular vein). For example, the area overlying the xiphoid and adjacent costal cartilage may be prepared and draped using standard techniques. A local anesthetic may be administered and skin incision may be made, typically about 2 cm in length. The percutaneous penetration passes beneath the costal cartilage, and a sheath may be introduced into the pericardial space. The pericardial space may be irrigated with saline, preferably with a saline-lidocaine solution to provide additional anesthesia and reduce the risk of irritating the heart. The occlusion device may thereafter be introduced through the sheath, and through an access pathway created through the wall of the LAA. Closure of the wall and access pathway may thereafter be accomplished using techniques understood in the art.

[0144] Depending upon the desired clinical performance, any of the LAA occlusion devices described herein may be provided with a drug or other bioactive agent, which may be injected via the deployment catheter, or impregnated within the open cell foam or coated on the implant. The bioactive agent may be eluted or otherwise released from the implant into the adjacent tissue over a delivery time period appropriate for the particular agent as is understood in the art. Useful bioactive agents can include those that modulate thrombosis, those that encourage cellular ingrowth, throughgrowth, and endothelialization, and potentially those that resist infection. For example, agents that may promote endothelial, smooth muscle, fibroblast, and/or other cellular growth into the implant including collagen (Type I or II), heparin, a combination of collagen and heparin, extracellular matrix (ECM), fibronectin, laminin, vitronectin, peptides or other biological molecules that serve as chemoattractants, molecules MCP-1, VEGF, FGF-2 and TGF-beta, recombinant human growth factors, and/or plasma treatment with various gases.

[0145] Anti-thrombotics can typically be separated into anti-coagulants and antiplatelet agents. Anti-Coagulants include inhibitors of factor(s) within the coagulation cascade and include heparin, heparin fragments and fractions as well as inhibitors of thrombin including hirudin, hirudin derivatives, dabigatran, argatroban and bivalirudin and Factor X inhibitors such as low molecular weight heparin, rivaroxaban, apixaban.

[0146] Antiplatelet agents include GP 2b/3a inhibitors such as eptifibatide, and abciximab, ADP Receptor agonists (P2/Y12) including thienopyridines such as ticlopidine, clopidogrel, prasugrel and ticagrelor and aspirin. Other agents include lytic agents, including urokinase and streptokinase, their homologs, analogs, fragments, derivatives and pharmaceutical salts thereof and prostaglandin inhibitors.

[0147] Antibiotic agents can include, but are not limited to penicillins, cephalosporins, vancomycins, aminoglycosides, quinolones, polymyxins, erythromycins, tetracyclines, chloramphenicols, clindamycins, lincomycins, sulfonamides, their homologs, analogs, derivatives, pharmaceutical salts and combinations thereof.

[0148] Biologic agents as outlined above may be added to the implant **204** and may be injected through the delivery catheter into the space between the proximal cap **206** and the foam plug **204**. This may serve as a reservoir to minimize thrombus formation during the initial implantation and reduce the need for systemic anticoagulation following device implantation.

[0149] An electronic pressure sensor may be embedded into the proximal end of the foam plug which may be used to transmit LA pressure to a remote receiver outside the body for the monitoring of LA pressure which is useful to monitor cardiac function. In addition, a cardiac pacer or defibrillator may be embedded into the foam plug and attached electrically to the distal anchor. A drug delivery reservoir may be embedded with connection to the LA for controlled delivery of biologic agents as outlined above.

[0150] Another means of anchoring is shown in FIG. **25A** where the foam plug **2500** is placed in the LAA. The distal screw lead **2502** is advanced and screwed into the LAA wall. Guide **2506** is pulled proximally as shown in FIG. **25B**. When this guide **2506** is pulled back, the screw lead wire, made of Nitinol, bunches up into a “birds nest” **2508** or forms a coil inside the foam plug **2500**. The screw lead wire **2502** is pushed distally from the guide catheter **2504** with a pusher **2510** and

continues to bunch up into the foam. The catheter system **2504**, **2506** and **2510** are then removed. [0151] Another means of anchoring the distal anchor element to the foam is shown in FIG. **26**. Two barbed leads **2604** are attached to anchor **2602** such that when advanced into place in the foam plug **2600**, the barbs **2604** dig into the foam plug.

[0152] FIGS. **27A-27G** are various views of an embodiment of a device **10** for occlusion of the LAA (LAA). The device **10** may include the same or similar features as other devices for occlusion of the LAA described herein, such as the plug **204**, device **1020**, device **3000**, etc., and vice versa. The device **10** includes an internal locking system **101** for securing the device **10** within the LAA. In some embodiments, the device **10** may not include the internal locking system **101** or other anchoring features, for example the device **10** may be anchored by tissue ingrowth alone. The occlusion device **10** comprises an expandable media such as an open cell foam body **15**, for example a plug. The body **15** enables collapse and expansion of the device **10** and also enhances ingrowth of tissue into the foam.

[0153] The body **15** of the device **10** shown in FIG. **27A** through FIG. **27F** is in its expanded configuration. The body **15** is in a compressed configuration in FIG. **27G**. The device **10** includes the foam body **15**, a skin **20**, a central lumen **25**, a finial **30**, and a dynamic internal locking system **101** which anchors the device **10** within the LAA. FIG. **27A** is a side cross-section view of the device **10** showing the body **15** and the internal locking system **101** in a deployed configuration. FIG. **27B** is an end view of the proximal end of the device **10** showing the body **15** and the internal locking system **101** in a deployed configuration. FIG. **27C** is a side view of the device **10** showing the body **15** and the internal locking system **101** in a deployed configuration. FIG. **27D** is a side cross-section view of the device **10** showing the body **15** in a deployed configuration and the internal locking system **101** in a constrained configuration. FIG. **27E** is an end view of the distal end of the device **10** showing the body **15** and the internal locking system **101** in a deployed configuration. FIG. **27F** is a cross-section view of the device **10** taken along the line **1F-1F** as shown in FIG. **27C**. FIG. **27G** shows the body **15** and internal locking system **101** loaded and compressed within a delivery sheath **1**. The device **10** may be delivered via a delivery catheter in the configuration shown in FIG. **27G**. The body **15** of the device **10** may then expand with the internal locking system **101** still constrained, as shown in FIG. **27D**. The internal locking system **101** may then deploy into the deployed configuration as shown in FIG. **27A**.

[0154] FIG. **27G** shows the body **15** and internal locking system **101** loaded and compressed within an embodiment of a delivery sheath **1**. In some embodiments, the delivery sheath **1** may be an outer delivery catheter. The body **15** and internal locking system **101** are loaded and compressed within a delivery catheter **5**. The device **10** may be entirely or partially inside the delivery catheter **5**. In some embodiments, the delivery catheter **5** may be an inner delivery catheter. The device **10** may be loaded and compressed with the delivery catheter **5** inside of the delivery sheath **1**. Removing the delivery sheath **1**, for example by retracting the delivery sheath **1** in the proximal direction, may allow the body **15** of the device **10** to expand. The body **15** expands while the internal locking system **101** is still constrained, for example by the delivery catheter **5**. FIG. **27D** shows the body **15** in its deployed state, with the internal locking system **101** in a constrained configuration within the delivery catheter **5**. This demonstrates the first step in the deployment process, specifically placement of the device **10** within the LAA where the body **15** is expanded and the internal locking system **101** is constrained and thus the anchors are not deployed. The second step of the deployment process is shown in FIG. **27A** where the internal locking system **101** has been deployed through the body **15**. In some embodiments, this second step is reversible to retract the anchors, for example if placement of the device **10** within the LAA is unacceptable. The internal locking system **101**, for example an anchoring component or system as further described herein, is deployed from within the body **15** to deploy at least one and in some implementations at least 2 or 4 or 6 or more anchors of the internal locking system **101** outside the body **15** to engage adjacent anatomy of the LAA.

[0155] The internal locking system **101** may be controllably deployed a period of time after the body **15** expands. For instance, the location, orientation, etc. of the device **10** may be verified with various imaging techniques such as by fluoroscopy with injection of contrast media via the central lumen before the internal locking system **101** is deployed and the anchors secure the device **10** within the LAA. In some embodiments, even after deployment of the internal locking system **101** and anchors thereof, the anchors may be retracted to a position within the body **15** for repositioning, and/or retrieval of the device **10** from, within the LAA.

[0156] FIG. 27F shows an embodiment of the device having slots **17**. The slots **17** are formed within the foam body **15**. For instance, material of the foam body **15** may be removed to facilitate deployment of the internal locking system **101**, such as outward expansion of anchors to engage the tissue.

[0157] The device **10** may have any or all of the same or similar features and/or functionalities as the other plugs described herein, for example the plug **204**, etc. For example, the device **10** is at least partially encapsulated within the skin **20**. In some embodiments, the skin **20** may cover the proximal end of the body **15**. The skin **20** may be a thin, strong outer layer. The skin **20** may be a thin, encapsulating layer. The skin **20** may be fabricated from ePTFE (expanded polytetrafluoroethylene), polyolefin, polyester, other suitable materials, or combinations thereof. In some embodiments, the skin **20** may be fabricated from bioabsorbable materials, for example polylactic acid (PLA), Polyglycolic acid (PGA), polycaprolactone (PCL), PHA, collagen, other suitable bioabsorbable materials, or combinations thereof. The skin **20** can be oriented or otherwise modified to be elastomeric in at least one direction, such as radially.

[0158] The body **15** may be made of polyurethane, polyolefin, PVA, collagen foams or blends thereof. One suitable material is a polycarbonate-polyurethane urea foam with a pore size of 100-250 μm and 90-95% void content. The body **15** may be non-degradable or use a degradable material such as PLA, PGA, PCL, PHA, and/or collagen. If degradable, the tissue from the LAA will grow into the foam body **15** and replace the foam over time. The body **15** may be cylindrical in shape in an unconstrained expansion, but it may also be conical with its distal end smaller than the proximal end, or vice versa. The body **15** may also be oval in cross section to better match the opening of the LAA.

[0159] The device **10** is oversized radially in an unconstrained expansion to fit snugly into the LAA. The device **10** may be 5-50 millimeters (mm) and generally at least about 10 mm or 15 mm in diameter in its unconstrained configuration, for example depending on the diameter of the target LAA. The length “L” of the device **10** may be less than, similar to or greater than its diameter “D” such that the L/D ratio is less than 1.0, about or greater than about 1.0, greater than about 1.5, or greater than about 2.0. The L/D ratio may be greater than 1.0 to maximize its stability. However, in some embodiments, the L/D ratio may be less than 1.0, for example, from about 0.2 to about 0.9, or from about 0.3 to about 0.8, or from about 0.4 to about 0.6. The compliance of the material of the device **10** is designed such that it pushes on the walls of the LAA with sufficient force to maintain the plug in place but without overly stretching the LAA wall. The foam body **15** and/or skin **20** also conforms to the irregular surfaces of the LAA as it expands, to provide a complementary surface structure to the native LAA wall to further enhance anchoring and promote sealing. Thus, the expandable foam body **15** conforms to the native irregular configuration of the LAA. In some embodiments, the structure of the foam body **15** may be fabricated such that axial compression on the opposing ends of the body **15** such as by proximal retraction of a pull wire or inner concentric tube causes the foam to increase in diameter.

[0160] The body **15** and/or skin **20**, for example the foam material and/or ePTFE, may be provided with one, two or more radiopaque markers, such as radiopaque threads **210** (see FIG. 2) or filled with or impregnated with a radiopaque filler such as barium sulfate, bismuth subcarbonate, or tungsten, which permit the operator to visualize under x-ray the device **10** for proper positioning in the anatomy. Visualization of the device **10** may be used to verify the position of the device **10**

before deployment of anchors to secure the device **10** in place.

[0161] The skin **20**, such as an outer ePTFE layer, may have a thickness between about 0.0001 inches and about 0.0030 inches. In some embodiments, the thickness of the skin **20** may be between about 0.0003 inches and about 0.0020 inches. In some embodiments, the thickness of the skin **20** may be between about 0.0005 inches and about 0.0015 inches. The thickness of the skin **20** may be uniform, for example the same or approximately the same no matter where the thickness is measured. In some embodiments, the thickness of the skin **20** may be non-uniform, for example the thickness may be different in different portions of the skin **20**.

[0162] The skin **20**, such as an outer ePTFE layer, may also serve as the blood contacting surface on the proximal end of the device **10** facing the LA. The skin **20** may have pores or nodes such that blood components coagulate on the surface and an intimal or neointimal covering of tissue grows across it and anchors tightly to the skin material. Pore sizes may be within the range of from about 4 μ to about 110 μ . In some embodiments, the pore sizes are within the range of from about 30 μ to about 90 μ . In some embodiments, the pore sizes are within the range of from about 30 μ to about 60 μ . Such ranges of pore sizes are useful for formation and adherence of a neointima. In some embodiments, the skin **20**, such as an outer ePTFE layer, may be formed from a tube with a diameter about the same diameter of the foam body **15**. and allows one to collapse and pull on the body **15** without tearing the foam material.

[0163] The skin **20** may be constructed of materials other than ePTFE such as woven fabrics, meshes or perforated films made of FEP, polypropylene, polyethylene, polyester or nylon. The skin **20** may have a low compliance (e.g. non-elastic), for instance a low compliance longitudinally, may be sufficiently strong as to permit removal of the plug, may have a low coefficient of friction, and/or may be thromboresistant. The skin **20** serves as a matrix to permit plug removal as most foams are not sufficiently strong to resist tearing when pulled. The body **15** can also be coated with or contain materials to enhance its ultrasonic echogenic profile, thromboresistance, lubricity, and/or to facilitate echocardiographic visualization, promote cellular ingrowth and coverage.

[0164] The skin **20** may include holes to permit contact of the LAA tissue with the foam body **15**. Exposure of the foam body **15** to the LAA or other tissue has benefits for example encouraging ingrowth of tissue into the foam plug pores and/or increasing friction to hold the body **15** in place. These holes may be 1 to 5 mm in diameter or may also be oval with their long axis aligned with the axis of the foam plug, the length of which may be 80% of the length of the foam plug and the width may be 1-5 mm. The holes may be as large as possible such that the outer covering maintains sufficient strength to transmit the tensile forces required for removal. The holes may be preferentially placed along the device **10**. In some embodiments, the holes are placed distally to enhance tissue ingrowth from the distal LAA wall.

[0165] In some embodiments, the device **10** includes an occlusion region and anchoring region. The proximal portion of the device **10** facing the LA after the device is implanted in the LAA may include the occlusion region. The occlusion region may be a blood contacting surface on the proximal end of the device **10** that is thromboresistant while promoting formation of a neointima at the occlusion region. The occlusion zone encourages thromboresistance and endothelialization from the blood and adjacent tissue. The anchoring zone promotes fast and tenacious tissue ingrowth into the device **10** from the adjacent non-blood tissue. The anchoring zone may be lateral surfaces of the device **10** that interface with tissue adjacent and/or within the LAA. The anchoring zone can also include the distal end of the device **10** that faces the distal wall of the LAA after implantation.

[0166] FIGS. **28A-28D** are various views of an embodiment of an internal locking system **101** that may be used with the device **10**. In some embodiments, multiple internal locking systems **101** may be used with the device **10**. FIG. **28A** is a side view of the internal locking system shown in a deployed configuration. FIG. **28B** is an end view of a distal end of the internal locking system **101** in a deployed configuration. FIG. **28C** is a side view of the internal locking system **201101** in a

constrained configuration. FIG. 28D is a side view of an embodiment of an anchor **120** of the internal locking system **101**.

[0167] Any of a variety of structures may be utilized as the dynamic internal locking system **101** with the device **10**. In general, at least about two or four or six or more tissue anchors **120** may be actively or passively advanced from the implantable device **10** into adjacent tissue surrounding the implantation site. Following deployment of the device **10** and expansion of the body **15**, a tissue engaging segment **121** of the tissue anchor **120** will extend beyond the skin by at least about one, and in some implementations at least about two or four 4 mm or more. The tissue engaging segment **121** is carried by a support segment **122** of the tissue anchor **120** which extends through the foam body **15**, and may be attached to a deployment control such as a pull wire, push wire, tubular support or other control structure depending upon the desired configuration.

[0168] The locking system **101** discussed primarily herein is a passive deployment construction. Removal of a constraint allows the tissue anchors **120** to laterally self-expand to deploy into adjacent tissue. Self-expansion may be accomplished by constructing the tissue anchor **120** using nitinol, Elgiloy, stainless steel or other shape memory or spring bias materials. The constraint may be removed by proximal retraction or distal advance until the tissue anchors **120** are no longer engaged by the constraint, depending upon the construction of the locking system **101**.

[0169] Alternatively, tissue anchors **120** may be deployed actively such as by distal advance, proximal retraction or rotation of a control, or inflation of a balloon positioned within the device **10** to actively drive the anchors **120** through the skin **20** or corresponding apertures on the skin **20** and into tissue. For example, a plurality of support segment **122**, such as struts, may be joined at a distal end to a central hub **111**, and incline radially outwardly in the proximal direction. Proximal retraction of the hub **111** will cause the tissue engaging segment **121** to advance along its axis beyond the skin **20** and into the adjacent tissue. The inclination angle of the support segment **122**, for example the struts, may be reversed, in another construction, such that distal advance of the hub **111** will deploy the tissue engaging segments **121** beyond the skin **20**. Proximal or distal advance of the hub **111** may be accomplished by proximal or distal advance of a control such as a control wire or inner tube releasably engaged with the hub **111**.

[0170] Depending upon the desired clinical performance, the tissue anchors **120** may be retractable, such as by axial distal or proximal movement of the control depending upon the inclination angle of the anchors **120**. In the embodiment primarily illustrated herein, re-sheathing the anchors **120** may be accomplished by advancing the tubular constraint along the ramped surface of the tissue anchor **120** to move the anchor **120** radially inwardly towards the central longitudinal axis of the device **10**. In the case of an anchor **120** which deploys by advance along its own longitudinal axis, the anchor **120** may be retracted by advancing the control in the opposite direction from the direction advanced to deploy the anchors **120**.

[0171] Referring to FIGS. 28A-28D, the internal locking system **101** includes a central tubular element or hub **111** and anchors **120**. The anchors **120** may be arms, segments, or other members extending from the hub **111**. Each anchor **120** may comprise a tissue engaging segment **121** and a support segment **122** which extends to the hub **111** or other control. The internal locking system **101** has a single central tubular hub **111** and a multitude of the anchors **120**. As shown, there are four anchors **120**. There may be two, three, four, five, six, seven, eight, or more anchors **120**. The anchor **120** may be rotatably, hingedly, or otherwise moveably coupled with the hub **111**. The anchor **120** may thus move relative to the hub **111**, for example after being released from a restraint holding the anchor **120** in a constrained configuration to deploy into a deployed configuration. As further example, the anchor **120** may move from the deployed configuration to a retracted position, as further described herein. The anchor **120** may be curvilinear as shown, for example to allow the anchor **120** to take the geometry shown in FIG. 28A when unconstrained.

[0172] The illustrated anchor **120** may have a distal region **130**, a hinge region **135**, and/or a proximal region **125**. The distal region **130** interacts with the hub element **111**. The hinge region

135 and the curvilinear geometry as shown allow the end of the proximal region **125** to extend beyond the body **15**, for example beyond a sidewall of the body **15**. The proximal region **125** includes a tissue engaging segment **121** configured to engage adjacent tissue. The tissue engaging segment **121** may be the entire proximal region **125** or a portion thereof, for example the tip, etc. The proximal region **125** may thus include a sharpened tissue engaging segment **121**, a shaped tissue engaging segment **121**, an angled tissue engaging segment **121**, a thickness configured for tissue engagement, and/or other suitable features. In some embodiments, the proximal region **125** may retract back within the body **15**, as further described herein. In the embodiment shown, the anchor **120** and central tube **111** are distinct elements which are affixed to one another as shown. In other embodiments, the anchor **120** and tube **111** are a single, integral unit.

[0173] The internal locking system **101** is made from biocompatible metallic wire such as Nitinol, implant grade stainless steel such as 304 or 316, or cobalt-chromium based alloys such as MP35N or Elgiloy. In some embodiments, the internal locking system **101** may be cut from a single tubular piece of metal fabricated via machining or laser cutting followed by a secondary forming or annealing step using similar materials.

[0174] The internal locking system **101** may be in a constrained configuration as the device **10** is placed in position in the LAA and the body **15** expands therein. Then, in a secondary step, the internal locking system **101** locks or otherwise secures the device **10** in the LAA by engaging the anchors **120**. If the position is not considered optimal, or if the device **10** otherwise needs to be repositioned within and/or removed from the LAA, the internal locking system **101** and anchors **120** thereof can be unlocked and the device **10** repositioned and/or removed.

[0175] FIGS. **29A-29B** are sequential side views of an axially movable loop type unlocking mechanism that may be used with the device **10** to release the tissue anchors. FIG. **29A** is a side cross-section view of the device **10** showing the tissue anchors of internal locking system **101** in a deployed configuration. FIG. **29B** is a side cross-section view of the device **10** showing the tissue anchors in a retracted configuration. An embodiment of an unlocking system **140** is shown. The unlocking system **140** includes a ring **145**. The ring **145** may be moved over the anchors **120** to move the anchors **120** to retracted configurations. The ring **145** may be moved by a pull rod **147**. The ring **145** may be releasably attached to the pull rod **147**. The pull rod **147** may extend through a catheter to engage the ring **145**. The unlocking system **140** may be utilized if, after the internal locking system **101** is deployed, it is desirable to unlock the device **10** from within the LAA in order to reposition and/or remove the device **10**.

[0176] In the illustrated construction, deployment of the tissue anchors by distal advance of the restraint enables reversible deployment, so that subsequent proximal retraction of the restraint will retract the tissue anchors. Alternatively, proximal retraction of the restraint to release the tissue anchors will irreversibly release the tissue anchors.

[0177] FIG. **30** is a side view of an embodiment of the device **10** having flexible anchors **401**. The device **10** shown in FIG. **30** may have the same or similar features and/or functionalities as the other devices for excluding the LAA described herein, and vice versa. The device **10** may be in the configuration shown in FIG. **30** adjacent to or within an LA **201**. The device **10** in FIG. **30** includes the expandable body **15**, such as an open cell foam body, which enables collapse and expansion of the device **10**, and at least partially encased within the skin **20**, which may be a thin, strong layer fabricated from ePTFE (expanded polytetrafluoroethylene), polyolefin, or polyester which assists with healing, anchoring, and retrieval. The device **10** may also be deployed, and, if desired, repositioned and/or retrieved, or the device **10** may be permanently fixated within the LAA by engaging an anchoring system, such as the internal locking system **101**, as described herein. The anchors **401**, which may be metallic, may be fabricated from Nitinol. The anchors **401** may be small diameter Nitinol wire approximately 0.001 inches to approximately 0.010 inches in diameter. In some embodiments, the anchors **401** may be approximately 0.0005 inches to approximately 0.020 inches in diameter. The anchors **401** may be deployed upon expansion of the body **15**. For

example, the anchors **401** may self-deploy upon deployment of the device **10** from a delivery catheter. The anchors **401** may be relatively short and extremely flexible. The anchors **401** may be unable to penetrate the tissue or cause any anchoring immediately after deployment of the device **10**.

[0178] FIG. **31** is a side view of an embodiment of a device **10** for occlusion of the LAA having anchors **401** with tubes **500**. The device **10** may be positioned adjacent to or within the LA **201**. The anchors **401** may be flexible anchors, or in some embodiments the anchors **410** may be relatively stiffer, as further described. The tubes **500** may be stationary or moveable tubes, as further described. In some embodiments the tubes **500** are hypotubes. The tubes **500** may be stainless steel, polyamide, or other suitable materials. The tubes **500** may surround a corresponding anchor **401**, as further described.

[0179] In some embodiments, the anchors **401** may be fixed such that they do not move axially. For example, the anchors **401** may have a portion, such as a tissue engaging segment **121**, of a fixed length extending outside the body **15**. The portion of the anchors **401** extending outside the body **15** may be bent when compressed within a delivery catheter and/or sheath, and these portions of the anchors **401** may then straighten out to the configuration shown in FIGS. **31** and **32** after deployment of the body **15**. The fixed length portion of the anchors **401** extending beyond the body **15** may be from about 1 mm to about 5 mm, or from about 1.5 mm to about 4 mm, or from about 2 mm to about 3 mm. This length of exposed anchor **401** outside the body **15** may be effectively shortened by deployment of a corresponding tube **500**, as further described. Deployment of the corresponding tube **500** about the corresponding anchor **400** may shorten the effective length of exposed anchor **401**, i.e. the length of the anchor **401** extending beyond the end of the tube **500** after deployment of the tube **500**, from about 0.5 mm to about 1 mm. These are merely examples of different lengths of the anchor **401** and other suitable lengths may be implemented.

[0180] In some embodiments, the anchors **401** may be moveable axially. For example, the anchors **401** may not deploy or otherwise extend outside the body **15** immediately upon expansion of the body **15**. Following acceptable positioning of the device **10** within the LAA, the flexible anchors **401** may then be advanced through a corresponding tube **500**. The anchors **401** may move axially in any suitable manner, including those described elsewhere herein. The anchor **401** may be moved through the tube **500** either before or after the tube **500** has been moved and deployed outside the body **15**, as described below.

[0181] In some embodiments, the tubes **500** are moveable and deploy outside the body **15**. The tubes **500** may be moveable in embodiments having either fixed or moveable anchors **401**. The tubes **500** may be pre-loaded over corresponding wire anchors **401** as shown in FIG. **31**, for example one tube **500** per anchor **401**. The tube **500** may then be moved over the corresponding anchor **401** as shown in FIG. **32**. The tube **500** may straighten the anchor **401** and add mechanical integrity. The tube **500** may also act as a perforation protector to prevent the anchor **401** from puncturing through the wall of the LAA. Movement of the tube **500** over the corresponding anchor **401** may shorten the exposed length of the anchor **401**, as described. This may provide for a stiffer tissue engaging segment of the anchor **401** due to the shortened exposed length.

[0182] In some embodiments, the tubes **500** extend from the delivery catheter to or near the outer surface of the body **15** but do not extend outside the body **15**. Instead, the tubes **500** just guide the anchor **401**, for example around the curve, and support the wire **401** right up to tissue penetration. The tubes **500** may set the launch angle so the anchor **401** does not buckle and hits the tissue at the right angle. In this embodiment, the anchor **401** may have relatively more stiffness than in the embodiments where the anchors **401** are relatively flexible, in order to provide a more secure anchoring of the device **10** to the tissue. It is understood the tube **500** may provide this guiding function to the corresponding anchor in any of the embodiments described herein having moveable anchors, such as the moveable anchors **401**, the anchors **120**, etc.

[0183] The flexible anchors **401** and/or the external stiffening tube **500** may be made from

biocompatible metallic materials such as Nitinol, implant grade stainless steel such as 304V or 316LVM, cobalt-chromium based alloys such as MP35N or Elgiloy, other suitable materials, or combinations thereof. The anchor **401** length can vary from 0.1 mm to 5 mm in length with an external stiffening tube **500** that covers from 10% to 90% of the exposed length of the anchor **401**.

[0184] The skin **20** at least partially surrounds the body **15** and portions of the skin **20** may or may not be attached to the body **15**. The various devices **10** described herein may have the body **15** at least partially encased within the skin **20**, which may be fabricated from a material such as ePTFE (expanded polytetrafluoroethylene), polyolefin, or polyester which assists with healing, anchoring, and retrieval. FIG. **33** is a side view of an embodiment of a device **10** for occlusion of the LAA having discrete points of attachment **700** of the skin **20** to the internal foam body **15**. For clarity, the points of attachment **700** are shown as dots in the figures. It is understood the points of attachment **700** may not be visible from outside the device **10**, for example the skin **20** may be bonded to the body **15** at the points of attachment **700**, etc. In some embodiments, in addition or alternatively to bonding, the skin **20** may be secured for example with sutures to the body **15** at the points of attachment **700**, and thus some or all of the points of attachment **700** may be visible from outside the device **10**. The device **10** may be in the configuration shown in FIG. **33** adjacent to or within the LA **201**. The skin **20** may be attached to the body **15** at the various separate points of attachment **700**. As shown in FIG. **33**, the skin **20** can be partially attached, with portions thereof not attached at all, to the body **15**. This may allow, for example, the skin **20** to move during expansion of the body **15** that occurs after deployment of the device **10** from the delivery catheter. The skin **20** may, in some embodiments, be attached at points of attachment **700** located near the proximal side of the device **10**, for example to help promote closure of the ostium of the LAA, such as with a rim **800** as described below. The skin **20** may, for example, be tacked in place in one or more points of attachment **700** near the proximal face so that any bunching of the skin **20** that occurs during implantation occurs near the ostium but within the LAA. This can be achieved using sutures, adhesive bonding, heat bonding, other suitable approaches, or combinations thereof.

[0185] The selective location of the points of attachment **700** may facilitate with formation of a circumferential rim **800** of the skin **20**. The rim **800** is shown schematically in FIG. **34** as a triangular rim for clarity. It is understood the rim **800** may be a variety of different shapes depending on the configuration of the device **10**, the shape of the LAA, etc. Further, the rim **800** may extend completely or partially around the device **10**. The rim **800** may surround the ostium of the LAA. The formation of the rim **800** may help to completely seal the entrance to the LAA around the device **10** and thereby prevent leakage. The attachment points **700** between the skin **20** and the body **15** can prevent irregular bunching of the fabric and instead guide any excess material to form the sealing rim **800** around or near the proximal face of the device **10**, as shown in FIG. **34**. The rim **800** may form upon expansion of the body **15** after deployment from the delivery catheter, as described herein. Alternatively the attachment points **700** can be designed so as to prevent any bunching at all of the fabric and provide a smooth surface, such as a smooth proximal surface.

[0186] FIGS. **35-36** are side views of embodiments of the device **10** having anchors **120** with V-tips **901** shown in the deployed configuration. The V-tips may be located in the proximal region **125** and/or may form all or part of the tissue engaging segment **121** of the anchor **120**, as described herein. The V-tips **901** form a V-shaped point. The V-tips **901** are generally in the shape of a “V” or an otherwise angled, segmented shape. The V-tips **901** may be sharp barbs or hooks. The V-tips **901** may be formed from wire or laser-cut tubing or other suitable methods. As shown in FIG. **35**, one or more of the V-tips **901** are attached to the body **15** encased in the skin **20**. The V-tips **901** can be attached to the body **15** and/or to the skin **20**. In some embodiments, the V-tips **901** are ends of anchors **1000**. For example, the V-tips **901** may be part of the anchors **1000** that are within the body **15** and skin **20**, as shown in FIG. **36**. The distal ends of the V-tips **901** may be free to slide along and collapse or expand. The distal ends of the V-tips **901** may be attached to the body **15**, skin **20**, and/or the anchors **1000** to allow the V-tips **901** to collapse and retract. During retrieval into a

catheter or sheath, the V-tips **901** can flatten out when engaging the inner diameter of the catheter or sheath. The V-tips **901** can be formed from Nitinol, implant grade stainless steel such as 304 or 316, cobalt-chromium based alloys such as MP35N or Elgiloy, other suitable materials, or combinations thereof. The V-tips **901** may then recover their pre-set shape after deployment or re-deployment.

[0187] FIGS. **37A-37C** are side views of various embodiments of V-tips that may be used with the anchors described herein. FIG. **37A** is a side view of an embodiment of the V-tip **901**. The V-tip **901** includes two angled segments. The segments may form the angle in a free state. The angle may be various angular amounts. In some embodiments, the angle formed by the V-tip **901** is no more than about 170°, 160°, 150°, 140°, 130°, 120°, 110°, 100°, 90°, 80°, 70°, 60°, or any smaller, greater or intermediate angular amount. FIG. **37B** is a side view of an embodiment of a wave V-tip **1101**. The wave V-tip **1101** may include a curved segment and an angled straight segment. FIG. **37C** is a side view of an embodiment of a two-wave V-tip **1103**. The two-wave V-tip **1103** may include two curved segments. The curved segments may promote engagement of the tip with the inner wall of the LAA. The end of the various V-tips may be smooth and rounded or sharp to promote tissue penetration. In some embodiments, all of the V-tips may have the same shape. In some embodiments, some of the V-tips may have a first shape and other V-tips may have a second shape different from the first shape. In some embodiments, some of the V-tips may be attached to the skin **20** and or body **15**. In some embodiments, some of the V-tips may be attached to the anchors **1000**.

[0188] FIG. **38** is a side view of another embodiment of the device **10** for occlusion of the LAA implanted inside an LAA **1201**. The device **10** includes a body **15** with a skin **20** and finial **30** placed within the LAA **1201**. The LAA includes a thicker proximal portion **1203** closer to the ostium. The internal locking system **101**, for example anchors thereof, may be configured to engage with the thicker proximal portion **1203** of the LAA. The various anchors, V-tips etc. described herein for the various embodiments of the device **10** may be used to secure the anchors in the thicker proximal portion **1203**. In some embodiments, the device **10** may be deployed from the catheter such that the body **15** expands. The location, orientation, etc. of the expanded body **15** within the LAA may be verified, for example by imaging, as described herein. The location, orientation, etc. of the expanded body **15** within the LAA may be verified to ensure engagement of the internal locking system **101**, for example anchors thereof, with the thicker proximal portion **1203**. Then, the internal locking system **101**, for example anchors thereof, may be deployed to engage the thicker proximal portion **1203**. If after deployment of the internal locking system **101**, for example anchors thereof, it is determined that the anchors did not engage with the thicker proximal portion **1203**, the anchors may be retracted, as described herein, in order to reposition and/or retrieve the device **10**.

[0189] In some embodiments, the internal locking system **101**, for example anchors thereof, may be preloaded surface elements releasably constrained or otherwise locked down in a collapsed or constrained position or configuration. The internal locking system **101**, for example anchors thereof, may be constrained using a restraint. The restraint may be a dissolvable polymer, a lasso, or wires that can be retracted to release the anchors. The restraint may be similar to a deadbolt. Other anchoring concepts include Velcro integral to the ePTFE, electrically orientable/ratcheting anchoring elements, unidirectional Gecko tape, or wires pre-attached to the finial **30**. In some embodiments, the body **15** with skin **20** may be secured within the LAA by texturing the body **15** and exposing the body **15** to the tissue through holes in the skin **20** to increase the friction with the cardiac surface to a high enough level to prevent implant migration.

[0190] FIGS. **39A-39B** are perspective views of an embodiment of a deployable anchor **1302** activated by a pull wire **1301** and shown, respectively, in the constrained and deployed configuration, that may be used with the various devices **10**, **1020**, **3000** etc. for occlusion of the LAA described herein. A two-stage anchoring system allows deployment of the anchors **1302** after

implantation and expansion of the body **15**. This embodiment incorporates one or more hinged anchors **1302**. The anchors **1302**, which be a barb or other anchoring element, may lie flat during delivery and during deployment of the body **15**. Next, when pulled or pushed, the anchors **1302** bend at a hinge **1306** and extend outward from the surface of the body **15** and into the LAA tissue. The anchors **1302** may bend at the hinge **1306** using a hollow constraining element **1304** which can be a thin, metallic, round or rectangular box such as a round or rectangular shaped tube, and the pull wire **1301**, for example a sliding element, which can be a wire or suture. The pull wire **1301** is attached to the proximal end of the anchor **1302** and extends back through the delivery catheter or sheath. When the pull wire **1301** is retracted, the anchor **1302** slides back though a slot **1308** in the tube **1304** and bends at the preformed hinge **1306**. A portion of the anchor **1302** then extends out through the slot **1308**.

[0191] FIGS. **40A-40B** are perspective views of an embodiment of a deployable anchor **1405** activated by a lock wire **1401** and shown, respectively, in the constrained and deployed configuration, that may be used with the various devices **10**, **1020**, **3000** etc. for occlusion of the LAA described herein. The anchor **1405**, which be a barb or other anchoring element, may be formed from wire or a flat sheet of Nitinol or other shape memory material and heat set to be in an expanded configuration. One or more of the anchors **1405** can be placed along the skin **20** or otherwise along an external surface of the body **15**. One or more corresponding guides **1402**, such as loops, may be located along the skin **20** or the body **15**. The guides **1402** may be located on both sides of the anchor **1405**, as shown. The guides **1402** on a first side of the anchor **1405** may fix the anchors **1405** in place. The guides **1402** on a second, opposite side of the anchor **1405** may act as a guide for the lock wire **1401**, which may be a restraining wire, suture, etc. The lock wire **1401** may be used to constrain the anchors **1405** in a constrained configuration, for example in the flat position as shown in FIG. **40A**. When the lock wire **1401** is retracted, the anchors **1405** deploy, as shown in FIG. **40B**. The anchors **1405** may extend perpendicular to the body **15**, or at an angle.

[0192] FIGS. **41A-41B** are perspective views of an embodiment of a deployable anchor **1506** activated by a sheath **1502** and shown, respectively, in the constrained and deployed configuration, that may be used with the various devices **10**, **1020**, **3000** etc. for occlusion of the LAA described herein. The anchor **1506**, which be a barb or other anchoring element, may be formed from wire or a flat sheet of Nitinol or other shape memory material and heat set to be in an expanded configuration.

[0193] One or more of the anchors **1506** may be placed along the skin **20** or otherwise along an external surface of the body **15**. One or more corresponding guides **1500** and locking loops **1504** may be located along the skin **20** or the body **15**. The guides **1500** may be located on a first side of the anchor **1506** and the locking loops **1504** may be located on a second, opposite side of the anchor **1506**, as shown. The anchors **1506** are held in the constrained or restrained configuration or position by a sheath cover **1502**. The sheath cover **1502** may be tubular or rectangular in shape. The sheath cover **1502** constrains the anchors **1506**. The sheath cover **1502** may constrain the anchors **1506** in a flat position as shown in FIG. **41A**. When the sheath cover **1502** is retracted, the anchors **1506** deploy, as shown in FIG. **41B**. The anchors **1506** may extend at an angle to the body **15**, or perpendicularly.

[0194] FIGS. **42A-42D** are various views of embodiments of the device **10** for occlusion of the LAA having external deployable anchors **1601**, **1604** which can be collapsed and expanded by retraction into or out of a sheath or outer catheter. FIG. **42A** is a side view of the device **10** having anchors **1601** constrained by a delivery sheath **1603**. FIG. **42B** is a side view of the device **10** unconstrained by the delivery sheath **1603** with the anchors **1601** deployed. FIG. **42C** is a side view of the device **10** having anchors **1604** constrained by the delivery sheath **1603**. FIG. **42D** is a side view of the device **10** unconstrained by the delivery sheath **1603** with the anchors **1604** deployed. The body **15** with skin **20** can contain the anchors **1601** or **1604** which are fixed to the surface of the skin **20** and are unconstrained and therefore expanded in a free state, as shown in FIGS. **42B**

and 42D. The delivery sheath **1603**, such as a catheter, may be used to constrain the anchors **1601** or **1604**. The anchors **1601** or **1604** may then expand when the body **15** is unconstrained by the delivery sheath **1603**, for example when the when the body **15** is released from the delivery sheath **1603**. The anchors **1601** may deploy into a curved shape as shown in FIG. 42B. The anchors **1604** may deploy into an angled shape as shown in FIG. 42D. The anchors **1603** or **1604** after deployment may point toward either the proximal or distal side of the body **15**.

[0195] FIGS. 43A-43C are sequential side views of an embodiment of the device **10** for occlusion of the LAA shown, respectively, constrained by a lasso **1707**, deployed, and adjusted with a mount **1705**. One or more anchors **1709** may be pre-mounted within the body **15** and attached distally to the mount **1705**. The mount **1705** may be a ring-like member having an opening extending therethrough. The mount **1705** is positioned over a rod **1701** with end **1711**. The mount **1705** can move, for example slide, over the rod **1701** in the proximal direction. In some embodiments, the mount **1705** may be pulled proximally, for example by a pull wire. In some embodiments, the mount **1705** may move when the rod **1701** is rotated. In some embodiments, the mount **1705** and/or end **1711** of the rod **1701** may be threaded. Movement of the mount **1705** causes the anchor **1705** to move. The device may include a tapered cone **1708**. The cone **1708** may be attached to the end of the rod **1701**. The mount **1705** may be moved toward the cone **1708** to adjust the height of the anchors **1709**. Thus, the anchors **1709** are angled more in FIG. 17C relative to FIG. 17B. The anchor **1709** may move through the body **15** and into the tissue. The anchors **1709** may be adjusted to increase or decrease the amount of tissue penetration, for example by moving the mount **1705** as described. For retrieval, this process can be reversed. In some embodiments, the lasso **1707**, attached to a wire **1703**, may extend, for example thread, through the threaded rod **1701** and be placed around the anchors **1709** to retract the anchors **1709** back into the body **15**. In some embodiments, the lasso **1707** may be used to initially constrain the anchors **1709** and then retract to allow the anchors **1709** to deploy.

[0196] FIGS. 44A-44C are side views of an embodiment of the device **10** for occlusion of the LAA having an adjustable two stage anchor system with anchors **1801** activated by moving a mount **1803** along a rod **1804**. The anchors **1801** may be internal grappling hook type structures placed within the body **15** and skin **20**. The anchors **1801** may be introduced through a central lumen **1003** that extends through the body **15**, as shown in FIG. 43A. The anchors **1801** may then travel through the body **15** and skin **20** to engage tissue, as shown in FIG. 43B. The anchors **1801** may be adjusted to increase or decrease the amount of tissue penetration. The anchors **1801** are attached at distal ends to the moveable mount **1803**. The mount **1803** is prevented from rotating, for example the mount **1803** may be notched to prevent rotation of the mount **1803** within the finial **30**, as shown in FIG. 43C. The mount **1803** is threaded onto the threaded rod **1804** that can be rotated clockwise or counter clockwise to change the linear position of the mount **1803**. The distal end of the rod **1804** may couple with a cap **1807**. The cap **1807** may rotate with the rotation of the rod **1804**. The mount **1803** may move proximally causing the anchors **1801** to extend past the surface of the body **15** and skin **20** as shown in FIG. 43B. The mount **1803** may move distally to pull the anchors **1801** back within or under the surface of the skin **20**. The depth of penetration of the anchors **1801** may be controlled, for example to account for the non-circular cross-section of the LAA. In some embodiments, the anchors **1801** may be deployed individually. Another option is to deploy the anchors **1801** distal to the body **15** with skin **20** and control the stiffness of the anchors **1801** such that they apply a reasonably uniform penetrating force to the tissue at contact.

[0197] Various features for LAA (LAA) occlusion may be included, such as those described, for example, in U.S. patent application Ser. No. 15/290,692, filed Oct. 11, 2016 and titled DEVICES AND METHODS FOR EXCLUDING THE LAA, in U.S. patent application Ser. No. 14/203,187, filed Mar. 10, 2014 and titled DEVICES AND METHODS FOR EXCLUDING THE LAA, in European Patent Application no. EP 14779640.3, filed Aug. 24, 2015 and titled DEVICES AND METHODS FOR EXCLUDING THE LAA, and in PCT patent Application no.

PCT/US2014/022865, filed Mar. 10, 2014 and titled DEVICES AND METHODS FOR EXCLUDING THE LAA, the entire disclosure of each of which is hereby expressly incorporated by reference for all purposes and forms a part of this specification. Further additions and improvements to these and other concepts are described below. The embodiments described in the sections below may include the same or similar features and/or functionalities as the embodiments described above, and vice versa, except as otherwise noted or indicated by context.

A. Basic Plug Design Components and Improvements

[0198] Various occlusion devices and associated features are described with respect to FIG. 45A to FIG. 77. The same or similar features and/or functionalities of the various devices as shown and described with respect to FIG. 45A to FIG. 77 may be present in the various devices as shown and described with respect to FIGS. 1-44C and 78-93B, and vice versa.

[0199] As shown in FIGS. 45A-45C, the device 1020 (sometimes referred to herein as “implant”) may utilize a cored-out foam “cup.” In some embodiments, this may be similar to the foam 1600 design shown in and described with respect to FIG. 16 and/or the foam body 3002 described with respect to FIGS. 85A-93B. The “cup” design may be contrasted with a solid or generally solid tubular foam plug, such as those described and shown in FIGS. 2 and 6. Regarding the cup design of FIGS. 45A-45C, the approximate thickness of the foam may be about 2.5 mm but can range from about 0.25 mm to about 10 mm. Note that the thickness may be significantly thicker than the typical stent coating or covering as utilized in other applications (e.g. coronary stents, peripheral stents, AAA liners, etc.). In this application, where we are occluding, the thickness of the foam adds some desired structure between the gaps of the internal support structure 1032, such as a stent. In some embodiments, the thickness may be at least about 0.25 mm; in some embodiments at least about 0.50 mm, 0.75 mm, 1.0 mm or 2.0 mm or greater in an unconstrained state, and in one implementation about 2.5 mm, with thickness selected depending upon the desired performance.

[0200] FIG. 45A is a cross-sectional view of the preferred embodiment showing the components of the implant 1020 having a proximal (atrial) end 1022, a distal (LAA) end 1024 and an interior cavity 1026 in the expanded configuration. An expandable tubular wall 1028 defines the interior cavity 1026, which may be enclosed at its proximal end 1022 by a tissue scaffold 1030 or other barrier configured to span the ostium and isolate the LAA from the atrium post deployment. The proximal edge of the tubular wall 1030 may be provided with a ramped reentry or recapture surface such as an annular chamfer 1031 extending circumferentially or otherwise around the proximal edge of the implant 1020, preferably continuously, to facilitate proximal retraction of the implant into the deployment sheath to permit repositioning or removal if desired. A distal extension 1029 of the tubular wall 1028 extends distally beyond the internal support (discussed below) to form an atraumatic leading edge.

[0201] The tissue scaffold 1030 may be integrally formed with the tubular wall 1028 or may be bonded thereto. The tissue scaffold 1030 and the tubular wall 1028 may have approximately the same thickness and pore characteristics, discussed below. Alternatively the tissue scaffold may comprise a different material such as ePTFE, PTFE, Dacron, or others known in the art, configured to support tissue ingrowth and isolate the LAA, but thinner than the tubular wall 1028.

[0202] An expandable, internal support structure 1032 such as a wave stent 1034 or other frame may be provided. The illustrated wave stent 1034 comprises a plurality of struts 1038, adjacent pairs of struts joining to form a plurality of proximal apices 1041 and distal apices 1042. The stent 1034 may be laser cut from tube stock as is known in the art. Each of at least three, and preferably at least 4 or 6 or 8 or more of the proximally facing apices 1040 is provided with a reentry or recapture strut 1044, which incline radially inwardly in the proximal direction to a central hub 1046. Recapture struts may be cut from the same tube stock as the stent. Hub 1046 may be provided with a central lumen, such as for delivery over a guidewire, or for releasable engagement with a deployment device (not illustrated). Alternatively, the hub 1046 may be provided with an attachment such as an eyelet 1048 for receiving a suture loop. A suture or other retention element

may extend distally from the deployment catheter, through the tissue scaffold **1028**, through the eyelet **1048** and proximally back through the tissue scaffold **1028** and into the deployment catheter. Following satisfactory positioning of the implant **1020**, the suture may be removed, releasing the implant **1020** from the deployment catheter leaving a homogeneous tissue scaffold **1030** as resilience of the material closes the suture tracks. In one preferred implementation the implant **1020** is deployed from the delivery catheter without advancing over the wire and the hub lacks a central lumen. In one embodiment, the implant is secured to the delivery system using any of a variety of means known in the art including screw mechanisms or ball in socket attachment mechanism, which can also pivot.

[0203] The tubular wall **1028** may be attached to the stent **1034** by adhesives, sutures or other bonding techniques known in the art. In the illustrated embodiment, the tubular wall **1028** is sutured to the stent **1034** and the tissue scaffold **1030** is sutured to the recapture struts **1044**, with the support structure **1032** carried within the cavity **1026**. Alternatively, at least a portion of the support structure **1032** may be carried on the outside surface of the tubular wall **1028** or tissue scaffold **1030**. The wave stent may be embedded within the tubular wall **1028** such as by sandwiching the stent between inner and outer layers of foam which are then bonded together. Similarly the recapture struts may be enclosed between inner and outer polymer layers. The polymeric material can also be foamed around the stent so that a secondary attachment process is not required.

[0204] FIG. **45B** is a distal end view of an embodiment of the implant **1020** showing the internal metallic structure having a central hub **1046** and eight recapture struts **1044** inclining radially inwardly to the hub **1046**. A plurality of suture retainers such as apertures **1050** are attached to or formed in the struts **1044** to receive the sutures used to secure the tissue scaffold **1030**. This reduces the tendency of material of the tissue scaffold **1030** to slide distally along the struts **1044** if the implant **1020** is proximally retracted into the deployment catheter.

[0205] The frame may be expandable from a contracted delivery configuration to an expanded deployed configuration. The frame may be retractable from an expanded deployed configuration to a contracted delivery configuration. The frame may be generally tubular, e.g. circular, rounded, segmented, polygonal, other shapes, or combinations thereof in the unconstrained, expanded configuration, and preferably presses the foam into conformance with the shape of the inside surface of the LAA. This permits the deployed implant to minimize leaks, the largest of which is no more than about 4 mm or 3 mm or 2 mm or less and in some deployments essentially no leaks as viewed on color Doppler.

[0206] FIG. **45C** is a proximal end perspective view of the outside of an embodiment of the implant **1020** having a unitary foam shell and showing the proximal chamfer **1028**, in an unconstrained expansion. Anchors deployed near the distal end of the frame are discussed in further detail below.

[0207] Some advantages of the cored-out foam “cup”, as compared to a solid foam plug, are as follows: it still behaves like a full foam plug with respect to conformability and sealing; it allows incorporation of an internal metallic frame which can be optimized to provide the desired amount of expansion for sealing and anchoring by providing an optimal radial force and an attachment point for the anchors, and a front face inside the proximal face to aid in collapse of the foam for retrieval; by sizing the metallic frame so that it is shorter in length than the foam cup, an atraumatic distal bumper is formed which is entirely foam and can be extruded from the sheath tip as the tip is advanced within the LAA; and the reduced overall volume of material aids with the following: it significantly reduces the delivery profile, it is easier to flush to remove air prior to delivery into the vascular system, and it makes the plug more porous to blood so that if embolized within the cardiovascular system, it allows more blood to flow through it.

[0208] In an embodiment, the proximally facing edges of the foam plug **1040** are chamfered to aid in loading and retrieval. Also, while there can still be a central location to allow tracking of the

implant **1020** over a guidewire-type device, in some embodiments there is no lumen or there is just a slit in the foam plug **1040**, as it is not desirable to have a significant residual central hole which may cause thrombus formation or allow leakage. The slit can be a single slit, double cross-shaped slit, or multiple slits. The objective is to still allow tracking over a guidewire but to be sure the hole closes completely once the guidewire is removed. In the case of a solid face, the implant **1020** may not be tracked over a guidewire and may instead, for example, be delivered through a long transseptal sheath.

[0209] Please note that the term “guidewire”, as utilized above, can mean an actual medical device sold as a guidewire or it can be a catheter, such as a pigtail catheter, which is initially placed in the LAA over which the LAAC (LAA Closure) implant **1020** is tracked.

[0210] Diameters: The LAA may vary in diameter from about 15 mm to about 33 mm and as such, the implant **1020** diameter must be able to accommodate this variation in sizes. The more the implant **1020** may accommodate various large ranges of diameters, the fewer predetermined sizes are needed, thereby simplifying the procedure for implantation. The construct of this implant **1020** is such that it may accommodate diameters less than 50% of the fully expanded diameter. The preferred plug **1040** diameters may be about 27 mm, 33 mm, and 35 mm. Ideally, only 1-2 sizes would be needed to close a large range of LAA diameters. This is a key advantage of the foam plug **1040** concept as compared to metal cage type devices with fabric tissue scaffolds.

[0211] Depth: The preferred plug **1040** length (the depth of the occluder within the LAA generally along a proximal-distal direction) is 20 mm and is independent of the diameter of the implant **1020**. This allows for good implant **1020** stability while still accommodating the majority of anatomies. The distal tip of the foam plug **1040** is very soft, providing an atraumatic tip as it enters the LAA when the distal tip of the implant **1020** is allowed to protrude past the distal dip of the delivery catheter or sheath. The short depth makes placement of the plug **1040** more robust, as there is less need to align delivery catheters with the LAA, as is needed with longer devices.

[0212] Foam and Porosity: The average pore size of the foam is 250-500 microns. The foam has a very high void content (90-95%) to promote quick and thorough tissue ingrowth. The open cell foam permits blood to flow through it. If the plug **1040** should embolize, it will be open enough to allow enough blood flow to be safe until it can be retrieved. Additionally, the large void content should be beneficial for proper flushing of the implant **1020** to prevent air introduction into the vascular system. The porosity and cell size may be as described with respect to the foam body **3002** of the device **3000**, shown and described for example with respect to FIGS. **85A-90D**.

[0213] The compliance and thickness of the foam are designed to provide a good seal against the tissue with minimal compression. While other devices require significant oversizing to obtain a seal, this implant **1020** may require only ≤ 1 mm of oversizing.

[0214] LA Facing Surface: ePTFE (expanded Polytetrafluoroethylene) or PTFE as the skin/layer for the implant **1020** such as over or partially over the plug **1040**, as described above, may be ideal for supporting neointimal formation without thrombus formation. While embodiments may be described with respect to ePTFE, it is understood that PTFE may also be used. It may, however, reduce safety due to the inability or reduced ability to allow blood flow through the surface of an embolized implant **1020** due to the low porosity of the ePTFE, even though blood flow may be enabled around the outer surfaces of the cup. ePTFE porosity is much lower than that of the open cell foam, so blood flow across the membrane may be negligible. It is, however, hydrophobic which is beneficial for thromboresistance. One option, to maintain the desired open porosity of the foam structure while adding the thromboresistance of ePTFE and/or PTFE, is to add a PTFE coating via vapor deposition to the foam. The thromboresistant coating may contain ePTFE or PTFE. This creates a highly porous surface that simulates the ePTFE morphology. Methods of attachment could include vapor deposition, as mentioned above or elastomeric glue (although this may eliminate porosity at the attachment points). If ePTFE is preferred, that could be attached by encasing the metallic frame in ePTFE then wrapping through the center, around the OD, and

attaching via sutures.

[0215] It may be desirable to reduce the pore size of the foam to between about 30-200 μm , as further described herein.

[0216] Barbs/Anchors: There are several options or types of barb designs that may be implemented for anchoring the implant **1020** within the LAA. The following are some examples: 1) Static: always engages tissue when the plug is deployed. This makes re-sheathing and repositioning of the implant **1020** more difficult. 2) Constrained: the implant **1020** can be deployed with barbs constrained. The implant **1020** can then be repositioned, as needed. The barbs are then released when the plug **1040** is in its final position. 3) Dynamic: barbs can be deployed or retracted as desired without dislodging plug. Dynamic barbs may allow for deployment and retraction, for example to reposition and/or remove the implant **1020**.

[0217] In some embodiments, the implant **1020** may have different features from other implants described herein. In some embodiments, the implant **1020** may include any or all of the following features: does not have a central lumen; has a spoked element against the proximal face inside the cup in addition to the wave stent; has anchors/barbs that can preferably be activated as a secondary step following placement of the plug itself; has a layer on the proximal face, which may be similar to the proximal face **1604'** shown in FIG. **16** or the layer **3100** shown in FIG. **85A**, although in some embodiments the PTFE may be applied via vapor deposition coating as opposed to expanded PTFE (ePTFE) attached as a secondary material.

B. Endoskeleton System with Proximal Spokes

[0218] The implant **1020** may include a foam plug **1040** with a central endoskeleton that includes a proximal spoke face **1080** with several radial struts. This configuration improves the ability to retrieve (re-sheath) the foam implant **1020**. The stent version depicted in FIG. **45A** and FIG. **45B** may be laser cut from a superelastic nitinol tube, however, numerous other biocompatible metallic materials can be utilized such as shape memory Nitinol, stainless steel, MP35N, or Elgiloy. While this embodiment is self-expandable, a balloon-expandable design could be utilized. Additionally the frame could be fabricated from drawn wire as opposed to being laser cut from a tube. Loops can be provided along each strut to allow for attachment to the foam via sutures, although other attachment processes could be utilized, such as adhesive bonding. The loops located mid strut can be oval in shape and staggered, to allow easier loading into the delivery catheter and ease of fabrication. Additionally, as shown in FIG. **46**, there may be no loops. While the embodiment shown in FIG. **45A** has 8 struts, anywhere from 4 to 32 struts may be utilized. In general, it is preferred that the foam be attached to the frame at numerous points including the center. This promotes retrieval without damage to the foam and the suture loops are beneficial for this. In other embodiments, the foam could be formed around the endoskeleton so that it is within the foam, eliminating the need for a secondary attachment step. As shown in FIG. **45A**, it is preferable that the proximal foam face have a chamfer at the edges to minimize the bulk of the material in this area to aid in re-sheathing.

[0219] FIGS. **47A-47B** are perspective and side views respectively of an embodiment of the LAA occlusion device **1020** having an internal frame **1032**, which may be a single piece. While the design shown in FIG. **45B** is fabricated from two separate pieces—the proximal spoke face **1080** with eight struts and a wave stent—in some embodiments there may be a single piece unit frame **1032**, such as that shown in FIGS. **47A** and **47B**. In some embodiments the proximal spoke face **1080** may support re-sheathing, the eight crown wave cage stent **1060** supports the foam cylinder plug **1040**, and the eight to sixteen, or fewer or more, barbs or anchors **1100** located within the cylinder provide anchoring to resist embolization. The anchors **1100** can be located proximal, distal, and/or centrally along the length of the cylinder. The anchors **1100** can preferably range in size, when fabricated from a Nitinol tube, from about 0.003" to about 0.009" in thickness and from about 0.007" to about 0.015" in width. In some embodiments the anchors **1100** may extend about 1 mm from the surface of the implant **1020** but can range from extending about 0.5 mm to about 2

mm, or less or more, from the surface of the implant **1020**. The anchors **1100** may be in a single location along the length of the cylinder or staggered, as shown in FIG. **48**. In vitro anchoring dislodgement resistance with these designs may be in the 0.5 lb to 1.5 lb force range. There may be a single row of anchors **1100** as shown. There may be multiple

[0220] FIG. **49** depicts the implant **1020** with central endoskeleton that includes a proximal spoke face **1080** in its fully expanded configuration as attached to the delivery catheter. FIG. **50** depicts the implant **1020** in an embodiment of its collapsed configuration. Variations include an outer sheath component of the delivery catheter that can be stretchable or slit at the tip, aiding in tapering collapse. A reduced coefficient of friction on the foam face may reduce the forces required to collapse the implant into the catheter. This may be done, for example, by applying a layer of PTFE via vapor deposition or other processes to the foam, or by a layer of expanded PTFE (ePTFE) attached to the proximal face using adhesives or mechanical methods such as suturing. Secure attachment of the foam face to the spoked system may be obtained by suture attachment or other methods including adhesive bonding, which will prevent the foam from bunching up during retrieval which can result in increasing forces and potentially tearing of the foam. If not securely attached with distributed forces, the metallic spoked element may pull through the foam during re-sheathing, destroying the implant.

C. Proximal (Blood-Contacting) Surface of the Foam

[0221] The foam implant **1020** may be comprised of porous open cell foam. The foam can be any of a variety of currently available materials including polyurethane-based biomaterials such as polyurethane or polycarbonate-polyurethane, or polyvinyl acetal (PVA) (Ivalon®). The foam can also be reticulated, such as a net. An embodiment utilizes a non-resorbable reticulated polyurethane-based biomaterial. Additionally, resorbable foams could also be utilized including a polyhydroxyalkanoate (PHA) such as poly-4-hydroxybutyrate (P4HB) or cross-linked resorbable polyester urethane-urea scaffolds.

[0222] Pore sizes in the material may be from about 50 micron to about 800 micron, preferably from about 250 micron to about 500 micron. Such a high void content (e.g. from about 90% to about 95%) material promotes quick and tenacious tissue ingrowth and effectively mimics the extracellular matrix. While such a high void content material is desirable for tissue ingrowth, it may not be ideal for thromboresistance, which is required for the left atrial (LA) surface. A thromboresistant surface is desired on the face that faces the LA. If modifications are made to the LA facing surface to promote blood compatibility, those modifications may be extended from about 1 mm to about 20 mm, preferably from about 1 mm to about 5 mm, onto the sides of the implant **1020** to assure thromboresistance in the event the implant **1020** is deployed with a portion of the implant's **1020** side surface projecting outside the LAA and into the blood environment. If there is a hole within the proximal face, such as a guidewire lumen, the thromboresistant surface may extend at least partially within that lumen. Additionally, this thromboresistant layer should promote tissue ingrowth and endothelialization.

[0223] Various methods may be used to create a thromboresistant proximal face of the implant **1020**, including but not limited to the following. For example, an expanded PTFE (ePTFE) skin or layer may be applied to the outside surface of the foam implant as described elsewhere herein. It could be attached by encasing the metallic frame in ePTFE then wrapping through the center, around the OD, and attaching to the frame. This can be done using numerous methods including suturing or adhesive bonding, including the use of elastomeric glue (although this may eliminate porosity at the attachment points). This ePTFE layer can extend into the guidewire lumen, if there is one. In addition to ePTFE, electrospun, melt blown, non-woven, knitted or woven fibers of PTFE, polyester, PGA, PLA, poly-4-hydroxybutyrate (P4HB) or other biocompatible fiber materials can be utilized to create a porous biocompatible surface.

[0224] In some embodiments, a coating of a hydrophobic material such as PTFE is applied over the proximal face using any of numerous processes known to one skilled in the art including vapor

deposition coating. Ideally this coated layer would also extend partially onto the sides of the implant. While some embodiments may not include a guidewire lumen, if a central lumen is present, the coating would preferably extend at least partially (e.g. about 1 mm) into it. To promote thromboresistance, this coated layer would decrease the porosity of the blood-contacting face to a porosity from about 30 microns to about 200 microns, preferably from about 100 microns to about 150 microns. Materials which could be utilized for this include conformal coatings such as PTFE applied in a thickness of about 50-100 microns, polyurethane spray or dip coatings, albumin, polyethylene glycol (PEG), or poly(ethylene oxide) (PEO), all with or without the incorporation of heparin or nitric oxide. The PEG or PEO would ideally be attached via a grafting process. In the preferred embodiment, the outer layer would also be lubricious to assist in re-sheathing of the implant. This can be achieved with both hydrophobic materials, such as ePTFE and PTFE, and hydrophilic ones, such as PEO and PEG. In order to produce a desirable combination of porosity and blood compatibility, a two-step process may be utilized where the foam is first coated with a base layer, such as a polyurethane-based biomaterial, then in a second step a more thromboresistant and lubricious surface is created utilizing PTFE, PEG, or PEO. Heparin or other anticoagulants may be added to the final blood-contacting surface.

[0225] Another option to create smaller pore sizes with an ePTFE-like material would be to attach an electrospun layer of PTFE to the face of the foam using suturing. A very thin layer (<1 mm) can be made and attached via suturing or adhesive bonding.

[0226] Another desired property of the foam of the plug **1040** is to provide echogenicity of the implant **1020**, which allows for visualization by echocardiography. To promote echogenicity, a porous surface may be adequate; however, in certain cases a hydrophilic surface may be beneficial. To promote blood compatibility and a hydrophilic surface, the preferred embodiment would be a foam implant which has a surface grafted with PEO or PEG.

D. Static Barb (Anchor) Designs

[0227] Static barbs engage tissue when the plug **1040** is deployed, e.g. expanded. While this simplifies fabrication, it makes re-sheathing and repositioning of the implant more difficult.

[0228] In some embodiments, as shown in FIG. **51**, a barb **2000** may be fabricated from wire and crimped onto the stent **1034**, in this example a wave stent. The barb may be made from Nitinol wire of any diameter, with a preferred range of about 0.005" to about 0.012". The tip can be sharpened for ease of penetration into the tissue. It can be attached to the stent frame using a crimp sleeve made from stainless steel, Nitinol, or titanium tubing. It may require filler wires inside the crimp tube to prevent rotation of the barb. It could also be attached by welding, using a laser or other energy source.

[0229] Referring to FIG. **52**, in some embodiments a laser cut double barb **2000** system can be utilized. This can be fabricated from a Nitinol tube **2002** cut to allow two crimp ends with one barb **2000** near each crimp **2004** with a continuous Nitinol connection that follows the curvature of the stent **1034**, in this example a wave form. An advantage of this embodiment is that it takes less labor to fabricate and shape the barbs, is easier to create sharp tips, and the curvature of the tube wall results in stiffening the barbs. FIG. **53** shows an embodiment of the laser cut part of FIG. **52** prior to forming the crown curvature.

[0230] Referring to FIG. **54**, in some embodiments the wire form follows the curvature of the wave support cage **1034** (stent), terminating in two barbs **2000**. It can be crimped to the wave cage or can be sewn with sutures, welded, or glued in place on the wave cage (stent). The advantage is that crimping is not required to prevent barb rotation.

[0231] Referring to FIG. **55**, in some static barb embodiments, a laser-cut wave support cage (stent) can be fabricated with integrated barbs **2000**. The advantage of this concept is that no secondary attachment step is required to attach the barbs to the cage. It may also be easier to add more barbs. A limitation is that it may be difficult to have 8 crowns (waves) with barbs on every strut as there may not be enough material available, therefore 6 crowns may be preferable.

E. Constrained Barb (Anchor) Designs

[0232] With constrained barbs **2010**, the implant **1020** can be deployed in the LAA with the barbs **2010** constrained and can therefore be repositioned, as needed, prior to barb release. The barbs **2010** are released when the implant **1020** is in its final position.

[0233] In one embodiment, as shown in FIG. **56**, a lasso-style constraint system can be added to static barbs to create constrained, deployable barbs **2010**. A suture material can be tucked between the barbs **2010** and struts to prevent spinning upon deployment. Removal of the lasso can release multiple barbs in a circular array.

[0234] As shown in FIG. **57**, barbs can be formed with a loop approximately midway along the barb **2010** which allows suture or thread to be placed through the loop, forming a lasso. This prevents full expansion of the barbs **2010** until the body of the implant is in its final desired position within the LAA.

[0235] A flipping-barb option is shown in FIG. **58**. In this embodiment, barbs **2010** can be elongated designs that are folded back inside a cored-out foam opening prior to loading into the delivery system. This may require drawstrings or other locking elements to hold the barbs in a constrained configuration. Once the implant **1020** is in its final desired location, the constraint is removed and the barbs are no longer constrained and flip into position, engaging the tissue.

F. Dynamic Barb (Anchor) Designs

[0236] Dynamic barbs **2020** can be deployed or retracted as desired without dislodging the implant. Dynamic barbs **2020** may be the preferred option for some procedures, for example for recapture of the implant **102** into the delivery catheter and/or sheath.

[0237] One embodiment is a tube within a tube. In this embodiment, as shown in FIG. **59**, a laser cut tube **1030** may be utilized to construct a one-piece front spoked face **1080** and wave stent **1032** from a single piece of preferably superelastic Nitinol tubing. Other materials, including shape memory Nitinol, stainless steel, MP35N, or Elgiloy, could be utilized. A second smaller laser cut tube **1040**, as shown in FIG. **60**, may be utilized to form an inner spoked barb array.

[0238] As shown in FIG. **61**, during deployment of the foam implant, the front spoked face and wave stent may be deployed while the spoked barb array remains in a constrained position. The implant **1020** can be repositioned, as needed, then distal motion of the inner tube **1040** with respect to the outer tube **1030** allows the barbs **2020** to expand and engage, as shown in FIG. **62**. The inner spoked barb array can be re-constrained and re-released repeatedly until disconnected from the delivery catheter transmission lines.

[0239] In another embodiment, as shown in FIG. **63**, a dynamic barb **2020** design which can preferably be fabricated from a single component is illustrated. This embodiment can be cut from a laser cut tube **1045**, preferably superelastic Nitinol, where half of the front spokes connect to wave points of the stent cage **1032** to support re-sheathing while the other half of the spokes are formed into the barbs **2020**.

[0240] Recapture or re-sheathing of the implant **1020** may initiate the barb retraction by collapsing the front spoke face, as shown in FIG. **64**, thereby simultaneously retracting the barbs **2020** from the tissue. This allows for safe repositioning of the implant **1020** with a minimal amount of re-sheathing required (limiting the length of implant required to be fully retracted into the sheath).

[0241] In another embodiment, as shown in FIG. **65**, a dynamic hinged barb system is illustrated. This is a laser cut wave form integrated barb **2020** with a living hinge and constraint. When the constrained end is heat-set, it may curl inward to ride over or under the wave strut. When positioned over the wave strut, its sharp barb end may point to the inner diameter of the assembly. When popped into position under the strut, the opposing barb end may see-saw upwards to engage tissue. Activation of the hinged barbs **2020** can be accomplished through the use of a lasso threaded in a circular array between each barb and the corresponding strut. Cinching inward on the lasso can pop the retention constraint from above the strut to below the strut, causing the opposing barb end to be raised above the surface of the strut where it can engage a tissue surface.

G. Embodiments with Foam Cup, Wave Stent, and ePTFE Layer

[0242] As shown in FIGS. **66A-66C**, the foam cup plug **1040** with internal wave or zig-zag anchor (for example, as shown in and described with respect to FIG. **16**) can be modified to be completely covered on the outer surface by an ePTFE layer **2060**. This layer **2060** can attach to the distal end **1024** of the foam via suture or adhesive bonding as opposed to just the proximal face **1022**. It can wrap over the entire external surface and into the central lumen on the proximal face, attaching to the proximal portion of the wave stent by lamination or adhesive bonding. Other biomaterials or coatings could be utilized including, but not limited to, PTFE or polyurethane.

H. Dynamic and Static Anchor Concepts

[0243] The embodiment of an implant **2300** shown in side cross-section view and end view in FIG. **67**, respectively, includes anchors **2310** that can be tied, welded, or crimped together at their proximal ends. The anchors **2310** may be made of Nitinol wire to permit loading into the delivery system without taking a set. There could be 8-16 wires formed in the curve. The anchors may act like a “grappling hook”. The assembly is loaded into the delivery catheter in a straightened position. As they are pushed out of the delivery catheter, they take the shape shown in FIG. **67** and pierce into the tissue through the foam **2320**. The foam **2320** is cored out as shown to reduce the volume of foam needed to compress into the delivery system and yet maintain sufficient radial force to seal the tissue.

[0244] The embodiment of the implant **2350** shown in FIGS. **68A** and **68B** includes a wave stent internal anchor with barbs within a foam plug which has been cored out to be thicker on the distal end to create a larger atraumatic foam tip bumper **2360** when collapsed within but partially deployed from the delivery catheter. The proximal face **2363** can have an internal lumen **2365** as shown in FIG. **68A**, but may not include an internal lumen as shown in FIG. **68B**.

I. Constrained Anchor Deployed in Secondary Step

[0245] The embodiment of an implant **2370** shown in FIG. **69** has barbs/anchors **2371** pre-attached to a wave stent **2372** inside the foam **2374**. A suture **2375** is looped around the distal end of the wave stent **2372** to compress the stent **2372** and pull the barbs **2371** inward. The suture **2375** is tied with a slip knot. After delivery of the device **2370** into place, one end of the suture **2375** is pulled and the knot comes free and the suture is removed, allowing the stent **2372** to open radially and the barbs **2371** to engage the tissue. The barbs **2371** can be located distal to the foam **2374** as shown, or pre-engaged with the foam **2374** and penetrating through the foam **2374**.

J. Internal Stent with Increased Embolization Resistance

[0246] A metallic stent-like frame **2380** with distal static barbs **2381** and a proximal “speed bump” **2382** is disclosed as shown in FIGS. **70-72**. It could be a zig-zag stent (as shown) or a wave stent or other similar expandable embodiment. The bumps **2382** placed on the external portion of the metallic stent frame **2380** can be rounded or pointed in design. Their purpose is to provide added resistance and stability for the implant when deployed in the LAA, to prevent embolization. In the preferred embodiment, the foam “cup” **2384** could be shaped as depicted in FIG. **71** to provide added material **2385** on the distal end for an atraumatic bumper when the distal end of the implant is partially deployed from the delivery catheter. It may or may not have an internal lumen for a guidewire.

K. Constrained Atraumatic Anchor

[0247] As shown in FIGS. **73** and **74**, a wave or zig-zag or other stent **2390** may be fabricated with distally located loops **2391** which engage the tissue in the LAA to anchor the implant in place and prevent dislodgement resistance. Because they are round and not sharp, this limits the risk of perforation, however, if a totally rounded loop does not provide adequate dislodgement resistance, the tips can be modified to incorporate a sharp feature. Loop anchors **2391** can be placed on each stent end or on just a few, so anywhere from 4-16 anchors can be placed.

[0248] During delivery of the implant through the vascular system and into the heart and LAA, the loops **2391** are folded toward the center of the constrained stent and are lined up next to each other,

as shown in FIG. 75. The inner catheter is placed through the loops **2391** to keep them constrained. When the system is in the LAA, the outer catheter/sheath is retracted, deploying the proximal portion of the implant. If positioning appears acceptable, the inner catheter can be removed, allowing the distal portion of the implant to fully expand and the loop anchors to engage the LAA tissue.

L. Perfusion Element & Barb Options

[0249] In the event of the implant being dislodged after deployment in the LAA, it is anticipated that the device would travel into the LA, across the mitral valve and into the left ventricle, then out the left ventricular outflow track, across the aortic valve and into the aorta and distal circulation. In this journey, it is possible that the device may get lodged in any of the above structures, interrupting flow and causing distal ischemia and possible hemodynamic collapse. It therefore would be desirable for an implant **2392** to have design features which would allow for distal perfusion, for example as shown in FIGS. 76 and 77.

[0250] FIG. 76 is a top view of an embodiment of the implant showing the left atrial surface. Within the surface is seen a valve, which in the shown embodiment is a simple cut (A) which would open up when sufficient pressure was applied, allowing for flow through the device **2392**. This flow could be bi-directional or uni-directional and depending on the specific conditions allowing for flow rates from 10 ml/min to 5 L/min. In the embodiment as shown, this would be accomplished without loss of structural integrity, the loading conditions would result in the device **2392** separating into a multiple sub sections each of which would be retrievable using standard technique. A specific device could have a single valvular element or an array (as shown).

[0251] As shown in FIG. 77, in some embodiments, a series of side ports (B) may be cut in the side wall of the foam **2393** of the implant **2392** to allow for blood flow in the case of dislodgement and distal embolization. These can vary in size and number from 1-20 ports of about 0.1 mm to about 5 mm in diameter. They can also vary in shape including but not limited to circular, oval, and rectangular.

[0252] Regarding barb designs, such as those described herein, in addition to having an array of barbs deployed at different distances from the LA surface, barbs can be incorporated which penetrate into the tissue to different depths. In one embodiment, the barbs placed most proximal on the implant can be longer so they penetrate deeper into the thicker proximal LAA tissue whereas those placed most distal on the implant are shorter, so they penetrate less deep into the delicate distal tissue, to maximize embolization resistance while minimizing the risk of perforation. In another embodiment the proximal and distal barbs can be the same length but can be fabricated from different diameters of wire or can be cut at different thicknesses from tubing material, so that the barbs that engage the more delicate distal LAA tissue are more flexible, or they are designed to primarily engage the internal trabeculations within the LAA, whereas the proximally located barbs penetrate the tissue. Alternatively, two barbs can be placed at each crown point of the stent, as opposed to one.

M. Multi-Functional Occlusion Devices

[0253] FIGS. 78-84 depict various features that may be used, either alone or in combination, with any of the LAA occlusion devices and methods described herein. In some embodiments, the features of FIGS. 78-84 may be incorporated into the device **3000** and associated features and methods shown and described with respect to FIGS. 85A-94B.

[0254] FIG. 78 is a side view of an embodiment of the LAA occlusion device **3000** with ablative features. The LAA occlusion device **3000** has an array of ablative elements **3005**. The ablative elements **3005** deliver energy to the tissue in and around the LAA ostium to electrically isolate the LAA. In the shown embodiment, the array of ablative elements **3005** is positioned at the proximal end **3004** of the body **3002**. The ablative elements **3005** may be electrically connected through a deployment catheter to an energy source. Energy may be provided via radiofrequency, ultrasonic, electrical, or other suitable methods. An inner lumen **3003** extends through the body **3002**. In some

embodiments, there may not be the lumen **3003**.

[0255] FIG. **79** is a side view of an embodiment of the LAA occlusion device **3000** with pressure sensing features. The device **3000** has a pressure sensor **3007** on its proximal surface **3008**. In some embodiments, the sensor **3007** may be on the proximal surface **3102** of the proximal cover **3100** (see FIG. **85A**). In some embodiment, the sensor **3007** does not protrude into the LAA, such as a flat sensor on the proximal surface **3008** or **3102**. The sensor **3007** is electrically connected via a wire **3009** to an electronic element **3011**. The electronic element **3011** has the capability of transducing and storing the signal generated by the sensor **3007**. This information may be transmitted with a signal **3013** remotely. The electronic element **3011** may be powered remotely or by an internal battery.

[0256] FIG. **80** is a side view of an embodiment of the LAA occlusion device **3000** with drug elution features. The device **3000** has a sensor **3007** on its proximal surface **3008**. The sensor **3007** may be on the proximal surface **3102** of the proximal cover **3100** (see FIG. **85A**). The sensor **3007** is electrically connected via a first wire **3009** to an electronic element **3011**. The electronic element **3011** is electrically connected via a second wire **3015** to a drug reservoir **3017** which is fluidly connected via a conduit **3019** to a drug exit port **3021**. This allows for drug to be delivered into the LA, as well as for monitoring concentration of specific chemical(s) and or states and to deliver agents in response, e.g., blood sugar level driving insulin delivery. The sensor **3007** may detect the level of various chemicals and the reservoir **3017** may be controlled, for example by the electronic element **3011**, to elute drug via the port **3021** in response.

[0257] FIG. **81** is a side view of an embodiment of the LAA occlusion device **3000** with pacing/defibrillatory features. The device **3000** has an electrical pacing element **3025**, such as an electrode. The pacing element **3025** extends circumferentially about the body **3002**. The pacing element **3025** may be in other configurations. The pacing element **3025** is connected by a wire **3027** to a pace generator **3029**. The generator **3029** is attached by a wire **3031** to a battery **3033**. This pacing system may pace the atrium and defibrillate the atrium when in atrial fibrillation. The generator **3029** and/or battery **3033** may include components for controls, communications, commands, etc.

[0258] In some embodiments, the LAA may be electrically isolated. The LAA may be electrically isolated with an occlusion device that incorporates one or more ablative elements, such as those shown and described with respect to FIGS. **78-81**, conduct the ablation to electrically isolate the LAA, then disengage the occlusion device **3000** leaving it in place within the heart. In some embodiments, the LAA may first be electrically isolated and then the device **3000** implanted, as described herein, for example with respect to FIGS. **82-84**. In some embodiments, conformable circumferential ablation via a foam plug with ablation elements may be incorporated.

[0259] FIGS. **82-84** depict various systems and methods for electrically isolating the LAA that may be used with the device **3000**. In some embodiments, the LAA may be electrically isolated followed by LAA closure. For example, the systems and methods shown in FIGS. **82-84** may be used to conduct the isolation, followed by LAA occlusion using the various LAA occlusion devices described herein, such as the device **3000**.

[0260] FIG. **82** is a side view of an embodiment of over the wire circumferential ablation balloon system. An over-the-wire balloon catheter **3035** is placed over a guidewire **3037** and into the LAA (LAA). The balloon **3039** can have one or more circumferential ablative elements **3041**, such as apposed radiofrequency (RF) elements, to electrically isolate the LAA using RF to treat atrial fibrillation. The ablative element **3041** extends circumferentially about the balloon **3039**. The ablative element **3041** may be in other configurations. This guidewire **3037** may have attached to its distal end a balloon **3043** which is inflated in the LAA and serves as a bumper to prevent guide catheter **1100** from perforating the wall of the LAA. These features may be similar to those described with respect to FIGS. **8-11**.

[0261] FIG. **83** is a side view of an embodiment of over the wire circumferential ablation

ultrasound balloon system. An over-the-wire balloon catheter **3035** is placed over a guidewire **3037** and into the LAA (LAA). A balloon **3045**, such as a circumferential ablation ultrasound balloon, is used to electrically isolate the LAA using ultrasound (US) to treat atrial fibrillation. The guidewire **3037** may have attached to its distal end the balloon **3043**, as described with respect to FIG. **82**. [0262] FIG. **84** is a side view of an embodiment of an over-the-wire circumferential ablation helical wire system with ablation elements. An over-the-wire circumferential ablation helical wire **3047** with one or more ablation elements is placed into the LAA (LAA). The wire **3047** can be used to electrically isolate the LAA using radiofrequency to treat atrial fibrillation.

N. Embodiments with Compressible Foam Body, Proximal Cover, and Compliant Frame Having Proximal Recapture Struts and Distal Tubular Body

[0263] FIGS. **85A-93B** show another embodiment of an LAA occlusion device **3000**. The device **3000** described herein may have the same or similar features and/or functionalities as other LAA occlusion devices described herein, and vice versa. Any of the features of the device **3000** described with respect to FIGS. **85A-93B** may therefore apply to features of the devices described with respect to FIGS. **1-84**, such as the implant **1020**, and vice versa.

[0264] FIGS. **85A-85C** show the LAA occlusion device **3000** having a foam body **3002**, an expandable support or frame **3040**, and a proximal cover **3100**. FIGS. **86A-86C** show the foam body **3002**, with the body **3002** shown in cross-section in FIGS. **86B** and **86C**. FIG. **86C** additionally includes the full view (i.e. non-cross section) of the frame **3040**. The device **3000** is shown in an expanded configuration in these figures. The device **3000** has a longitudinal axis as shown, which may be defined by the foam body **3002**, as further described.

1. Compressible Foam Body

[0265] The body **3002** is formed from a compressible material, such as foam. The body **3002** may be a foam formed from reticulated (e.g. net-like) polycarbonate polyurethane-urea. The body **3002** may be cut, formed or assembled into a cup shape, as further described. The body **3002** may have a thickness and compressibility sufficient to engage the surrounding tissue and conform to the anatomic irregularities under radial force applied by the inner frame, as further described. The use of a compressible material such as foam for the body **3002** provides a complete seal of the LAA and superior performance for LAA occlusion over existing devices, as further described. The structure of the foam of the body **3002** comprises a three dimensional network of interconnected reticulations, spaced apart to form a network of interconnected open pores, as further described. The reticulations can carry a coating, such as PTFE, while preserving the open pores, as further described.

[0266] The foam material of the body **3002** has a high porosity. “Porosity” as used herein has its usual and customary meaning and refers to open void content between the interconnected reticulations of the foam. The porosity of the body **3002** may be at least about 65%, at least about 70% at least about 75%, at least about 80%, at least about 85%, at least about 90%, at least about 95% or more. The porosity may be within the range of approximately 90-95%. The porosity may be approximately 90%. The porosity may be approximately 95%. The porosity may be 90%, 91%, 92%, 93%, 94%, or 95%. The high porosity promotes quick and tenacious tissue ingrowth, allows it to be compressed into a small catheter, and/or allows blood to pass if the implant embolizes, among other advantages.

[0267] The foam body **3002** has pores or cells formed between the interconnected reticulations of the foam material. The foam body **3002** has cells with sizes in the range of from about 250 μm to about 500 μm . The foam may have a cell size from about 125 μm to about 750 μm , from about 175 μm to about 650 μm , from about 200 μm to about 600 μm , from about 225 μm to about 550 μm , from about 275 μm to about 450 μm , less than 125 μm , or greater than 750 μm . These sizes may refer to the size of the cell prior to application of any coating, such as PTFE. The cell size may thus change, e.g. decrease, after application of the coating. The desired porosity and/or cell size may be determined based on allowing the passage of blood while blocking debris of a size capable of

potentially causing ischemic stroke. The allowable size of such debris may drive the selection of the particular porosity and/or cell size. For example, the cell size from about 250 μm to about 500 μm may be based on prevention of debris of a particular size from passing through the body **3002**. [0268] In an embodiment, the foam body **3002** is made from a non-resorbable, reticulated, cross-linked, polycarbonate polyurethane-urea matrix, structurally designed to support fibrovascular tissue ingrowth, with a fully interconnected, macroporous morphology with over 90-95% void content and cell sizes ranging from 250 to 500 μm .

[0269] The body **3002** has a proximal end **3004** and a distal end **3006**. In some embodiments, the axial length of the device **3000** from the proximal end to the distal end in a free, unconstrained state is 20 mm. As used herein, the “free, unconstrained” state, and the like, refers to a state of the device **3000** without any external forces applied to the device **3000** other than a normal or reactive force from a surface (e.g. table top) on which the device **3000** is placed. In some embodiments, this axial length may be from about 10 mm to about 30 mm, from about 12 mm to about 28 mm, from about 14 mm to about 26 mm, from about 16 mm to about 24 mm, from about 18 mm to about 22 mm, or about 20 mm. The body **3002** may have any of these lengths regardless of outer diameter of the body **3002**.

[0270] The proximal end **3004** of the body **3002** has a proximal end wall or face **3008**. The proximal face **3008** faces generally toward the LA when the device **3000** is implanted into the LAA. The device **3000** may be implanted off-axis, as further described, in which case the proximal face **3008** may not reside at a perpendicular to a longitudinal axis of the LA. The proximal face **3008** thus provides a closed proximal end **3004** of the body **3002**. The closed proximal end **3004** is configured to span the ostium but the porosity, as further described, is sufficient to permit the passage of blood while blocking debris of a size capable of potentially causing ischemic stroke. This membrane may be formed by the body **3002** and/or the cover **3100**. In some embodiments, the proximal face **3008** or portions thereof may be open. For example, there may not be a proximal face **3008**, there may be a partial proximal face **3008**, there may be a proximal face **3008** with portions removed, etc. In some embodiments, the proximal face **3008** or portions thereof is/are not included and any opening or openings is/are covered by the cover **3100**. The size of any such openings in the proximal face **3008** may be driven by the desired size of embolic debris to be prevented from escaping the LAA, as further described.

[0271] The proximal face **3008** is flat or generally flat and generally perpendicular to the longitudinal axis of the device **3000**. The proximal face **3008** has a circular or generally circular shape as viewed from the proximal end **3004** in an unconstrained expansion. In some embodiments, the proximal face **3008** may be flat, rounded, segmented, angled with respect to the longitudinal axis, other shapes, or combinations thereof. The proximal face **3008** may have a non-circular, polygonal, other rounded shape, other shapes, or combinations thereof, as viewed from the proximal end **3004**.

[0272] The proximal face **3008** has an outer surface **3010** and an opposite inner surface **3012**. The outer surface **3010** faces proximally away from the device **3000** and the inner surface **3012** faces distally toward the frame **3040**. The surfaces **3010**, **3012** may define outer and inner sides of the proximal face **3008**. The thickness of the proximal face **3008** may be measured axially between the outer surface **3010** to the inner surface **3012**. This thickness in a free, unconstrained state (e.g. uncompressed and expanded) may be from about 0.5 mm to about 5 mm, from about 1 mm to about 4 mm, from about 2 mm to about 3 mm, about 2.5 mm, or 2.5 mm. In some embodiments, the thickness may be less than 0.5 mm or greater than 5 mm. The thickness of the proximal face **3008** may be uniform or non-uniform. Thus the thickness may be greater or smaller in different regions of the proximal face **3008**.

[0273] The body **3002** includes a sidewall **3014** extending distally from the proximal face **3008**. The sidewall **3014** extends circumferentially about a perimeter of the proximal face **3008** to form a closed cross-section (i.e. extends circumferentially 360 degrees about the axis). The sidewall **3014**

extends axially to define a tubular body concentric about the longitudinal axis of the device **3000**. The longitudinal axis extends through a geometric center of the tubular body defined by sidewall **3014**. The sidewall **3014** is tubular or generally tubular, e.g. cylindrical, along the axis. In some embodiments, the sidewall **3014** may be conical or frustoconical, for example where the proximal end is wider than the distal end or vice versa. The sidewall **3014** may have an outer profile at the proximal end thereof, and as viewed from the proximal or distal end, to match that of the outer perimeter of the proximal face **3008**.

[0274] In some embodiments, the cross-section of the sidewall **3014** may not be closed, for example where there are openings in the sidewall **3014**. Thus cross-sections taken at various locations along the longitudinal axis may or may not show a closed section. In some embodiments, the sidewall **3014** may be non-tubular, non-cylindrical, non-circular, polygonal, other rounded shapes, other shapes, or combinations thereof. In some embodiments, as shown, the sidewall **3014** may extend continuously for the entire length from the proximal end **3004** to the distal end **3006**. In some embodiments, the sidewall **3014** may not extend continuously for the entire length from the proximal end **3004** to the distal end **3006**. For example, the sidewall **3014** may include a plurality of disconnected sections, such as annular portions of the sidewall, located and spaced along the longitudinal axis and connected to the frame **3040**.

[0275] The sidewall **3014** has an outer surface **3016** and an opposite inner surface **3018**. The outer surface **3016** faces radially outward from the axis. The inner surface **3018** faces radially inward toward the axis. The thickness of the sidewall **3014** may be measured radially between the outer surface **3016** to the inner surface **3018**. This thickness in a free, unconstrained state (e.g. uncompressed) may be from about 0.5 mm to about 5 mm, from about 1 mm to about 4 mm, from about 2 mm to about 3 mm, about 2.5 mm, or 2.5 mm. In some embodiments, the thickness may be less than 0.5 mm or greater than 5 mm. The thickness of the sidewall **3014** may be uniform or non-uniform. Thus the thickness may be greater or smaller in different regions of the sidewall **3014**. The thickness of the sidewall **3014** may be the same or different as the thickness of the proximal face **3008**. In some embodiments, the thickness of the proximal face **3008** is 2.5 mm and the thickness of the sidewall **3014** is 2.5 mm. In some embodiments, the thickness of the proximal face **3008** is about 2.5 mm and the thickness of the sidewall **3014** is about 2.5 mm.

[0276] The sidewall **3014** has a distal free end **3020** having a distal surface **3022**. The distal surface **3022** is flat or generally flat and perpendicular to the longitudinal axis of the device **3000**. In some embodiments, the distal surface **3022** is non-flat, angled with respect to the axis of the device **3000**, curved, rounded, segmented, other shapes, or combinations thereof.

[0277] The body **3002** may have a distal opening **3024**. The opening **3024** is formed by the distal free end **3020** of the sidewall **3014**. The opening **3024** is at a distal end of an internal central volume or cavity **3028** of the body **3002** that is formed at least partially by the sidewall **3014**, the proximal face **3008** and/or the shoulder **3030**. The frame **3040** may reside within the cavity **3028**, as further described. The distal opening **3024** may be completely open. In some embodiments, the distal opening **3024** may be mostly open, partially open, or closed, for example where the body **3002** has a distal face similar to the proximal face **3008** to enclose or partially enclose the cavity **3028**.

[0278] The body **3002** has a shoulder **3030**, shown as a bevel, that extends between the proximal face **3008** to the sidewall **3014**. The shoulder **3030** may be an intersection of a proximal end of the sidewall **3014** and the proximal face **3008**. The shoulder **3030** extends circumferentially about the entire perimeter of the intersection. The shoulder **3030** has an outer surface **3032**. The outer surface **3032** may be a beveled surface. The outer surface **3032** is flat or generally flat in an axial direction. The outer surface **3032** extends circumferentially about the entire perimeter of the shoulder **3030**. In some embodiments the shoulder **3030** and/or outer surface **3032** may be non-flat, rounded, other shapes in an axial direction, or combinations thereof. The shoulder **3030** and/or outer surface **3032** may extend circumferentially less than the entire perimeter of the shoulder **3030**. The thickness of

the shoulder **3030** may be measured inward perpendicularly to the outer surface **3032**. The thickness of the shoulder **3030** may be the same as the thicknesses of the proximal face **3008** and/or the sidewall **3014**, as described herein. In some embodiments, the thickness of the shoulder **3030** may be different from the thicknesses of the proximal face **3008** and/or the sidewall **3014**. The shoulder **3030** may function as a recapture ramp, to facilitate drawing the implant proximally into the deployment catheter.

[0279] The compressibility of the body **3002** contributes to the superior sealing capability of the device **3000**. The foam may be compressible to provide a larger radial “footprint” and spread out the radial forces from struts on the frame **3040**, as further described. The foam body **3002** may have a compressive strength of at least 1 pound per square inch (psi) or within a range of about 1 psi to about 2 psi, or no more than about 2 psi. The “compressive strength” here refers to the pressure to compress the foam to 50% strain. With some foam materials for the body **3002**, the pressure may not change from 50% strain through at least 80% strain, and the relation of pressure versus strain may be flat or generally flat. Thus, even with thicker foams for the body **3002**, the body **3002** will not exert much more outward force on the tissue due to the increased thickness by itself. In an embodiment, the foam body **3002** is a reticulated, cross linked matrix having at least about 90% void content, an average cell size within the range of from about 250-500 microns, a wall thickness of at least about 2 mm and a compressive strength of at least about 1 psi. In an embodiment, the body **3002** is formed from a foam material having or substantially having the material properties indicated in Table 1. In some embodiments, the body **3002** is formed from materials described in, for example, U.S. Pat. No. 7,803,395, issued Sep. 28, 2010, and titled “Reticulated elastomeric matrices, their manufacture and use in implantable devices,” or U.S. Pat. No. 8,337,487, issued Dec. 25, 2012, and titled “Reticulated elastomeric matrices, their manufacture and use in implantable devices,” the entire disclosures of which are incorporated herein by reference.

TABLE-US-00001 TABLE 1 Example material properties for an embodiment of foam material that may be used for the foam body 3002. Material Property Value Permeability 311 Darcy Average Cell Size 377 μ m Density 2.7 lb/ft.^{sup.3} Compressive Strength 1.1 psi Tensile Strength Parallel 68 psi Tensile Strength 32 psi Perpendicular Elongation Parallel 219% Elongation Perpendicular 243%

[0280] The device **3000** may include markers **3023** (see FIG. 85B; for clarity only some of the markers **3023** are labelled in the figures) to facilitate visualization during delivery. The markers **3023** may be radiopaque marker bands sewn into the distal free end **3020** of the body **3002**. The markers **3023** may be for visualization using fluoroscopy imaging of the distal end **3006** of the device **3000** during delivery. There may be a series of the markers **3023** located circumferentially along the distal surface **3022** of the body **3002** (for clarity, only some of the markers **3023** are labelled in FIG. 85B). In some embodiments, the markers **3023** may additionally or alternatively be located in other areas of the body **3002** and/or on other parts of the device, such as the cover **3100** or frame **3040**.

[0281] In some embodiments, four platinum iridium (PtIr) radiopaque (RO) tubular markers **3023** are sewn onto the distal end **3006** of the foam body **3002** to enable visualization of the distal edge of the device **3000** under fluoroscopy. In some embodiments, a PtIr marker **3023** is attached to the foam body **3002** at the location of the proximal shoulder **3030** to use as a marker during recapture of the device **3000**. Visualization of the proximal and/or distal markers **3023** may facilitate with identifying the amount of recapture. If the device **3000** is recaptured up to but not including the anchors proximal **3090** inside the access sheath, the device **3000** can be redeployed and reused. If the proximal anchors **3090** are recaptured into the access sheath, the device **3000** may be removed and discarded due to permanent deformation of the anchors **3090**. In some embodiments, other materials may be used for the markers **3023**, such as gold or other suitable materials.

[0282] The foam body **3002** may be attached to various features of the device **3000**. The body **3002**

may be attached to the frame **3040** at numerous points, including for example the center of the proximal end of the frame **3040**, as further described herein. Attachment can be done using suture, such as polypropylene monofilament suture, although other methods known in the art such as adhesive bonding could be utilized. The proximal row of proximal anchors **3090** may be individually attached to (e.g. inserted through) the foam body **3002** to prevent relative movement between the foam body **3002** and the frame **3040**. In other embodiments, the foam body **3002** could be formed around the endoskeleton so that the metallic frame is within the foam body **3002**, eliminating the need for a secondary attachment step. Attachment of the body **3002** to the frame **3040** promotes retrieval without damage to the foam body **3002**, among other advantages. The attachment also ensures that a bumper **3026**, further described herein, extends beyond the frame **3040** at all times, including during initial exposure of the device **3000** upon proximal retraction of the delivery sheath.

[0283] The foam body **3002** may include a coating. The coating is applied to the interconnected reticulations of the foam material. The body **3002** may be coated with pure polytetrafluoroethylene (PTFE). The PTFE coating minimizes the thrombogenicity of the LA surface, while also reducing the friction of the foam body **3002** against the delivery system to facilitate ease of deployment and retrieval. The body **3002** may be coated with conformable, vacuum deposited, pure PTFE. In addition or alternatively, the body **3002** may be coated with a coating other than PTFE. The coating, whether PTFE or otherwise, may be about 0.5 μm thick, and covers at least a portion of the surface of the interconnected reticulations of the foam without occluding the pores. The coating may be applied to some or all of the foam body **3002**. The coating may be applied to some or all of the outer surfaces of the foam body **3002**.

[0284] In some embodiments, the thickness of the coating is from about 0.1 μm to about 1 μm , from about 0.2 μm to about 0.9 μm , from about 0.3 μm to about 0.8 μm , from about 0.4 μm to about 0.7 μm , about 0.4 μm to about 0.6 μm , or about 0.5 μm thick. In some embodiments, greater or smaller thicknesses of the coating may be applied. The coating has a uniform or substantially uniform thickness. In some embodiments, the coating may have a non-uniform thickness. For example, the portion of the body **3002** facing the LA when implanted, such as the proximal face **3008** and/or shoulder **3030**, may have a thicker coating relative to a coating along the sidewall **3014** of the body **3002**. In some embodiments, the outer surface **3010** of the proximal face **3008** has a PTFE coating and the proximal face **3008** also has a ePTFE cover **3100**.

[0285] The coating is applied using a vapor deposition process. In some embodiments, the coating is applied through coating, vapor deposition, plasma deposition, grafting, other suitable processes, or combinations thereof. The coating is applied to the outer surfaces **3010**, **3032** and **3016** of, respectively, the proximal face **3008**, the shoulder **3030** and the sidewall **3014**. In some embodiments the coating is applied to the outer surfaces **3010**, **3032** and only partially on the outer surface **3016**. In some embodiments the coating is applied to outer and inner surfaces of the body **3002**.

[0286] In some embodiments, other biocompatible, thromboresistant and/or lubricious materials could be applied to the surface(s) of the foam body **3002** and/or the cover **3100**. These materials may encourage tissue ingrowth. Such materials may include, for example, heparin, albumin, collagen, polyethylene oxide (PEO), hydrogels, hyaluronic acid, materials that release nitric oxide, oxygen, nitrogen, amines, bioabsorbable polymers, and other biomaterials, pharmacologic agents, and surface modification materials. Additionally, the surface(s) of the body **3002** could be roughened, textured, or otherwise modified or coated to promote healing or to make it more echogenic.

2. Proximal Cover

[0287] The device **3000** may include a cover **3100**, which may be an ePTFE cover as further described. The cover **3100** is a generally flat material applied over and covering at least a portion of the body **3002**. The cover **3100** is on the proximal end **3004** of the device **3000**. The cover **3100**

covers the proximal face **3008** of the body **3002** and at least part of the sidewall **3014**. The cover **3100** covers a proximal portion of the sidewall **3014**. The cover **3100** has a proximal surface **3102** that at least partially faces the LA when implanted. The cover **3100** has an outer edge **3104** forming outer vertices **3106** (for clarity, only some of the outer edges **3104** and outer vertices **3106** are labelled in the figures). In some embodiments, the cover **3100** may cover only the proximal face **3008** or portions thereof. In some embodiments, the cover **3100** may extend over more of the sidewall **3014**, such as the middle or distal portion thereof, or the entire sidewall **3014**.

[0288] The cover **3100** may have a thickness measured perpendicularly from the proximal surface **3102** to an opposite distal surface of the cover **3100** that faces the body **3002**. The cover **3100** may have a thickness of 0.001". In some embodiments, the cover **3100** may have a thickness from about 0.00025" to about 0.005", from about 0.0003" to about 0.004", from about 0.0004" to about 0.003", from about 0.0006" to about 0.002", from about 0.0008" to about 0.0015", or about 0.001".

[0289] The cover **3100** may be attached to the frame **3040** through the foam body **3002**. The cover **3100** may in addition or alternatively be attached to the body **3002**. The cover **3100** may be attached at at least two or four or six or more of the outer vertices **3106**. The cover **3100** may be attached to the frame **3040** and/or body **3002** at various locations, including at the outer vertices **3106**, through the proximal surface **3100**, at the proximal face **3008** of the body **3002**, other locations, or combinations thereof. The cover **3100** is attached using mechanical attachments, such as sutures. In some embodiments, polypropylene 6-0 sutures are used throughout the device to attach the foam body **3002**, proximal cover **3100**, and RO markers **3023** to the foam body **3002** and/or frame **3040**. In some embodiments, the cover **3100** is attached to the frame **3040** via standard braided or monofilament suture material, such as polypropylene, ePTFE, or polyester. In some embodiments, a polypropylene monofilament is utilized. Proximal anchors **3090** of the frame **3040** (further described herein) may extend through the outer vertices **3106** of the cover **3100**. Such penetrating anchors **3090** may further secure the cover **3100** in place relative to the body **3002**. In some embodiments, the cover **3100** may be attached to the various parts of the device **3000** with mechanical attachments, fasteners, adhesives, chemical bonds, other suitable techniques, or combinations thereof.

[0290] As shown, the cover **3100** is formed from expanded Polytetrafluoroethylene ("ePTFE"). An ePTFE cover **3100** provides many advantages. For example, the ePTFE cover **3100** may enhance the ability to recapture the device **3000** in vivo by distributing the proximal retraction forces applied by the catheter. The cover **3100** may be an ePTFE material approximately 0.001" thick, with the appropriate porosity to encourage healing and minimize thrombus formation, similar to the underlying PTFE coated foam.

[0291] An ePTFE cover **3100** may assist in recapture of the implant into the access sheath while providing a smooth, thromboresistant surface which encourages tissue coverage and integration. The ePTFE may cover the entire proximal face and partially covers the sides, as shown in FIG. **85C**. The ePTFE cover **3100** is fabricated from a previously laminated sheet comprised of two or more sheets of oriented material, offset to form a biaxially orientated material. Alternatively, one could use a tube, preferably biaxially oriented, that is then cut to form a sheet. The thickness of the final construct can be from 0.0005"-0.005" but is preferably about 0.001".

[0292] In some embodiments, the cover **3100** is fabricated from other thromboresistant, high strength, biocompatible materials, such as knitted or woven polyester fabrics, polypropylene, polyethylene, non-woven vascular scaffolds, porous films, or bioabsorbable scaffolds such as polylactic acid, polyglycolic acid, and co-polymers. The shape of the cover prior to attachment with the device **3000**, such as shown in FIG. **88**, minimizes wrinkling and provides a smooth surface following attachment to the implant. This shape may be a star shape, an outer pointed shape, or other shapes.

[0293] The cover **3100** may be perforated with a series of openings **3120** (for clarity, only some of the openings **3120** are labelled in the figures). The openings **3120** are perforations or holes formed

in the cover **3100** via laser or mechanical cutting. The openings **3120** include proximal openings **3122** and side openings **3124** (for clarity, only some of the proximal openings **3122** and side openings **3124** are labelled in the figures). When the cover **3100** is assembled with the body **3002**, the proximal openings **3122** are located over the proximal face **3008** and/or shoulder **3030**, and the side openings **3124** are located over the sidewall **3014**. In some embodiments, the cover **3100** includes forty proximal openings **3122**. In some embodiments, the cover **3100** includes forty side openings **3124**. The number of openings **3120** located over the proximal face **3008** and/or shoulder **3030** when assembled with the body **3002** may range from ten to eighty, from twenty to seventy, from thirty to sixty, from thirty five to fifty, or forty openings **3120**. The number of openings **3120** located over the sidewall **3014** may range from ten to eighty, from twenty to seventy, from thirty to sixty, from thirty five to fifty, or forty openings **3120**.

[0294] The openings **3120** may have a variety of sizes. The openings **3120** are 0.070" in width, e.g. minor axis, or diameter for circular openings. The openings **3120** may have a width from about 0.010" to about 0.200", from about 0.020" to about 0.150", from about 0.030" to about 0.110", from about 0.040" to about 0.100", from about 0.050" to about 0.090", from about 0.060" to about 0.080", or about 0.070". In some embodiments, the width may be less than 0.010" or greater than 0.200", such as 0.25", 0.5" or greater. These widths may apply to circular as well as non-circular shaped openings **3120**.

[0295] In some embodiments, the openings **3120** may be various shapes. The openings **3120** may be elongated slots. The openings **3120** may extend radially along the cover **3100** from or near a center portion of the proximal surface **3102** toward and/or to the outer edge **3104**. The openings **3120** may be annular openings extending circumferentially along the cover **3100** and having varying radial positions. The openings **3120** may be of uniform size and shape. Some of the openings **3120** may have varied sizes and/or shapes with respect to other of the openings **3120**. The openings **3120** may have various distributions or concentrations about the cover **3100**. For example, the openings **3120** may be more densely located in various areas, such as along the proximal surface **3102** that faces the LA, along the shoulder **3030**, etc.

[0296] The openings **3120** enable blood to flow through the device **3000**. The openings **3120** may allow blood to adequately flow through the device **3000** and thereby mitigate the risk of occlusion in the bloodstream should the device **3000** embolize within the vasculature system. In some embodiments, should the device **3000** embolize, it may act as a stationary filter at low pressures but may pass through the bloodstream at higher pressures. In some embodiments, the device **3000** allows for about two to about fourteen liters, from about four to about twelve liters, from about six to about ten liters, or from about eight liters per minute of blood to pass at <30 mmHg pressure drop to prevent shock in the event of a device embolization. In some embodiments, there are forty circular openings **3120** each having a diameter of 0.070", and allowing for approximately eight liters per minute of blood to pass at <30 mmHg pressure drop. In some embodiments, the proximal end of the device **3000** may be a foam layer such as the foam proximal face **3008** or a membrane such as the cover **3100** or both, enclosing the cavity **3028** defined within the tubular side wall **3014** of the body **3002**. In one implementation, having both the foam proximal face **3008** and the cover **3100**, the foam body **3002** has the open cell structure further discussed herein that can permit the passage of blood but block escape of embolic debris. The cover **3100** may be occlusive to blood flow, and is present to provide structural integrity and reduced friction for retracting the expanded body **3002** back into the deployment catheter. In one implementation, the cover **3100** is ePTFE in a form that is substantially occlusive to blood flow, as described. In this embodiment, the cover **3100** is therefore provided with a plurality of perfusion windows or openings **3120**, so that blood can pass through the open cell foam and cover **3100** but the device **3000** still benefits from the other properties of the cover **3100**.

[0297] FIGS. **87A-87C** depict an embodiment of the LAA occlusion device **3000** having another embodiment of a cover **3300**. The device **3000** includes the foam body **3002** and the frame **3040**,

and features thereof, as described herein, and additionally includes the cover **3300**. The cover **3300** may have the same or similar features and/or functionalities as the cover **3100**, and vice versa. The cover **3300** is on the proximal end **3004** of the device **3000**. The cover **3300** covers the proximal face **3008** of the body **3002** and a proximal part of the sidewall **3014**. The cover **3300** has a proximal surface **3302**. The cover **3300** has an outer edge **3304** forming a plurality of at least two or four or six or eight or ten or more outer vertices **3306** (for clarity, only some of the outer vertices **3306** are labelled in the figures). The cover **3300** is attached to the body **3002** at the outer vertices **3306**. The proximal anchors **3090** extend through side openings **3324** in the outer vertices **3106** of the cover **3100**.

[0298] The cover **3300** includes a series of openings **3320**. The openings **3320** include proximal openings **3322**, shoulder openings **3323**, and the side openings **3324**. The proximal openings **3322** are located over the proximal end **3004** of the body **3002**. The shoulder openings **3323** are located over the shoulder **3030**, e.g. a bevel, of the body **3002**. The side openings **3324** are located over a proximal portion of the sidewall **3014** of the body **3002**. The proximal anchors **3090** may extend through the side openings **3324** that are located in the outer vertices **3106**. The openings **3320** may have the same or similar features and/or functionalities as the openings **3120**, and vice versa. In some embodiments, the proximal anchors **3090** may extend through the cover **3300** material at or near the outer vertices **3106**.

[0299] FIG. **88** shows another embodiment of a cover **3150** that may be used with the device **3000**. The cover **3150** may have the same or similar features and/or functionalities as the cover **3100** and/or cover **3300**, and vice versa. The cover **3150** may be used to cover the proximal face **3008** of the body **3002** and part of the sidewall **3014**. The cover **3150** has a proximal surface **3152**. The cover **3150** has an outer edge **3154** forming outer vertices **3156**. The cover **3150** may be attached to the body **3002** at the outer vertices **3156**. The proximal anchors **3090** may extend through the outer vertices **3156** of the cover **3100**. The cover **3150** includes a series of openings **3170**. The openings **3170** include proximal openings **3172** and side openings **3174** (for clarity, only some of the openings **3170**, **3172**, **3174** are labelled in the figures). When the cover **3150** is assembled with the body **3002**, the proximal openings **3172** are located over the proximal end **3004** and the side openings **3174** are located over the sidewall **3014**. As shown, the openings **3174** may be substantially uniformly located along the cover **3150** except for a center region of the proximal surface **3152**.

3. Compliant Frame

[0300] The expandable and compliant support or frame **3040** is shown in FIGS. **85B**, **86C** and **87C**. FIGS. **89A** and **89B** are side and proximal perspective views, respectively, of the frame **3040** shown in a deployed configuration and in isolation from the rest of the device **3000**. The frame **3040** provides a compliant structure with anchors to facilitate delivery, anchoring, retrieval and to enable the foam body **3002** to compress against the LAA tissue to facilitate sealing, among other things, as further described. The frame **3040** is located inside the cavity **3028** formed by the foam body **3002**. In some embodiments, the frame **3040** may be located partially or entirely inside one or more portions of the body **3002**, e.g. within the proximal face **3008** and/or the sidewall **3014**, as further described. For example, the frame **3040** may be partially located within the sidewall **3014** as shown in FIG. **87C**.

[0301] The frame **3040** has a proximal end **3042** and an opposite distal end **3004**. The frame **3040** may be tubular, e.g. cylindrical, in a free, unconstrained state. Thus the width of the proximal end **3042** may be the same or similar to the width of the distal end **3004** in the free, unconstrained state. In some embodiments, the frame **3040** or portions thereof may be conical or frustoconical, e.g. where in the free, unconstrained state the width of the proximal end **3042** is greater than the width of the distal end **3004** or vice versa.

[0302] At the proximal end **3042**, the frame **3040** has a proximal hub **3050**, shown as a cylindrical nipple. The hub **3050** is a rounded, structural end piece. The hub **3050** may be tubular, e.g. circular

and having the cylindrical shape as shown, or may be rounded, non-circular, segmented, other shapes, or combinations thereof. The hub **3050** extends axially and may have a central lumen. The hub **3050** may be wider than it is long, or vice versa. The hub **3050** is hollow and has a sidewall defining a space therethrough, such as a longitudinal opening. In some embodiments, the hub **3050** may be partially hollow, solid, or other configurations. The hub **3050** facilitates delivery and retrieval of the device **3000**, as further described. The hub **3050** may provide a central structural attachment, as further described herein. The hub **3050** may be located within the cavity **3028** at a proximal end thereof. In some embodiments, the hub **3050** may be located partially or entirely within the foam body **3002**, e.g. within the proximal face **3008**.

[0303] A pin **3051** is located within the hub **3050** (shown in FIGS. **89A** and **89B**). The pin **3051** is an elongated, rounded structural element extending laterally across the central lumen. “Lateral” here refers to a direction perpendicular or generally perpendicular to the longitudinal axis. The pin **3051** has a cylindrical shape. The pin **3051** provides a rounded outer surface configured to provide a smooth engagement surface with a tether, as further described. The pin **3051** provides a high strength connection with the frame **3040** to allow for pulling on the device **300** with sufficient force to re-sheath the device **3000**. The pin **3051** may be formed from Nitinol. The pin **3051** is secured across the width, e.g. diameter, of the proximal hub **3050**. The pin **3050** may be secured at its two opposite ends with the sidewall of the hub. The pin **3051** is configured to be engaged by a tether **3240**, which is wrapped around the pin **3051** in sliding engagement for temporary attachment to a delivery catheter, as further described. In some embodiments, the pin **3051** is assembled with a cap **3180**, as further described herein, for example with respect to FIGS. **90A-90C**.

[0304] The frame **3040** at the proximal end **3042** includes a proximal face **3060**. The proximal face **3060** may be located within the cavity **3028** at a proximal end thereof. In some embodiments, the proximal face **3060** may be located partially or entirely within the foam body **3002**, e.g. within the proximal face **3008** and/or sidewall **3014**. The proximal face **3060** includes a series of recapture or reentry struts **3061**. The struts **3061** are located at a proximal end of the cavity **3028**. In some embodiments, the struts **3061** or portions thereof may be located partially or entirely within the foam body **3002**, e.g. within the proximal face **3008** and/or sidewall **3014**.

[0305] The struts **3061** are elongated structural members. The struts **3061** may have rectangular, circular or other shaped cross-sections. In some embodiments, the struts **3061** have a cross-section, e.g. rectangular, with a width that is greater than a thickness such that the struts **3061** are stiffer in one direction compared to another direction. This width may be in the lateral direction or a direction generally perpendicular to the longitudinal axis of the device **3000** when the device **3000** is in the expanded configuration, with the thickness perpendicular to the width. The struts **3061** may be less stiff in the direction of flexing or bending, for example to facilitate contraction and expansion of the device **3000** in the delivery and expanded configurations. The struts **3061** may be elongated pins. The struts **3061** may extend from the hub **3050**, for example, and incline radially outwardly in the distal direction from the hub **3050**. The struts **3061** may be attached inside, outside, and/or at the end of the sidewall of the hub **3050**. The struts **3061** may be separate parts that are then attached to the hub **3050**, for example welding, bonding, fastening, other suitable means, or combinations thereof. In some embodiments, some or all of the struts **3061** and the hub **3050** may be a single, continuous structure formed from the same raw material such as a laser cut hypotube. Some or all of the struts **3061** may be attached, e.g. with sutures as described herein, to the body **3002** and/or the cover **3100** at one or more attachment locations.

[0306] Each recapture strut **3061** may include an inner curved portion **3062** connected to a distal end of the hub **3050**, a middle straight portion **3064**, and/or an outer curved portion **3066** (for clarity, only some of the portions **3062**, **3064**, **3066** are labelled in the figures). In the deployed configuration, the inner curved portion **3062** extends from the hub **3050** primarily in a distal direction and then curves to face more outwardly radially. The middle straight portion **3064** extends from the inner curved portion **3062** primarily radially but also slightly distally. The outer curved

portion **3066** extends from the middle straight portion **3064** primarily in the radial direction and then curves toward the distal direction. The portions may have different shapes in the delivery configuration inside a delivery catheter. In the delivery configuration, the portions may extend primarily distally. The portions may then take the deployed configuration as described upon deployment from the delivery catheter. In some embodiments, the struts **3061** may include fewer or more than the portions **3062**, **3064**, **3066**.

[0307] The device **3000** may include ten of the proximal recapture struts **3061**. Such configuration may accompany a device **300** having a foam body **3002** with an outer diameter of 27 mm in the free, unconstrained state. Such configuration may accompany a device **300** having a foam body **3002** with an outer diameter of 35 mm in the free, unconstrained state. In some embodiments, the device **3000** may have from about two to about thirty, from about four to about twenty, from about six to about eighteen, from about eight to about sixteen, from about ten to about fourteen, or other numbers of struts **3061**.

[0308] In the deployed configuration, each strut **3061** extends radially outward and distally at an angle to the axis. This angle, measured relative to a portion of the axis that extends distally from the device **3000**, may be from about 60° to about 89.9°, from about 65° to about 88.5°, from about 70° to about 85°, from about 72.5° to about 82.5°, from about 75° to about 80°, or other angular amounts. This angle may be much smaller when the device **3000** is in the delivery catheter. The struts **3061** may bend or flex when transitioning between, or when positioned in, the delivery and expanded configurations. The struts **3061** may bend or flex at the inner curved portion **3062**, the middle straight portion **3064**, and/or the outer curved portion **3066**.

[0309] The proximal end **3042** of the frame **3040**, such as the proximal face **3060**, may therefore have a conical shape in the expanded configuration. The conical proximal face **3060** may facilitate with recapture of the device **3000** back into the delivery catheter. For example, the orientation of the struts **3061** inclining distally and radially outward from the hub **3050** in the expanded configuration provides an advantageous conical shape to the proximal face **3008** such that distal advance of the delivery sheath over the device **3000** will bias the struts **3061** inward and cause the device **300** to stow back toward the delivery configuration and size for retrieval within the catheter.

[0310] The proximal face **3060** foreshortens considerably upon expansion of the device **3000** relative to the delivery configuration. “Foreshortening” here refers to the difference in axial length of the proximal face **3060** between the reduced delivery configuration and the expanded configuration (expanded either freely or as implanted). This length may be measured axially from the distal or proximal end of the hub **3050** to the distal ends of the outer curved portions **3066** of the recapture struts **3061**. The proximal face **3060** may foreshorten by 50%, 60%, 70%, 80%, 90% or more. The proximal face **3060** has significantly more foreshortening upon expansion than the tubular body **3080**, the latter of which may be referred to as the “working length” or “landing zone.” The landing zone is further described with respect to the tubular body **3080** herein.

[0311] As shown, the struts **3061** are angularly spaced about the axis in even angular increments. That is, looking at the frame **3040** from the distal or proximal end, the angles between the struts may be equal. In some embodiment, the struts **3061** may not be evenly angularly spaced about the axis as described. The struts **3061** may or may not be symmetrically disposed about the axis or about a plane that includes the axis.

[0312] The frame **3040** includes a tubular body **3080**. The body **3080** provides a mechanical base structure for the device **3000**, as further described. The tubular body **3080** is attached to a distal end of the proximal face **3060** of the frame **3040**. The tubular body **3080** extends to the distal end **3044** of the frame **3040**. The tubular body **3080** is attached at a proximal end to the outer curved portions **3066** of the recapture struts **3061**, as further described. The tubular body **3080** may be attached to other portions of the recapture struts **3061**. The tubular body **3080** of the frame **3040** may be attached to the body **3002** and/or the cover **3100**, e.g. with sutures as described herein, at one or more attachment locations, as further described. The tubular body **3080** may be located within the

cavity **3028**. In some embodiments, the tubular body **3080** may be located partially or entirely within the foam body **3002**, e.g. within the sidewall **3014**.

[0313] The tubular body **3080** includes a series of proximal struts **3082** and distal struts **3086** (for clarity, only some of the struts **3082**, **3086** are labelled in the figures). The proximal struts **3082** and/or distal struts **3086** may have rectangular, circular or other shaped cross-sections. In some embodiments, the proximal struts **3082** and/or distal struts **3086** have a cross-section, e.g. rectangular, with a width that is greater than a thickness, or vice versa, such that the struts **3061** are stiffer in one direction compared to another direction. The struts **3061** may be less stiff in the direction of flexing or bending, for example to facilitate contraction and expansion of the device **3000** in the delivery and expanded configurations. Proximal ends of pairs of adjacent proximal struts **3082** join at proximal apexes **3084**. Each proximal strut **3082** is connected at a respective proximal apex **3084** to a respective outer curved portion **3066** of one of the recapture struts **3061**. Each distal end of the proximal struts **3082** connects to a distal end of an adjacent proximal strut **3082** and to proximal ends of two distal struts **3086** at an intermediate vertex **3087**. Pairs of adjacent distal struts **3086** extend distally to join at a respective distal apex **3088**. A repeating pattern **3089**, shown as a diamond shape, may be formed by adjacent pairs of proximal struts **3082** and adjacent pairs of distal struts **3086**. Some or all of the proximal struts **3082** and/or distal struts **3086** may be attached, e.g. with sutures as described herein, to the body **3002** and/or the cover **3100** at one or more attachment locations. Some or all of the proximal struts **3082** and/or distal struts **3086** may be located within the cavity **3028**. In some embodiments, some or all of the proximal struts **3082** and/or distal struts **3086** may be located partially or entirely within the foam body **3002**, e.g. within the sidewall **3014**.

[0314] There are the same number of proximal apexes **3084** as distal apexes **3088**. As shown, there are eleven proximal apexes **3084** and eleven distal apexes **3088**. The number of proximal and distal apexes **3084**, **3088** may each be at least seven, at least eight, at least nine, at least ten, at least eleven, at least twelve, at least thirteen, at least fourteen, at least fifteen, at least sixteen, or fewer or more apexes. In some embodiments, there may not be the same number of proximal apexes **3084** as distal apexes **3088**. In some embodiments, there may be more than one row of the pattern, e.g. diamond pattern, formed by the proximal struts **3082** and distal struts **3086**. There may be two, three, four or more rows of the pattern. Some or all of the proximal apexes **3084** and/or distal apexes **3088** may be attached, e.g. with sutures as described herein, to the body **3002** and/or the cover **3100** at one or more attachment locations.

[0315] The body **3080** may be tubular, e.g. cylindrical or generally cylindrical, in the expanded configuration. The tubular body **3080** may be cylindrical, rounded, segmented, polygonal, tube-like, other shapes, or combinations thereof, all of which are subsumed non-exhaustively under the category “tubular.” The tubular shape is formed by the proximal struts **3082** and distal struts **3086** in the expanded configuration. The tubular shape may also be formed by the outer curve portions **3066** of the recapture struts **3061** in the expanded configuration. The tubular shape may also be formed by the foam body **3002** exerting an outward radial force on the frame **3040**. The frame **3040** may therefore have a proximal conical section and a cylindrical working length. In some embodiments, the body **3080** may be conical or frustoconical, for example where the distal end is wider than the proximal end or vice versa.

[0316] The tubular body **3080** may be referred to as a “landing zone,” as described. This landing zone may refer to the axial length of the body **3080**, from a distal-most end to a proximal-most end at the transition to recapture struts **3061**, in the expanded configuration. The landing zone may have an axial length as measured from the proximal apex **3084** to the distal apex **3088**. The length of the landing zone may be 10 mm or about 10 mm. The landing zone may have a length from about 5 mm to about 15 mm, from about 6 mm to about 14 mm, from about 7 mm to about 13 mm, from about 8 mm to about 12 mm, from about 9 mm to about 11 mm, or other lengths. The tubular body **3080** may foreshorten slightly upon expansion of the device **3000** relative to the delivery

configuration. The tubular body **3080** has significantly less foreshortening upon expansion than the length of the proximal face **3060**. The tubular body **3080** may foreshorten by no more than about 5%, 10%, 15%, 20% or 30%.

[0317] The frame **3040** self-expands upon delivery from the sheath. The proximal face **3060** and the tubular body **3080** will self-expand. Upon expansion, the radially outward portions of the tubular body **3080** will contact and compress the foam body **3002** against tissue of the LAA wall. The tubular body **3080**, for example the proximal struts **3082** and distal struts **3086**, will contact the inner surface **3018** of the sidewall **3014** and press against the sidewall **3014** so that the outer surface **3016** of the sidewall **3014** contacts and compresses against the LAA wall.

[0318] When compressed against the LAA wall, the foam body **3002** provides a larger “footprint” than the skeletal frame **3040** components and forms a complete seal. Thus, the sidewall **3014** acts as a force dissipation layer, spreading radial force out from the struts **3082**, **3086** of the frame **3040** over a larger area than just the area of the individual struts **3082**, **3086** (e.g. a larger area than just the area of the radially outer surfaces of the struts **3082**, **3086**). The use of the foam material in the body **3002** and the thickness of that foam, such as 2.5 mm, provide advantages in this regard over devices with thinner and less resilient materials than foam. For example, thin fabrics or similar materials that are pressed against the LAA wall with a skeletal frame will not spread the radial force out, and may even sag or otherwise bend, creating gaps and an unsealed portion of the LAA wall. The foam body **3002** as described herein will take the shape of the LAA wall to create a complete circumferential seal and will also spread out the radial forces from the frame **3040** to create a stronger seal and retention with the foam body **3002**.

[0319] Further, the device **3000** described herein with the compressible body **3002** allows for a structural frame **3040** that is compliant due to the smaller required radial force from the frame **3040**. For example, existing devices with a non-compressible fabric material will have a less effective seal, and so the structural elements of those devices must provide larger radial forces to compensate and ensure an effective seal, resulting in a less compliant device. In contrast, the current device **3000** provides advantages in this regard by having the compressible foam body **3002**, allowing for among other things smaller radial forces from, and thus better compliance of, the frame **3040**, while still providing an effective seal. This structural configuration has a cascading effect in terms of performance advantages. For instance, the compliance of the device **300** allows for delivery off-axis while still providing an effective seal, among other advantages as further described herein.

[0320] The frame **3040** includes a series of proximal anchors **3090**. Each proximal anchor **3090** extends from a respective intermediate vertex **3087**. The proximal anchors **3090** may extend from other portions of the tubular body **3080**. As shown, in the deployed configuration, the proximal anchors **3090** extend from the tubular body **3080** radially and proximally. The proximal anchors **3090** may extend into an adjacent region of the sidewall **3014**. The proximal anchors **3090** may extend through the outer surface **3016** of the sidewall **3014** to penetrate tissue adjacent the device **3000**.

[0321] The frame **3040** includes a series of distal anchors **3094**. Each distal anchor **3094** extends from a respective distal apex **3088**. The distal anchors **3094** may extend from other portions of the tubular body **3080**. As shown, in the deployed configuration, the distal anchors **3094** extend from the tubular body **3080** radially and proximally. The distal anchors **3094** may extend into an adjacent region of the sidewall **3014**. The distal anchors **3094** may extend through the outer surface **3016** of the sidewall **3014** to penetrate tissue adjacent the device **3000**. The anchors **3090**, **3094** may incline radially outward in a proximal direction to engage the tissue to resist proximal movement of the device **3000**.

[0322] The anchors **3090**, **3094** are elongated structural members. The tips of the anchors **3090**, **3094** may be sharpened to facilitate tissue engagement and penetration. The anchors **3090**, **3094** may be straight, extending generally along a local axis thereof. The anchors **3090**, **3094** may have a

curved or other non-straight proximal portion where they attach to the tubular body **3080**. In some embodiments, the anchors **3090**, **3094** or portions thereof may be non-straight, curved, rounded, segmented, other trajectories, or combinations thereof. In some embodiments, the tissue engaging tips may be curved. In some embodiments, the anchors **3090**, **3094** may have engagement features extending radially away from the anchor **3090**, **3094**, such as barbs, hooks, or other features.

[0323] The cross-section of the anchors **3090**, **3094** may be rectangular. In some embodiments, the cross-section may be circular, rounded, non-rounded, square, rectangular, polygonal, other shapes, or combinations thereof. The cross-sections may or may not be uniform along the length of the anchor **3090**, **3094**. The anchors **3090**, **3094** may be about 0.006" thick and about 0.008" wide. The anchors **3090**, **3094** may range from about 0.003" to about 0.009" in thickness and from about 0.003" to about 0.015" in width. The cross-section of the anchors **3090**, **3094** may reduce in size, for example taper, toward the distal tip.

[0324] In some embodiments, the anchors **3090**, **3094** in the deployed configuration are inclined at an incline angle of about 30° relative to a portion of the central axis that extends proximally from the device **3000**. This incline angle may be from about 10 degrees to about 50°, from about 15° to about 45°, from about 20° to about 40°, from about 25° to about 35°, or about 30°. This incline angle of the anchors **3090**, **3094** in the delivery configuration may be smaller than in the deployed configuration.

[0325] The anchors **3090**, **3094** may have various lengths. The length of the anchor **3090**, **3094** is measured from a proximal end that connects to the tubular body **3080** to a distal tissue engaging tip of the anchor. In some embodiments, the length of the anchors **3090**, **3094** may be from about 0.5 mm to about 10 mm, from about 1 mm to about 9 mm, from about 2 mm to about 8 mm, from about 3 mm to about 7 mm, from about 4 mm to about 6 mm, about 5 mm, or other greater or lesser lengths. In some embodiments, the anchors **3090**, **3094** are 5 mm long. In some embodiments, the anchors **3090**, **3094** are about 5 mm long. In some embodiments, the anchors **3090**, **3094** have a length of at least 2.5 mm, at least 3 mm, at least 3.5 mm, at least 4 mm, at least 4.5 mm, at least 5 mm or more. The anchors **3090**, **3094** may each be the same or similar length. In some embodiments, the anchors **3090**, **3094** may not be the same length. In some embodiments, some or all of the proximal anchors **3090** may have lengths that are less than or greater than some or all of the lengths of the distal anchors **3094**.

[0326] In the expanded configuration, the anchors **3090**, **3094** extend for a length outside of the uncompressed sidewall **3014**. This length of the anchor **3090**, **3094** is measured along a local longitudinal axis of the anchor from the outer surface **3016** of the body **3002** to the distal tip of the anchor. The anchors **3090**, **3094** may extend through the sidewall **3014** and/or the cover **3100**, and then be trimmed so that the anchors **3090**, **3094** extend beyond the sidewall **3014** and/or cover **3100** by the desired length. In a free, unconstrained state, the anchors **3090**, **3094** extend about 0.5 mm beyond the outer surface **3016** of the sidewall **3014**. In some embodiments, in the free, unconstrained state, the anchors **3090**, **3094** extend beyond the outer surface **3016** of the sidewall **3014** for a length of from about 0.1 mm to about 1.5 mm, from about 0.2 mm to about 1.25 mm, from about 0.3 mm to about 1.0 mm, from about 0.4 mm to about 0.8 mm, from about 5 mm to about 0.6 mm, or other greater or lesser lengths. In a compressed state, such as in the delivery configuration or after implantation, the anchors **3090**, **3094** extend about 1.0 mm beyond the outer surface **3016** of the sidewall **3014**. In some embodiments, in the compressed state, the anchors **3090**, **3094** extend beyond the outer surface **3016** of the sidewall **3014** for a length of from about 0.25 mm to about 2.5 mm, from about 0.5 mm to about 2 mm, from about 0.75 mm to about 1.5 mm, from about 0.875 mm to 1.125 mm, or other greater or lesser lengths.

[0327] The geometry of the anchors **3090**, **3094** provides several advantages. For example, the relatively long length allows for flexibility of the anchors **3090**, **3094**. This provides for potentially less trauma to the LAA tissue should the device **3000** need to be unanchored and/or retrieved. The anchors **3090**, **3094** are less susceptible to loss of strength with off-axis orientation within the

LAA. Further, the anchors **3090**, **3094** provide high resistance to pull out. For instance, the device **3000** may provide at least about 0.5 lb-force of dislodgment resistance from the LAA. Such pullout tests may be simulated with in vitro or benchtop models, as further described below.

[0328] The anchors **3090**, **3094** in the illustrated embodiment are located in two circumferential rows. One row is located proximal to the other distal row. Each row has ten anchors each. This configuration may be incorporated, for example, in the device **3000** having a foam body **3002** with a free, unconstrained outer diameter of 27 mm. Each row may have fourteen anchors each. This configuration may be incorporated, for example, in the device **3000** having a foam body **3002** with a free, unconstrained outer diameter of 35 mm. In some embodiments, a single row of anchors **3090**, **3094** may have from two to twenty-four, from four to twenty-two, from five to twenty, from six to eighteen, from seven to sixteen, from eight to fifteen, from nine to fourteen, from ten to thirteen anchors, or greater or fewer amounts of anchors **3090** or **3094**. In some embodiments, there may only be one row or greater than two rows of anchors. The anchors **3090**, **3094** may be spaced circumferentially in a single row.

[0329] In embodiments with multiple rows of anchors **3090**, **3094**, the rows may be circumferentially offset, as shown. That is, as viewed from the proximal or distal end of the device **3000**, the anchors **3090**, **3094** are angularly spaced apart from each other about the axis. The anchors **3090**, **3094** may not be circumferentially offset, e.g. they may be evenly angularly spaced when viewed as described. The anchors **3090**, **3094** are located axially at or near a middle portion of the sidewall **3014**. The anchors **3090**, **3094** may be located such that the tips of the anchors **3090**, **3094** extend to adjacent tissue at a middle portion of the sidewall **3014**. The offset and middle locations of the anchors **3090**, **3094** may ensure engagement with the LAA tissue distal to the ostium. Having the anchors **3090**, **3094** located at the largest width, increases the stability of the device **3000**. With a generally cylindrical shaped device **3000**, the anchors **3090**, **3094** effectively sit on the largest diameter of the device **3000**. In some embodiments, the anchors **3090**, **3094** may be located proximal, distal, or centrally along the length of the frame body **3080**. In some embodiments, the anchors **3090**, **3094** may not be offset and/or may not be angularly evenly spaced.

[0330] The anchors **3090**, **3094** may provide advantageous flexibility, as demonstrated by pullout tests and in comparison to existing devices. For example, the device **3000** was tested to determine the force required to dislodge the device **3000** from a simulated tissue model by pulling the device **3000** proximally outward from the model. A low durometer silicone tube with a circular inner diameter (ID) was used as the model. For the device **3000** having a foam body **3002** with a 27 mm outer diameter in a free unconstrained state, tubes with ID's of 16.5 mm, 21 mm and 25 mm were tested. The pullout forces for existing devices drop off significantly going up to a 21 mm model, whereas the forces for the device **3000** drop only slightly.

[0331] In the largest diameter (25 mm) model, where there is not a lot of interference in the fit, the forces for the existing devices approach zero as the device does not engage the model wall because the anchors are sitting at a smaller diameter on a trailing edge of the device. The device **3000** consistently resists dislodgment with about 0.7 lbs of force. Since there is very little friction resisting pullout, that force is almost entirely resisted by the anchors **3090**, **3094**. When examining failure modes, all devices eventually begin to slide out of the model. Upon failure, the anchors **3090**, **3094** fold backward or sideways before slipping starts. Assuming 0.7 lbs force is required to cause all twenty anchors **3090**, **3094** to fold backward, then the force per anchors is estimated to be about 0.035 lbs.

[0332] The frame **3040** may be laser cut. The tubular body **3080** may be laser cut from a single tube. The body **3080** may be cut from a tube having a thickness from about 0.002" to about 0.014", or about 0.008". The tube may have an outer diameter (OD) from about 0.05" to about 0.30". The tube may have an outer diameter (OD) of 0.124" for the 27 mm device **3000** (i.e. the embodiment of the device **3000** having a foam body **3002** with an OD of 27 mm in the unconstrained, free

state). The tube may have an OD of 0.163" for the 35 mm device **3000** (i.e. the embodiment of the device **3000** having a foam body **3002** with an OD of 35 mm in the unconstrained, free state). [0333] In some embodiments, the body **3080** is laser cut from a superelastic nitinol tube, however, numerous other biocompatible metallic materials can be utilized such as shape memory Nitinol, stainless steel, MP35N, or Elgiloy. The frame **3040** is self-expandable. In some embodiments, a balloon-expandable frame **3040** could be utilized. Additionally, the body **3080** could be fabricated from drawn wire as opposed to being laser cut from a tube.

[0334] As shown, an embodiment of the device **3000** includes the frame **3040** having ten proximal recapture struts **3061** and twenty total anchors **3090**, **3094**, with the foam body **3002** having an outer diameter of 27 mm. In some embodiments, the device **3000** may include the frame **3040** having fourteen proximal recapture struts **3061** and twenty-eight total anchors **3090**, **3094**, with the foam body **3002** having an outer diameter of 35 mm.

[0335] In one embodiment, the frame **3040** includes a proximal hub **3050**, tether pin **3051**, front face with ten or fourteen recapture struts **3061**, a diamond pattern cylindrical body **3080**, and twenty or twenty-eight anchors **3090**, **3094**. The frame proximal face **3060** supports recapture, the frame body **3080** supports the foam cylinder body **3002**, and the anchors **3090**, **3094** located on the cylinder provide resistance to embolization.

[0336] The design of the device **3000** provides numerous advantages, some of which have been described. As further example, the frame **3040** provides many advantages, including but not limited to: 1) implant radial stiffness/compliance—the frame **3040** provides enhanced radial stiffness while still being sufficiently compliant to allow for off-axis implantation, recapture, etc.; 2) dislodgement resistance—the frame **3040** provides for high pullout strength, as described; 3) transcatheter delivery—the frame **3040** can be compressed into a delivery catheter and then fully expand when delivered; 4) recapture—the frame **3040** allows for recapture/retrieval into the delivery catheter after deployment or even after implantation in the LAA; and 5) mechanical integrity—the frame **3040** has acute and long term structural integrity, for example the ability to withstand loading into the delivery catheter, deployment from the catheter, and cyclic loading/fatigue. The frame **3040** also provides a conformable structure to enable the foam body **3002** to compress against the LAA tissue to facilitate sealing and anchoring with minimal compression (oversizing). The resulting compliance of the frame **3040** provides better anchoring than existing solutions, as described.

[0337] As further example, the device **3000** seals against irregularly shaped LAA ostia and necks. For instance, a combination of a Nitinol frame **3040** with a foam body **3002** having a coating of PTFE and cover **3100** of ePTFE contribute to ability of the device **3000** to conform to the anatomy and seal against irregular projections and shapes, while providing a smooth thromboresistant LA surface.

[0338] As further example, the device **3000** provides for controlled & safe delivery. The design of the combined frame **3040** and foam body **3002** facilitates delivery in a controlled fashion by slowing the speed of expansion. The bumper **3026** acts as an atraumatic leading edge portion when delivering the implant into the LAA mitigating the risk of injury. The user has the ability to recapture and redeploy the device **3000**, if necessary. A flexible tether **3240** attachment, as further described, from the delivery catheter to the device **3000** permits the device **3000** to sit tension free immediately following implantation so the user can ensure final appropriate positioning prior to release of the device **3000**.

[0339] As further example, the device **3000** provides for simplified placement. The foam-covered cylindrical design makes alignment of the device **3000** with the central axis of the LAA during delivery non-critical (by allowing deployment up to, for example, 45 degrees off-axis), which is designed to simplify the implantation procedure, as further described.

[0340] As further example, the device **3000** provides for simple sizing. The foam and frame design contributes to the ability to need only two diameters (e.g., 27 mm and 35 mm) to seal the range of expected LAA configurations and diameters (e.g. targeting LAA diameters of 16 to 33 mm). The

conformability of the foam and frame allow the 20 mm long implant to fit into LAA's as short as 10 mm deep. The short landing zone requirement (LAA depth) of the device **3000**, combined with the need for only two implant diameters, enables treatment of a wide range of LAA anatomies with minimal need for burdensome echo and CT sizing. The conforming nature of the implant is key to facilitating a simple to use product platform that is adaptable to a variety of anatomic structures. [0341] As further example, the device **3000** provides thromboresistant materials and design. The removable tether leaves a smooth, metal-free surface in the LA. Thromboresistant materials (PTFE-coated foam and an ePTFE cover) create a smooth LA face (no metal attachment connection) to reduce anticoagulation needs, enhance thromboresistance, and encourage endothelialization. [0342] As further example, the device **3000** provides thin, low profile anchors **3090**, **3094** around the midpoint of the device **3000** to provide secure yet atraumatic anchoring.

4. Distal Bumper

[0343] The foam body **3002** has a distal bumper **3026**. The bumper **3026** may be a foam distal region of the body **3002**, such as a distal portion of the sidewall **3014**. The bumper **3026** may be a portion of the foam body **3002** that extends beyond the distal end **3044** of the frame **3040**. The bumper **3026** may extend beyond the distal end **3044** of the frame **3040** in the delivery configuration and in the deployed configuration. The body **3002** may be attached to the frame **3040** in various locations such that the body **3002** may stretch in some embodiments, for example in the delivery configuration, to ensure the bumper **3026** extends beyond the frame **3040** upon initially retracting the sheath during delivery.

[0344] The device **3000** can conform both in length and diameter due to conformability of both the foam body **3002** and the frame **3040**. This allows for the device **3000** to accommodate most patient LAA anatomies with only a couple or few different sizes of the device **3000**, such as 27 mm and 35 mm outer diameter body **3002** as described herein, and one length, such as 20 mm. The frame **3040** may thus be shorter than the foam body **3002**, resulting in some embodiments in about 5 mm of foam bumper **3026** distal to the distal-most end of the frame **3040**. The distal bumper **3026** acts as an atraumatic tip during delivery of the device **3000** and can be compressed following implantation to allow the device **3000** to conform to appendages with a depth (landing zone) as short as 10 mm. This ability to conform both in length and diameter is due to the conformability of both the foam body **3002** and the frame **3040**.

[0345] The length of the bumper **3026** may be measured axially from the distal-most end of the frame **3040** to the distal surface **3022** of the body **3002**. For example, the bumper **3026** may extend from the distal apexes **3088** to the distal surface **3022**. The bumper **3026** may have a length of 5 mm or about 5 mm. The bumper **3026** may have a length of about 1 mm, 2 mm, 3 mm, 4 mm, 5 mm, 6 mm, 7 mm, 8 mm, 9 mm or more. The bumper **3026** may have a length from about 2.5 mm to about 7.5 mm, from about 3 mm to about 7 mm, from about 3.5 mm to about 6.5 mm, from about 4 mm to about 6 mm, from about 4.5 mm to about 5.5 mm.

5. Cap & Pin

[0346] FIGS. **90A-90C** are proximal perspective views of the frame **3040** having a cap **3180**. FIG. **90D** is a distal perspective view of the cap **3180**. In some embodiments, the pin **3051** is placed across the proximal hub **3050** diameter and serves to engage the delivery catheter tether **3240** (e.g. a suture), which is wrapped around the pin **3051** for temporary attachment to the delivery catheter **3220**, as described further herein for example with respect to FIGS. **89A-89B**. As shown, the hub **3050** has a pair of opposite side openings **3053** extending through a sidewall of the hub **3050**. The cap **3180** has a corresponding pair of opposite side openings **3190** extending through a sidewall **3184** of the cap **3180**. When the cap **3180** is assembled with the hub **3050**, the pin **3051** may be inserted through the aligned pairs of openings **3053**, **3182**. The assembly can be further secured by welding the ends of the pin **3051** to the hub **3050**.

[0347] As shown in FIG. **90D**, the cap **3180** includes a proximal end **3182** and a distal end **3184**. The cap **3180** includes a rounded sidewall **3186** extending from the proximal end **3182** to the distal

end **3184**. The sidewall **3186** defines a longitudinal opening **3188** through the cap **3180**. The sidewall **3186** includes a pair of lateral openings **3190** located opposite each other. The cap **3180** includes a flange **3192** at the proximal end **3182** extending radially outward.

[0348] The cap **3180** is formed from titanium and the pin **3051** is formed from Nitinol or superelastic Nitinol. In some embodiments, the cap **3180** and/or pin **3051** may be formed from other materials, for example numerous biocompatible metallic or polymeric materials such as shape memory Nitinol, stainless steel, MP35N, Elgiloy, polycarbonate, polysulfone, polyether ether ketone (PEEK), or polymethyl methacrylate (PMMA) or other materials.

[0349] The cap **3180** and pin **3051** facilitate attachment to the tether **3240**. The cap **3180** and pin **3051** also mitigate damage to the foam body **3002** during recapture of the device **3000**. The cap **3180** also creates an atraumatic surface for the hub **3050** of the frame **3040**. For example, the cap **3180** may prevent the hub **3050** from cutting through the foam body **3002** as the device **3000** is collapsed into an access sheath. Without the cap **3180**, the sharp edges of the hub **3050** may shear through the foam body **3002** during recapture of the device **3000** into the access sheath.

6. Loading System

[0350] FIG. **91** is a side view of an embodiment of a loading system **3200** for loading the device **3000** into a delivery catheter **3220**. The system **3200** includes a loading tool **3210**. The loading tool **3210** has a conical portion **3212**, having a distal opening **3213**, and a cylindrical portion **3214**. The delivery catheter **3220** extends through the cylindrical portion **3214** with a distal end **3222** of the delivery catheter **3220** located within the cylindrical portion **3214**. A pusher **3230** extends through the delivery catheter **3220**. A tether **3240** (see FIG. **92**) is attached to the device **3000** and extends through the loading tool **3210**, the delivery catheter **3220** and the pusher **3230**. The tether **3240** and pusher **3230** are pulled in the proximal direction while the delivery catheter **3220** and the loading tool **3210** are held stationary. The device **3000** is compressed laterally by the conical portion **3212** as the device **3000** is pulled proximally by the tether **3240** through the loading tool **3210**. A distal end **3232** of the pusher **3230** remains adjacent to the proximal end **3004** of the device **3000** as the device **3000** is loaded into the delivery catheter **3220**. The removable tether **3240**, which may be fabricated from ultra-high molecular weight polyethylene (UHMWPE), is used to attach the implant to the delivery catheter.

[0351] In some embodiments, the conical portion **3212** of the loading tool **3210** has a chamfered distal edge of approximately 45°-75° (degrees), preferably 60°. In some embodiments, the conical portion **3212** has a distal inner diameter (ID) greater than the outer diameter (OD) of the device **3000** and an angle of ideally between 15° and 25°, and in one implementation about 20°, to appropriately collapse the anchors **3090**, **3094** which may protrude off the foam body **3002** surface at an angle of 30° or about 30°. The gradual taper of the loading tool **3210** ensures, for example, that the frame **3040** folds evenly without crossovers or extra strain.

7. Delivery System

[0352] FIG. **92** is a side view of a schematic of a delivery system **3201** for delivering the device **3000**. The delivery system **3000** includes the delivery catheter **3220** having a distal end **3222** and a proximal end **3224**. The delivery system **3000** includes the pusher **3230** having the distal end **3232** and a proximal end **3234**. The tether **3240** includes a first end **3242** and a second end **3244**. A restraint **3246** secures the first and second ends **3242**, **3244**.

[0353] To deliver the device **3000** to the LAA, an access sheath is placed across the interatrial septum into the LAA through which the delivery catheter **3220**, containing the device **3000**, is placed. The device **3000** is loaded into the distal end **3222** of the delivery catheter **3220** using the loading tool **3210**, either at the point of manufacture or at the treatment site. To load the implant, the pusher and tether are pulled proximally, collapsing the implant as it enters the tip of the delivery catheter. Once the loaded delivery catheter is placed through the sheath into the LAA, the pusher rod is held steady as the delivery catheter and access sheath are simultaneously retracted proximally, deploying the implant

[0354] The tether **3220** passes from the proximal end of the delivery catheter **3220**, through the catheter pusher **3230**, around the implant tether pin **3051**, and back through the delivery catheter **3220**. When both ends of the tether **3230** (held together by the restraint **3246** at the proximal end of the catheter) are pulled, the device **3000** is pulled into the delivery catheter **3220**. Once the device **3000** is properly placed in the anatomy, one end of the tether **3240** is cut and the entire tether **3240** can be removed from the system, disengaging from the pin **3051**.

8. Tether Release System

[0355] FIGS. **93A** and **93B** are proximal and distal perspective views respectively of a tether release system **3400**. The release system **3400** includes a tube **3420** and a lock **3402**. The tube **3420** has a proximal end **3422** and a distal end **3424**. An opening **3426** extends through the tube **3420**.

[0356] The lock **3402** includes a proximal end **3404** and a distal end **3406**. An opening **3408** defined by a sidewall **3409** extends through the lock **3402** from the proximal end **3404** to the distal end **3406**. The tube **3420** extends through the opening **3408** at the proximal end **3404** of the lock **3402** and to the distal end **3406** of the opening **3408**. The sidewall **3409** of the lock **3402** has a first groove **3410** extending longitudinally from the proximal end **3404** to the distal end **3406** and extending radially partially through the thickness of the sidewall **3409**. The sidewall **3409** of the lock **3402** has a second groove **3412** extending longitudinally from the proximal end **3404** partially along the sidewall **3409** toward the distal end **3406** and radially partially through the thickness of the sidewall **3409**.

[0357] The tether **3240** includes a first end **3243** and the second end **3245**. The tether **3240** extends distally from the first end **3243** within the opening **3426** of the tube **3240** and out through the distal end **3424** of the tube **3420** to the cap **3180**. The tether **3240** extends distally into the opening **3188** of the cap **3180** and around the pin **3051** and back in the proximal direction. The tether **3240** then extends proximally into the first groove **3410** of the lock **3402**, around the proximal end **3404** of the lock **3402**, and then distally into and through the second groove **3412**. The tether **3240** terminates at the second end **3245** in a knot **3247**.

[0358] In use, the knot **3247** may be secured due to the relative location of the tube **4320** and the pusher **3230** inside the delivery catheter **3230**. The knot **3247** may be prevented from advancing distally due to the inner diameter of the distal end of the delivery catheter **3230** fitting tightly about the outer diameter of the lock **3402**. The pusher **3420** may be advanced distally to expose the second groove **3412** and free the knot **3247** from restraint. When the proximal end of the tether **3240** is pulled proximally, the knot **3247** slides distally through the second groove **3412**, advances around the proximal end **3404** of the lock **3402**, advances distally through the first groove **3410**, into the cap **3180** and around the pin **3051**, and then distally through the opening **3408** of the lock **3402** and can be retrieved with the pusher **3230**.

9. Off-Axis Delivery and Deployment

[0359] The device **3000** may be deployed off-axis within an LAA while still providing a complete, stable, and atraumatic seal. In some embodiments, the device **3000** may be deployed at an angle of at least about 15° or 25° and in some embodiments as much as 35° or 45°, for example, relative to a central longitudinal LAA axis and still provide an effective seal. The LAA axis here is defined as the geometric center of the ostium to the LAA, and tracks the best fit geometric center of the LAA cavity.

[0360] This ability of the device **3000** to be deployed off-axis is due in part to the relatively thick, compressible foam body **3002** material, the compliant frame **3040** and the cylindrical shape of the device **3000** with the foam bumper **3026**. The device **3000** is stable within the LAA despite having a length that is less than the diameter, or having $L/D < 1$. As described, the length may be 20 mm for the device **3000** having an OD of both 27 mm and 35 mm. Thus not only is flexibility and simplicity allowed with manufacturing processes by having one length, but also stability and effectiveness of the device in use. Further, the axial compressibility of the bumper **3026**, combined with the axially compliant frame **3040**, allows a 20 mm long device **3000** to be placed within a 10

mm deep LAA, whereas existing LAA closure devices require longer landing zones, or at least landing zones equal to the size of the length of the metallic frame.

[0361] Various modifications to the implementations described in this disclosure will be readily apparent to those skilled in the art, and the generic principles defined herein can be applied to other implementations without departing from the spirit or scope of this disclosure. Thus, the disclosure is not intended to be limited to the implementations shown herein, but is to be accorded the widest scope consistent with the claims, the principles and the novel features disclosed herein. The word “example” is used exclusively herein to mean “serving as an example, instance, or illustration.” Any implementation described herein as “example” is not necessarily to be construed as preferred or advantageous over other implementations, unless otherwise stated.

[0362] Certain features that are described in this specification in the context of separate implementations also can be implemented in combination in a single implementation. Conversely, various features that are described in the context of a single implementation also can be implemented in multiple implementations separately or in any suitable sub-combination. Moreover, although features can be described above as acting in certain combinations and even initially claimed as such, one or more features from a claimed combination can in some cases be excised from the combination, and the claimed combination can be directed to a sub-combination or variation of a sub-combination.

[0363] Similarly, while operations are depicted in the drawings in a particular order, this should not be understood as requiring that such operations be performed in the particular order shown or in sequential order, or that all illustrated operations be performed, to achieve desirable results. Additionally, other implementations are within the scope of the following claims. In some cases, the actions recited in the claims can be performed in a different order and still achieve desirable results.

[0364] It will be understood by those within the art that, in general, terms used herein are generally intended as “open” terms (e.g., the term “including” should be interpreted as “including but not limited to,” the term “having” should be interpreted as “having at least,” the term “includes” should be interpreted as “includes but is not limited to,” etc.). It will be further understood by those within the art that if a specific number of an introduced claim recitation is intended, such an intent will be explicitly recited in the claim, and in the absence of such recitation no such intent is present. For example, as an aid to understanding, the following appended claims may contain usage of the introductory phrases “at least one” and “one or more” to introduce claim recitations. However, the use of such phrases should not be construed to imply that the introduction of a claim recitation by the indefinite articles “a” or “an” limits any particular claim containing such introduced claim recitation to embodiments containing only one such recitation, even when the same claim includes the introductory phrases “one or more” or “at least one” and indefinite articles such as “a” or “an” (e.g., “a” and/or “an” should typically be interpreted to mean “at least one” or “one or more”); the same holds true for the use of definite articles used to introduce claim recitations. In addition, even if a specific number of an introduced claim recitation is explicitly recited, those skilled in the art will recognize that such recitation should typically be interpreted to mean at least the recited number (e.g., the bare recitation of “two recitations,” without other modifiers, typically means at least two recitations, or two or more recitations). Furthermore, in those instances where a convention analogous to “at least one of A, B, and C, etc.” is used, in general such a construction is intended in the sense one having skill in the art would understand the convention (e.g., “a system having at least one of A, B, and C” would include but not be limited to systems that have A alone, B alone, C alone, A and B together, A and C together, B and C together, and/or A, B, and C together, etc.). In those instances where a convention analogous to “at least one of A, B, or C, etc.” is used, in general such a construction is intended in the sense one having skill in the art would understand the convention (e.g., “a system having at least one of A, B, or C” would include but not be limited to systems that have A alone, B alone, C alone, A and B together, A and C together, B

and C together, and/or A, B, and C together, etc.). It will be further understood by those within the art that virtually any disjunctive word and/or phrase presenting two or more alternative terms, whether in the description, claims, or drawings, should be understood to contemplate the possibilities of including one of the terms, either of the terms, or both terms. For example, the phrase “A or B” will be understood to include the possibilities of “A” or “B” or “A and B.”

Claims

1-5. (canceled)

6. A left atrial appendage occlusion device, comprising: a foam body having a closed proximal end; and an expandable frame comprising a pin configured to engage a tether, the pin having a longitudinal axis with first and second opposed pin sides transverse to the pin longitudinal axis, wherein the pin is configured to engage the tether such that the tether extends distally on the first pin side, wraps around the pin, and extends back proximally on the second pin side.

7. The left atrial appendage occlusion device of claim 6, wherein the pin is located within a proximal hub of the expandable frame on a distal side of the proximal end of the foam body.

8. The left atrial appendage occlusion device of claim 6, wherein a distal end of the foam body comprises one or more radiopaque markers.

9. The left atrial appendage occlusion device of claim 6, wherein the pin extends laterally across a central lumen of the expandable frame.

10. The left atrial appendage occlusion device of claim 6, wherein the closed proximal end is configured to be penetrated by the tether.

11. The left atrial appendage occlusion device of claim 6, wherein the left atrial appendage occlusion device is configured to be implanted such that a central longitudinal axis of the device is oriented non-parallel to a primary longitudinal axis of the left atrial appendage when the left atrial appendage occlusion device is deployed in the left atrial appendage.

12. The left atrial appendage occlusion device of claim 11, wherein the central longitudinal axis of the left atrial appendage occlusion device is configured to form an angle with the primary longitudinal axis of the left atrial appendage of at least about 15 degrees.

13. The left atrial appendage occlusion device of claim 6, wherein the left atrial appendage occlusion device in an unconstrained state has an axial length (L) and an outer diameter (D), and wherein a ratio L/D is less than 1.

14. The left atrial appendage occlusion device of claim 6, wherein the left atrial appendage occlusion device further comprises a thromboresistant coating or layer on a proximal end of the foam body.

15. The left atrial appendage occlusion device of claim 6, wherein the expandable frame further comprises: plurality of reentry struts configured to extend radially outwardly from a central hub and a tubular body configured to extend distally from radially-outer ends of the plurality of reentry struts, and a plurality of anchors extending radially outwardly from the tubular body and penetrating the foam body.

16. The left atrial appendage occlusion device of claim 6, wherein the foam body of the left atrial appendage occlusion device comprises a foam side wall that extends distally beyond a distal end of the frame to form a distal, atraumatic bumper.

17. A method for left atrial appendage occlusion, the method comprising: providing a left atrial appendage occlusion device comprising: a foam body having a closed proximal end; and an expandable frame comprising a pin, the pin having a longitudinal axis with first and second opposed pin sides transverse to the pin longitudinal axis; wrapping a tether around the pin of the expandable frame such that the tether extends distally on the first pin side and extends back proximally on the second pin side.

18. The method of claim 17, wherein the pin is located within a proximal hub of the expandable

frame on a distal side of the proximal end of the foam body.

19. The method of claim 17, wherein a distal end of the foam body comprises one or more radiopaque markers.

20. The method of claim 17, wherein the pin extends laterally across a central lumen of the expandable frame.

21. The method of claim 17, further comprising penetrating the closed proximal end with the tether.

22. The method of claim 17, further comprising implanting the left atrial appendage occlusion device such that a central longitudinal axis of the device is oriented non-parallel to a primary longitudinal axis of the left atrial appendage when the left atrial appendage occlusion device is deployed in the left atrial appendage.

23. The method of claim 22, wherein the central longitudinal axis of the left atrial appendage occlusion device is configured to form an angle with the primary longitudinal axis of the left atrial appendage of at least about 15 degrees.

24. The method of claim 17, further comprising expanding the left atrial appendage occlusion device to an unconstrained state such that the left atrial appendage occlusion device has an axial length (L) and an outer diameter (D), and wherein a ratio L/D is less than 1.

25. The method of claim 17, wherein the left atrial appendage occlusion device further comprises a thromboresistant coating or layer on a proximal end of the foam body.

26. The method of claim 17, wherein the expandable frame further comprises: plurality of reentry struts configured to extend radially outwardly from a central hub and a tubular body configured to extend distally from radially-outer ends of the plurality of reentry struts, and a plurality of anchors extending radially outwardly from the tubular body and penetrating the foam body.

27. The method of claim 17, wherein the foam body of the left atrial appendage occlusion device comprises a foam side wall that extends distally beyond a distal end of the frame to form a distal, atraumatic bumper.
